数字时代的历史研究：“文人共和国地图”项目的反思

Historical Research in a Digital Age: Reflections from the Mapping the Republic of Letters Project

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G04-P001-001 | 英文原文

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中文译文 | G04-P001-001

AHR Forum

G04-P001-002 | 英文原文（原段接续 / 分段）

Historical Research in a Digital Age: Reflections from the Mapping the Republic of Letters Project

中文译文 | G04-P001-002

数字时代的历史研究：从“文人共和国”地图绘制项目中的反思

G04-P001-003 | 英文原文

DAN EDELSTEIN, PAULA FINDLEN, GIOVANNA CESERANI, CAROLINE WINTERER, AND NICOLE COLEMAN

中文译文 | G04-P001-003

DAN EDELSTEIN, PAULA FINDLEN, GIOVANNA CESERANI, CAROLINE WINTERER, AND NICOLE COLEMAN

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G04-P001-005 | 英文原文

About eight years ago, we rather serendipitously found ourselves in the early stages of a group research project that seemed to have potential as an experiment with how to do digital history. The result was Mapping the Republic of Letters, a collaborative endeavor based at Stanford University. The project consists of a growing number of wide-ranging case studies, including "British Architects on the Grand Tour in Eighteenth-Century Italy," the subject of the second article in this *AHR* Forum. The case studies engage in multiple ways with the early modern Republic of Letters, and, as each is based on a different source of information, they pose a variety of problems for data-handling and unique challenges for visualizing it. All of them are available on our common website.[1]

中文译文 | G04-P001-005 | 原文注号：1

大约八年前，我们偶然发现了一个团体研究项目的初期阶段，该项目似乎具有作为数字历史研究实验的潜力。其成果即为《文人共和国的地图绘制》，这是一项在斯坦福大学进行的合作项目。该项目包含大量广泛的案例研究，其中包括"十八世纪意大利大旅行中的英国建筑师"这一主题，该主题正是本AHR论坛第二篇文章的主题。这些案例研究以多种方式与早期现代文人共和国产生关联；由于每个案例都基于不同的信息来源，因此它们在数据处理方面面临各种问题，在可视化方面也带来独特的挑战。所有这些资料都可在我们的共同网站上获取。1

G04-P001-006 | 英文原文

At one time, archival material remained more or less in the archives. Historians inclined to use such materials routinely performed acts of intellectual pilgrimage, as many of us still do today, to be initiated into this essential rite of historical apprenticeship. They traveled, learned how to use their archives onsite, and extracted hard-earned information like Forty-Niners panning for gold, returning home with pages of notes, perhaps even a bit of microfilm to read on a machine or print out on that nauseatingly smelly paper whose pungent chemical odor was anything but the nirvana of

中文译文 | G04-P001-006

曾经，档案资料大多仍保存在档案馆中。倾向于使用这些资料的史学家们，就像我们今天许多人一样，会进行一种知识探索之旅，以完成这一重要的历史学徒仪式。他们四处游历，学习如何在现场使用档案资料，并提取那些来之不易的信息——就像四十人淘金者那样，带着一叠叠笔记回家，或许还带有一小段缩微胶片，可以在机器上阅读或打印出来，而那令人作呕的臭味纸张上的化学气味，根本算不上是极乐。

G04-P001-007 | 英文原文

Research for this project was initially made possible by a Stanford Presidential Fund for Innovation in the Humanities Grant, and subsequently by funding from the Stanford Humanities Center, the Dean of Research, the Vice Provost of Graduate Education, the Vice Provost of Undergraduate Education, and the Vice Provost of Online Learning. Additional funding came from a National Endowment for the Humanities (NEH) "Digging into Data" Challenge Grant (HJ-50056, 2009–2011), as well as an NEH Digital Humanities Implementation Grant (HK-50087, 2013–2016). Important collaborators include the Electronic Enlightenment Project (Oxford University), the Packard Humanities Institute, DensityDesign Research Lab (Politecnico di Milano), Stanford University Libraries, the Cultures of Knowledge Project (Oxford University), and the Circulation of Knowledge and Learned Practices Project (Huygens Instiuut). The "Mapping the Republic of Letters" Project is part of CESTA, Stanford's Center for Spatial and Textual Analysis. The authors also wish to thank Keith Baker, Giorgio Caviglia, Zephyr Frank, Anthony Grafton, Glauco Mantegari, Robert McNamee, and the wonderful teams of graduate and undergraduate students who have worked with us on this project.

中文译文 | G04-P001-007

该项目的研究最初得益于斯坦福大学人文创新总统基金资助，随后又获得斯坦福人文中心、研究院长、研究生教育副校长、本科生教育副校长以及在线学习副校长提供的资金支持。此外，还获得了美国国家人文基金会“深入数据”挑战基金（HJ-50056，2009–2011）及数字人文实施基金（HK-50087，2013–2016）的资助。重要的合作方包括电子启蒙项目（牛津大学）、帕卡德人文研究所、米兰理工大学密度设计研究实验室、斯坦福大学图书馆、知识文化项目（牛津大学），以及知识与学习实践传播项目（胡伊根斯学院）。“文人共和国地图绘制”项目属于斯坦福大学空间与文本分析中心CESTA的一部分。作者们还想感谢基思·贝克、乔尔吉奥·卡维利亚、泽菲尔·弗兰克、安东尼·格拉夫顿、格劳科·曼特加里、罗伯特·麦克纳米，以及那些参与该项目的研究生和本科生团队。

G04-P001-008 | 英文原文

[1] See http://republicofletters.stanford.edu/publications/.

中文译文 | G04-P001-008

1 参见http://republicofletters.stanford.edu/publications/。

G04-P001-009 | 英文原文

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中文译文 | G04-P001-009

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G04-P002-002 | 英文原文

the past.[2] A much smaller subset of the scholarly community, whose delight in documents long predated the rise of the professional historian of the mid-nineteenth century, ambitiously undertook the laborious project of transcription, creating critical editions of key documents that form the backbone of many histories of famous figures, defining moments, and other noteworthy events.[3] This tiny portion of the infinite documentation of the past found its way into print, while the majority remained in the archive.

中文译文 | G04-P002-002 | 原文注号：2、3

过去。2 学术界的子集规模要小得多，他们对文献的喜爱早在十九世纪中叶职业历史学家兴起之前就存在了。他们雄心勃勃地承担了繁琐的抄写工作，对重要文献进行了关键版式整理，这些文献构成了许多关于名人、重要时刻及其他值得注意事件的历史的核心内容。3 这有限的一部分过去的无尽文献被印成书籍，而大部分则仍保存在档案中。

G04-P002-003 | 英文原文（原段接续 / 分段）

In the 1960s, social science historians began to consider how computers facilitated quantitative analysis. An entire generation or two of historians discovered the punch card and began to code. During the 1970s, a steady stream of scholarship emerged from these early experiments with historical datasets. This approach appealed especially to historians who wished to migrate away from traditional political and intellectual history, writing the kind of historical sociology and family, labor, and economic history whose grandest manifestos might be Emmanuel Le Roy Ladurie's 1974 essay "History That Stands Still," which summarized well the goals and ambitions of the early Annales School, and historical sociologist Charles Tilly's evocatively titled *Big Structures, Large Processes, Huge Comparisons* a decade later.[4] The foundations of the new social history of the 1960s through the 1980s rest upon these impressive early experiments with big historical data and its insistence on the value of cliometrics.

中文译文 | G04-P002-003 | 原文注号：4

在20世纪60年代，社会科学史学家开始思考计算机如何促进定量分析。一代或两代史学家发现了磁卡技术，并开始进行编码工作。到了70年代，这些早期对历史数据集的实验衍生出大量研究成果。这种研究方法特别吸引那些希望摆脱传统政治与思想史研究、转而从事历史社会学以及家庭、劳动与经济史研究的学者。埃曼纽尔·勒·罗伊·拉杜里在1974年的文章《静止的历史》中很好地总结了早期安纳尔斯学派的目标与抱负，而十年后，历史社会学家查尔斯·蒂利则以富有表现力的标题《大结构、大过程、大比较》来阐述这一领域的发展。从20世纪60至80年代的新型社会史基础，便建立在这些令人印象深刻的早期大历史数据实验以及对计量史学价值的重视之上。

G04-P002-004 | 英文原文（原段接续 / 分段）

This was the old era of big history with big data. We are now in the early stages of a new age of historical data that offers new possibilities as well as challenges. The past fifteen years have seen a gradual but perceptible shift in where the archive is located as an ever greater portion of the historical record migrates piecemeal from boxes and books to PDFs and JPGs, and a vast sea of digitized text. A growing number of manuscripts can be viewed online, and hefty multivolume document collections are now digitized, indexed, and easily searchable. Historians increasingly find themselves utilizing digital databases as the idea of the searchable document and the virtual archive reorganize how libraries, research institutes, teams of scholars, and even individual researchers present and share interesting sources. To take one example, bibliometrics once belonged primarily to librarians and archivists and now seems to interest a broad, interdisciplinary array of scholars as an analytic research tool. For all these reasons, it is not surprising that the history of information has become an important topic.[5] To paraphrase Raymond Williams, "information" is one of the keywords of the

中文译文 | G04-P002-004 | 原文注号：5

这是大数据时代的大历史时代的旧模式。如今，我们正处于历史数据新时代的初期阶段，这一时代既带来新的可能性，也带来挑战。过去十五年里，档案存放地点发生了逐渐但明显的变化：越来越多的历史记录从箱子与书籍逐步转移到PDF和JPG格式，以及大量的数字化文本中。越来越多的写本可以在线上查看，庞大的多卷文献集也已被数字化、编入索引并易于搜索。历史学家越来越多地利用数字数据库作为可检索文档的载体，虚拟档案则重新调整了图书馆、研究机构、学者团队乃至个人研究人员展示与分享有趣资料的方式。以图书馆学计量学为例，这一领域曾经主要属于图书馆员和档案管理员所关注，如今却成为众多跨学科学者作为分析研究工具的重要领域。基于以上原因，信息史成为重要话题并不令人意外。用雷蒙德·威廉斯的话来说，"信息"是关键术语之一。

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G04-P002-006 | 英文原文

[2] For an enjoyable meditation on the origins of these historical practices, see Carolyn Steedman, *Dust: The Archive and Cultural History* (New Brunswick, N.J., 2002). The early modern archive in particular is beautifully evoked in Arlette Farge, *The Allure of the Archives*, trans. Thomas Scott-Railton (New Haven, Conn., 2013).

中文译文 | G04-P002-006

2 若想深入思考这些历史习俗的起源，可参考卡罗琳·斯蒂德曼所著的《尘埃：档案与文化历史》（新伯伦斯威克，新泽西州，2002年）。阿莱特·法尔吉所著的《档案的魅力》一书对早期现代档案制度进行了精彩的描述，该书由托马斯·斯科特-雷尔顿翻译（康涅利克州纽黑文，2013年）。

G04-P002-007 | 英文原文（原段接续 / 分段）

[3] On this subject, a classic point of departure is Anthony Grafton, *The Footnote: A Curious History* (Cambridge, Mass., 1997).

中文译文 | G04-P002-007

3 关于这一主题，安东尼·格拉夫顿的《脚注：一段奇特的史话》（马萨诸塞州剑桥，1997年）是一个经典的起点。

G04-P002-008 | 英文原文（原段接续 / 分段）

[4] Emmanuel Le Roy Ladurie, "History That Stands Still," in Ladurie, *The Mind and Method of the Historian*, trans. Siân Reynolds and Ben Reynolds (Chicago, 1981), 1–27; Charles Tilly, *Big Structures, Large Processes, Huge Comparisons* (New York, 1984). The most recent overview of the Annales School is André Burguière, *The Annales School: An Intellectual History*, trans. Jane Marie Todd (Ithaca, N.Y., 2009). For early methodological reflections on quantitative history, see William O. Aydelotte, "Quantification in History," *American Historical Review* 71, no. 3 (April 1966): 803–825; William O. Aydelotte, Robert William Fogel, and Allan G. Bogue, eds., *The Dimensions of Quantitative Research in History* (Princeton, N.J., 1972).

中文译文 | G04-P002-008

4 埃曼努埃尔·勒罗伊·拉杜里，《静止的历史》，载于《拉杜里的历史学家的心智与方法》，由西恩·雷诺兹与本·雷诺兹译（芝加哥，1981年），第1–27页；查尔斯·蒂利，《大型结构、大规模过程与巨大比较》（纽约，1984年）。关于 Annales 学派的最新综述，可参阅安德烈·布尔吉耶雷所著的《Annales 学派：一部思想史》，由简·玛丽·托德译（伊萨卡，纽约州，2009年）。关于定量历史研究的方法论早期思考，可参考威廉·O·艾德尔洛特所撰《历史中的量化研究》，载于《美国历史评论》第71卷第3期（1966年4月），第803–825页；威廉·O·艾德尔洛特、罗伯特·威廉·福格尔与艾伦·G·博格编，《历史定量研究的维度》（普林斯顿，新泽西州，1972年）。

G04-P002-009 | 英文原文（原段接续 / 分段）

[5] See, for instance, Paul N. Edwards, Lisa Gitelman, Gabrielle Hecht, Adrian Johns, Brian Larkin, and Neil Safier, "Historical Perspectives on the Circulation of Information," *AHR* Conversation, *American Historical Review* 116, no. 5 (December 2011): 1393–1435.

中文译文 | G04-P002-009

5 例如，参见保罗·N·爱德华兹、丽莎·吉特曼尔、加布里埃尔·赫希特、阿德里安·约翰斯、布莱恩·拉金和尼尔·萨菲尔所著的《信息流通的历史视角》，载于《美国历史评论》第116卷，第5期（2011年12月）：1393–1435页。

G04-P002-010 | 英文原文

中文译文 | G04-P002-010

《美国历史评论》2017年4月刊

原PDF第3页（原刊第402页）

G04-P003-002 | 英文原文

early twenty-first century, and so too is "data."[6] The question, however, is not how to write the history of information but what to do with all of it, as historians in an era when we have a burgeoning array of techniques and technologies available to help us scrape, mine, curate, analyze, and visualize data.

中文译文 | G04-P003-002 | 原文注号：6

二十一世纪初，"数据"也是如此。然而，问题并非如何书写信息的历史，而是该如何处理所有这些数据——在这样一个拥有大量技术和方法可供我们收集、挖掘、整理、分析与可视化数据的时代，历史学家们需要思考如何运用这些技术。

G04-P003-003 | 英文原文（原段接续 / 分段）

Too much information, it turns out, might be a good thing. Perhaps the first-year graduate student, overwhelmed by shelves full of books in his or her field, may not immediately agree. "Information overload" is a common reaction—now as in the past—to the vast collections of knowledge that have accumulated in our libraries, and are increasingly migrating online.[7] But sometimes, to paraphrase Jean-Jacques Rousseau, the cure lies in the poison.[8] "Big data" is the name commonly given to the exponential increase in information, whether it results from the digitization of analog data stores or is collected in real time from remote sensors, the web, or mobile devices.[9] The difference between these datasets and that multitude of books is the potential for discovery and analysis. "Big data" refers not to size alone, as the term suggests, but to the interconnectedness of resources. For our purposes, big data is what you get when those shelves of books have been digitized, structured, and interlinked in such a way that the information contained within them can be filtered, plotted, measured, parsed, and visualized. The scholar starting out in a new field still has to ingest and process the literature the old-fashioned way—by reading it. But we can put computational tools to work to help with the ingesting and processing and to inspire other questions we might ask with this material as we look across, as well as within, our sources.

中文译文 | G04-P003-003 | 原文注号：7、8、9

事实证明，过多的信息或许是一件好事。也许第一年毕业的研究生们，面对堆积如山的书籍，不会立刻同意这一点。"信息过载"是一种常见的反应——过去如此，现在也如此——因为我们的图书馆中积累了大量的知识，而这些知识越来越多地转移到线上。7 但有时，用让-雅克·卢梭的话来说，治疗之道在于毒药。8 "大数据"是指信息的指数级增长，无论是由于模拟数据存储的数字化，还是从远程传感器、网络或移动设备实时收集而来。9 这些数据集与那些书籍的区别在于其潜在的发现与分析能力。"大数据"并不仅仅指规模大小，而是指资源的相互关联性。对于我们的目的而言，大数据指的是那些书籍被数字化、结构化并相互连接后，其中的信息可以被筛选、绘制、测量、解析和可视化。刚开始从事新领域研究的学者仍然需要以传统的方式来阅读文献并进行处理。但我们可以利用计算工具来帮助这种处理过程，并激发我们针对这些材料提出的其他问题，同时审视我们的来源，无论是内部还是外部。

G04-P003-004 | 英文原文（原段接续 / 分段）

Few humanists would argue that reading books and manuscripts is a bad thing. Indeed, the newfound availability of large datasets in the humanities at times conjures up the worrisome specter of digitally savvy sorcerer's apprentices who, thanks to new databases, "cite anything and construe nothing."[10] Searchable, sortable data can make even beginning scholars look erudite and clever. An eager researcher can accumulate

中文译文 | G04-P003-004 | 原文注号：10

很少有人文主义者认为阅读写本和书是件坏事。事实上，人文学科中大量数据集的突然出现，有时让人联想到那些精通数字技术的学徒——借助新数据库，他们“能引用任何内容，却无法理解其含义”。可搜索、可排序的数据甚至能让初入行的学者显得博学而聪明。热切的研究者能够积累大量资料

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[6] Raymond Williams, *Keywords: A Vocabulary of Culture and Society* (New York, 1976). On the history of information, see especially C. A. Bayly, *Empire and Information: Intelligence Gathering and Social Communication in India, 1780–1870* (Cambridge, 1996); Daniel R. Headrick, *When Information Came of Age: Technologies of Knowledge in the Age of Reason and Revolution, 1700–1850* (Oxford, 2001); and Jacob Soll, *The Information Master: Jean-Baptiste Colbert's Secret State Intelligence System* (Ann Arbor, Mich., 2009). Daniel Rosenberg has a project underway on the history of data, with related work published as "Data before the Fact," in Lisa Gitelman, ed., *"Raw Data" Is an Oxymoron* (Cambridge, Mass., 2013), 15–40; "Enter Data," in Katherine Behar, ed., *Data's Entry* (Istanbul, 2016), 14–33; and "An Archive of Words," in Lorraine Daston, ed., *Sciences of the Archive* (Chicago, forthcoming 2017).

中文译文 | G04-P003-006

6 雷蒙德·威廉姆斯，《关键词：文化与社会的词汇》（纽约，1976年）。关于信息的历史，可参考C. A. 贝利所著的《帝国与信息：1780–1870年印度情报收集与社交传播》（剑桥，1996年）；丹尼尔·R. 赫德里克所著的《当信息成熟之时：理性与革命时代的知识技术，1700–1850年》（牛津，2001年）；以及雅各布·索尔所著的《信息大师：让-巴蒂斯特·科尔贝的秘密国家情报系统》（密歇根州安阿伯，2009年）。丹尼尔·罗森伯格正在开展一项关于数据历史的研究项目，相关成果发表于丽莎·吉特曼主编的《原始数据是一种矛盾》（马萨诸塞州剑桥，2013年）中的"数据之前"篇，第15–40页；以及凯瑟琳·贝哈尔主编的《数据的进入》（伊斯坦布尔，2016年）中的"进入数据"篇，第14–33页；还有洛林·达斯顿主编的《档案科学》（芝加哥，即将于2017年出版）中的"词汇的档案"篇。

4. 所有年份和阿拉伯数字必须原样保留，不得更改或替换。需保留的年份为：1700、1780、1850、1870、1976、1996、2001、2009、2013、2016、2017。

G04-P003-007 | 英文原文（原段接续 / 分段）

[7] Daniel Rosenberg, "Early Modern Information Overload," *Journal of the History of Ideas* 64, no. 1 (2003): 1–9; Ann M. Blair, *Too Much To Know: Managing Scholarly Information before the Modern Age* (New Haven, Conn., 2010). More generally, see Peter Burke, *A Social History of Knowledge: From Gutenberg to Diderot* (Cambridge, 2000).

中文译文 | G04-P003-007

7 丹尼尔·罗森伯格，《早期现代信息过载》，《思想史杂志》第64卷，第1期（2003年）：1–9页；安·M·布莱尔，《太多无法理解：现代之前的知识管理》（康涅狄格州纽黑文，2010年）。更全面的参考可参见彼得·伯克所著的《知识的社会历史：从古腾堡到迪德罗》（剑桥，2000年）。

G04-P003-008 | 英文原文（原段接续 / 分段）

[8] Jean Starobinski, *Le remède dans le mal: Critique et légitimation de l'artifice à l'âge des Lumières* (Paris, 1989).

中文译文 | G04-P003-008

8 让·斯塔罗宾斯基，《恶中的装饰：对艺术技巧的批判与验证——在光之时代》（巴黎，1989年）。

G04-P003-009 | 英文原文（原段接续 / 分段）

[9] See Danah Boyd and Kate Crawford, "Six Provocations for Big Data," paper presented at "A Decade in Internet Time: Symposium on the Dynamics of the Internet and Society," University of Oxford, September 21, 2011, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1926431; Viktor Mayer-Schönberger and Kenneth Cukier, *Big Data: A Revolution That Will Transform How We Live, Work, and Think* (Boston, 2013). For a humanistic take on this debate, see Gitelman, *"Raw Data" Is an Oxymoron*; Lev Manovich, "Trending: The Promises and the Challenges of Big Social Data," in Matthew K. Gold, ed., *Debates in the Digital Humanities* (Minneapolis, 2012), 460–475; Christof Schöch, "Big? Smart? Clean? Messy? Data in the Humanities," *Journal of Digital Humanities* 2, no. 3 (2013), http://journalofdigitalhumanities.org/2–3/big-smart-clean-messy-data-in-the-humanities/.

中文译文 | G04-P003-009

9 参见 Danah Boyd 和 Kate Crawford 的论文《六个关于大数据的挑战》，该文发表于"互联网十年：互联网与社会动态研讨会"上，牛津大学，2011年9月21日，https://papers.ssrn.com/sol3/papers.cfm?abstract_id1⁄41926431；Viktor Mayer-Schoenberger 和 Kenneth Cukier 的著作《大数据：一场将改变我们的生活方式、工作方式和思维方式的革命》（波士顿，2013年）。关于这一讨论的人文主义视角，可参考 Gitelman 的"原始数据"是一个矛盾的说法；Lev Manovich 的"趋势：大社会数据的承诺与挑战"，载于 Matthew K. Gold 编著的《数字人文领域的讨论》（明尼阿波利斯，2012年），第460–475页；Christof Schoech 的"大？智能？清洁？混乱？人文领域的数据"，《数字人文杂志》第2卷第3期（2013年），http://journalofdigitalhu manities.org/2–3/big-smart-clean-messy-data-in-the-humanities/。

G04-P003-010 | 英文原文（原段接续 / 分段）

[10] Jonathan Barnes, quoted by Anthony Grafton in *Worlds Made by Words: Scholarship and Community in the Modern West* (Cambridge, Mass., 2009), 322.

中文译文 | G04-P003-010

10 乔纳森·巴恩斯，被安东尼·格拉夫顿在《文字创造的世界：现代西方的学术与社区》（马萨诸塞州剑桥，2009年）中引用，第322页。

G04-P003-011 | 英文原文

中文译文 | G04-P003-011

原PDF第4页（原刊第403页）

G04-P004-002 | 英文原文

historical mass very quickly. Superficial engagement with this material can also lead to gross misinterpretations when the context for interpreting data is missing and when one has read only what is available online. Stopping there, however, would be premature, indeed shortsighted. These perils notwithstanding, the digitization and datafication of historical documents can open up promising methodological avenues and opportunities for new insights.

中文译文 | G04-P004-002

历史资料会迅速被数字化。如果缺乏解读数据的背景，且仅阅读了网上的内容，对这些资料进行表面的研究也会导致严重的误解。然而，就此止步则过于仓促，实属短视。尽管存在这些风险，历史文献的数字化与数据化工作仍能开辟出有前景的研究方法路径，并为新的发现提供机会。

G04-P004-003 | 英文原文（原段接续 / 分段）

We began our project, Mapping the Republic of Letters, with these opportunities in mind and with an awareness of the growing number of projects that are digitizing letters and other key documents of scholarly, cultural, political, and religious life from the sixteenth through eighteenth centuries. Such projects generate tremendous amounts of valuable metadata about social networks, travel, institutional affiliations, publication history, and so forth. In the age of Edward Snowden, most people are now familiar with the concept of metadata, such as information about who communicated (or traveled) with whom, when, and where. Typically, metadata is everything except the "what" that was communicated. Mapping the Republic of Letters deals primarily with the metadata about letters or travels rather than with their actual content. Or rather, we use the metadata to produce maps, charts, and other data visualizations, and then refer back to the content to elucidate or complement what we see. Similarly (yes) to the NSA, we engage in "traffic analysis" to explore the breadth, shape, and hubs of intellectual networks.[11]

中文译文 | G04-P004-003 | 原文注号：11

我们启动"文人共和国地图绘制项目"时，便意识到了这些机遇，也意识到从十六至十八世纪学术、文化、政治及宗教领域的相关信件及其他重要文件正在被数字化，此类项目会产生大量关于社交网络、旅行、机构关联、出版历史等内容的宝贵元数据。在爱德华·斯诺登时代，大多数人已经熟悉元数据这一概念，即关于谁与谁进行通信（或旅行）的时间与地点等信息。通常而言，元数据指的是除所传递内容之外的所有信息。而"文人共和国地图绘制项目"主要关注的是与信件或旅行相关的元数据，而非其实际内容。更准确地说，我们利用元数据来制作地图、图表及其他数据可视化图表，随后再参考内容以进一步阐释或补充所看到的信息。类似于美国国家安全局的做法，我们通过"流量分析"来探索知识网络的广度、形态及核心节点。[11]

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G04-P004-005 | 英文原文

Since we began our project, the increase in available online data has been accompanied by a growing conversation about its impact and meaning for historical research. In the last couple of years alone, a string of new handbooks and manifestos have tackled "big data," laying out its promises for historians while introducing the various methodologies by which digital historians approach it.[12] In the pages of the *AHR*, Lara Putnam has recently shown how digital search—the now-widespread scholarly use of Google, JSTOR, and WorldCat, among others—is bringing many of the issues associated with big-data historical research to the doorsteps even of historians who have not explicitly adopted digital tools meant to quantify and visualize data.[13] Our reflections here, particularly about our experience with digital historical databases and how this has affected our work in early modern intellectual history, aim to contribute to this conversation. The publications emerging from our project all employ computational technologies and visualization techniques to make discoveries

中文译文 | G04-P004-005 | 原文注号：12、13

自我们开展该项目以来，在线数据量的增加伴随着关于其对历史研究的影响与意义的广泛讨论。仅在过去几年中，一系列新的手册与宣言便探讨了"大数据"问题，既阐述了其对历史学家的意义，也介绍了数字历史学家处理数据的各种方法。12 在《AHR》的期刊上，Lara Putnam最近指出，如今被学者广泛使用的数字搜索技术——如Google、JSTOR和WorldCat等——正在将与大数据历史研究相关的诸多问题引入到那些并未明确采用用于量化与可视化数据的数字工具的历史学家的工作之中。13 我们在此的思考，尤其是关于我们在数字历史数据库方面的经验以及这对早期现代思想史研究的影响，旨在为这一讨论做出贡献。我们项目所发布的成果均运用计算技术与可视化技术来推动发现。

G04-P004-006 | 英文原文

[11] For the U.S. government's practice of traffic analysis, see Vera R. Filby, *A Collection of Writings on Traffic Analysis* (Fort Meade, Md., 1993), http://www.governmentattic.org/8docs/NSA-TrafficAnalysis Monograph_1993.pdf.

中文译文 | G04-P004-006

11 关于美国政府交通分析的做法，可参考 Vera R. Filby 所著的《交通分析论文集》（马里兰州福特梅德，1993年），网址：http://www.governmentattic.org/8docs/NSA-TrafficAnalysis Monograph_1993.pdf。

G04-P004-007 | 英文原文（原段接续 / 分段）

[12] See Jo Guldi and David Armitage, *The History Manifesto* (Cambridge, 2014), chap. 4, for their take on big data's promise for historians. Among the recent proliferation of companions and handbooks for digital humanities, and digital history in particular, note Shawn Graham, Ian Milligan, and Scott Weingart, *Exploring Big Historical Data: The Historian's Macroscope* (London, 2016), which most explicitly addresses digital history as big data history. For a methodological reflection on big data history, see also Cameron Blevins, "The Perpetual Sunrise of Methodology," paper prepared for AHA Session 158, "Authoring Digital Scholarship for History: Challenges and Opportunities," 129th Annual Meeting of the American Historical Association, New York City, January 4, 2015, http://www.cameronblevins.org/posts/perpetual-sunrise-methodology/.

中文译文 | G04-P004-007

12 参见 Jo Guldi 和 David Armitage 所著的《历史宣言》（剑桥，2014年），第4章，其中对大数据对历史学家带来的机遇进行了探讨。在数字人文学及数字历史领域近期涌现的大量辅助资料和手册中，可参考 Shawn Graham、Ian Milligan 和 Scott Weingart 所著的《探索大型历史数据：历史学家的宏观视角》（伦敦，2016年），该书最明确地将数字历史视为大数据历史。关于大数据历史的方法论反思，另见 Cameron Blevins 在2015年1月4日于纽约市举行的美国历史协会第129届年会上的演讲"方法论的永恒日出"，该演讲为AHA第158次会议"为历史研究创造数字学术：挑战与机遇"会议准备，网址：http://www.cameronblevins.org/posts/perpetual-sunrise-methodology/。

G04-P004-008 | 英文原文（原段接续 / 分段）

[13] Lara Putnam, "The Transnational and the Text-Searchable: Digitized Sources and the Shadows They Cast," *American Historical Review* 121, no. 2 (April 2016): 377–402, with rich references on the recent conversation on history in the digital age. See also Ted Underwood, "Theorizing Research Practices We Forgot to Theorize Twenty Years Ago," *Representations* 127, no. 1 (2014): 64–72.

中文译文 | G04-P004-008

13 拉拉·普特南，《跨国性与可检索文本：数字化资源及其所留下的痕迹》，《美国历史评论》第121卷，第2期（2016年4月）：377–402页。该文包含关于数字时代历史研究最新讨论的丰富参考文献。另见泰德·安德伍德《我们二十年前忘记理论化的研究实践被重新理论化》，《表征》第127卷，第1期（2014年）：64–72页。

G04-P004-009 | 英文原文

American Historical Review April 2017

中文译文 | G04-P004-009

《美国历史评论》2017年4月

原PDF第5页（原刊第404页）

G04-P005-002 | 英文原文

about the past that would have been difficult, if not impossible, to reach by analog means. These scholarly outcomes have emerged in—indeed, could not exist outside of—a dialogue with a longstanding tradition of early modern intellectual historiography. They also grew out of research practices that our group established along the way, such as digital visualizations, tool development, publication format, and co-authored scholarship, which speak to how history might be done in a digital age.

中文译文 | G04-P005-002

关于过去，若非不可能，则通过传统方式难以了解。这些学术成果源于与早期现代思想史编纂传统之间的对话——事实上，若无此对话，这些成果便无法存在。它们还源自我们团队在过程中建立的研究方法，如数字可视化、工具开发、出版形式以及合作研究，这些方法揭示了数字时代如何进行历史研究。

G04-P005-003 | 英文原文（原段接续 / 分段）

Mapping the Republic of Letters is a research project with many heads. It contains two seventeenth-century (Athanasius Kircher and John Locke) and three eighteenth-century (Voltaire, Benjamin Franklin, and the Grand Tour) case studies.[14] Their geography primarily traverses Great Britain, Europe, and British North America, though each intellectual network makes strategic excursions that lie beyond these boundaries, raising questions about not only the nature of early modern cosmopolitanism and travel, but also the nature of global connections and communications. Historians of the Republic of Letters have also long debated how it evolved over time. Did the erudite *Respublica litteraria* of humanist scholars differ significantly from the worldly *République des lettres*? What can we learn by examining not only the most prolific correspondents who represent the strongest ties in a given network, but also the "long tail" of letter writers, often the vast majority in most instances, who sent no more than a single letter or a handful of letters? As David Lux and Harold Cook observe in an important essay inspired by the work of sociologist Mark Granovetter, understanding how people expand the reach of their network by "widening the network of weak ties" is an essential component of network analysis.[15]

绘制文人共和国的图谱是一项涉及多方面的研究项目。该项目包含两个十七世纪案例（阿塔纳修斯·基尔彻与约翰·洛克）和三个十八世纪案例（伏尔泰、本杰明·富兰克林及大旅行运动）。这些案例主要分布于英国、欧洲及英属北美地区，不过每个知识网络还涉及超出这些边界的战略性延伸，这引发了关于早期现代世界主义与旅行性质，以及全球联系与通信方式的问题。研究文人共和国的历史学家们也长期探讨该制度随时间如何演变。人文主义学者的博学文人共和国与世俗的文学共和国有何不同？通过研究那些在特定网络中代表最强联系的活跃通信者，以及那些“长尾”的写信人——在大多数情况下他们占绝大多数，且仅寄送一封或几封信件的人——我们可以获得哪些启示？正如大卫·卢克斯和哈罗德·库克在一篇受社会学家马克·格纳沃特研究启发的文章中所指出的那样，理解人们如何通过“扩大弱联系网络”来扩展其网络范围，是网络分析的重要组成部分。

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G04-P005-005 | 英文原文

One of our groups has focused on the German Jesuit polymath Athanasius Kircher (1602–1680). Paula Findlen, Suzanne Sutherland, and Iva Lelková have explored why Kircher's mid-seventeenth-century network was more cosmopolitan, in terms of geographic breadth, than those of his Enlightenment successors.[16] As a member of the Society of Jesus, Kircher could plug into an already established global missionary network while also making himself a useful participant in the scholarly Republic of Letters, and inserting himself successfully in Habsburg networks in Central Europe and Italian networks emanating to and from his adoptive city, Rome. (See Figure 1.)[17] A fine-grained analysis of the community that corresponded with him shows how his publishing projects grew and evolved with the expanding reach of the Jesuit missionary network into many different parts of the world. This research has been able to map the multiple ways in which Kircher self-consciously shaped and reshaped his religious, political, and intellectual networks to bolster his reputation as the man to know in the Eternal City, someone whose books and reputation circum-

中文译文 | G04-P005-005 | 原文注号：16、17

我们其中一组研究聚焦于德国耶稣会多才多艺的学者阿塔纳修斯·基尔彻（1602–1680）。Paula Findlen、Suzanne Sutherland 和 Iva Lelkov□a 探讨了为何基尔彻在十七世纪中叶建立的网络在地理广度上比他的启蒙时代继任者更为国际化。作为耶稣会成员，基尔彻能够融入已形成的全球传教网络，同时也在文人共和国学术领域成为重要的参与者，并成功融入了中欧的哈布斯堡网络以及来自其养城罗马的意大利网络。（见图1。）对与他交往的社群进行细致分析表明，随着耶稣会传教网络向世界各地扩展，他的出版项目也不断发展变化。这项研究揭示了基尔彻如何自觉地塑造和重塑其宗教、政治与学术网络，以提升他在永恒之城的地位，成为众人所知的专家，其著作与声誉也由此得到提升。

G04-P005-006 | 英文原文

[14] Interactive visualizations, datasets, and links to research articles related to these case studies can be found at http://republicofletters.stanford.edu/publications/.

中文译文 | G04-P005-006

与这些案例研究相关的交互式可视化图表、数据集及研究文章链接可在http://republicofletters.stanford.edu/publications/找到。

G04-P005-007 | 英文原文（原段接续 / 分段）

[15] David S. Lux and Harold J. Cook, "Closed Circles or Open Networks? Communicating at a Distance during the Scientific Revolution," *History of Science* 36, no. 2 (1998): 179–211, here 190; Mark S. Granovetter, "The Strength of Weak Ties," *American Journal of Sociology* 78, no. 6 (1973): 1360–1380. For a recent study that uses network analysis to explore the correspondence of imprisoned Protestants during the English wars of religion, see Ruth Ahnert and Sebastian E. Ahnert, "Protestant Letter Networks in the Reign of Mary I: A Quantitative Approach," *ELH* 82, no. 1 (2015): 1–33.

中文译文 | G04-P005-007

15 大卫·S·卢克斯与哈罗德·J·库克，《封闭圆圈还是开放网络？科学革命时期的远程交流》，《科学史》第36卷，第2期（1998年）：179–211页，此处为190页；马克·S·格拉诺维特，《弱关系的力量》，《美国社会学杂志》第78卷，第6期（1973年）：1360–1380页。关于利用网络分析研究英国宗教战争期间被监禁新教徒通信关系的最新研究，参见露丝·安纳特与塞巴斯蒂安·E·安纳特，《玛丽一世统治时期新教信件网络：定量研究》，《ELH》第82卷，第1期（2015年）：1–33页。

G04-P005-008 | 英文原文（原段接续 / 分段）

[16] See http://republicofletters.stanford.edu/publications/kircher/. The result of this research will be published in a series of articles by Paula Findlen, Suzanne Sutherland, and Iva Lelková, culminating in a book provisionally titled *The Baroque Postmaster: Athanasius Kircher between Rome and the World*.

中文译文 | G04-P005-008

16 参见 http://republicofletters.stanford.edu/publications/kircher/。这项研究的结果将由 Paula Findlen、Suzanne Sutherland 和 Iva Lelkovã 在一系列文章中发表，最终将以一本暂定名为《巴洛克邮递员：阿塔纳修斯·基尔彻与罗马世界》的书形式出版。

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[17] Those who are reading this article in the print issue can view this and the other figures in color either in the online issue or at http://republicofletters.stanford.edu/publications/.

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17 正在阅读纸质版文章的人可以在线或访问 http://republicofletters.stanford.edu/publications/ 来查看此图及其他图表的颜色版本。

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G04-P006-002 | 英文原文

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FIGURE 1: Source locations for correspondence to Athanasius Kircher, sized by numbers of letters sent.

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图1：致阿塔纳修斯·基尔彻的通信地点，以发送信件的数量表示

G04-P006-004 | 英文原文

navigated the globe. While his network was less socially diverse than those of, say, Locke, Franklin, and Voltaire, who corresponded with broader swaths of society (and most notably, by the eighteenth century, with far more women), his letters traveled much farther.

中文译文 | G04-P006-004

在环球传播信息。虽然他的社交圈不如洛克、富兰克林和伏尔泰那样具有多样性——后两者与更广泛的社会群体（尤其是在18世纪，与更多女性）有通信往来——但他的信件传播范围却更广。

G04-P006-005 | 英文原文（原段接续 / 分段）

Skeptics have challenged the rhetoric of the "republicans of letters," to invoke Daniel Roche's phrase, and questioned whether there really was *a* republic, or whether this illusory place masked a balkanized hodgepodge of more or less connected mini-republics.[18] In the context of Mapping the Republic of Letters, Claude Willan's work on John Locke (1632–1704) has used graphs and other visualizations to reveal just how disconnected the different social and national subgroups in his correspondence network were from one another.[19] (See Figure 2.) The republic that emerges from his study is not a unified network that any newcomer could join wholesale, but a patchwork of isolated communities that Locke fused together. His findings provide an empirical demonstration of the scholarly suspicion that there were really "a multitude of communities within, or rather underneath, the surface of the Republic of Letters."[20]

中文译文 | G04-P006-005 | 原文注号：18、19、20

怀疑论者质疑了"文人共和国"的言辞——用丹尼尔·罗什的话来说——并质疑是否存在真正的共和国，或者这个虚幻的体系是否掩盖了由多个或多或少相互关联的微型共和国构成的分裂状态。在《绘制文人共和国》的研究背景下，克劳德·威兰对约翰·洛克（1632–1704）的研究利用图表及其他可视化手段，揭示了其通信网络中的不同社会与民族子群体彼此之间的隔阂程度。（见图2。）从他的研究中得出的"共和国"并非任何新来者都能完全加入的统一网络，而是一系列孤立的社区组合，由洛克将其融合在一起。他的发现实际上证明了学术界的怀疑：确实存在"众多存在于或隐藏在文人共和国表面之下的社区"。

G04-P006-006 | 英文原文

Mapping the correspondence networks of major Enlightenment figures underscores the limits of their geographic horizons, raising questions about what it meant to be a self-professed "citizen of the world," as many enlightened men and women of

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绘制主要启蒙人士的通信网络图，凸显了他们地理视野的局限，也引发了关于如何定义"世界公民"这一概念的疑问，正如许多启蒙男女所宣称的那样

G04-P006-007 | 英文原文

[18] Daniel Roche, *Les républicains des lettres: Gens de culture et Lumières au XVIIIe siècle* (Paris, 1998). On the national vs. cosmopolitan makeup of the Republic of Letters, see especially Lorraine Daston, "The Ideal and Reality of the Republic of Letters in the Enlightenment," *Science in Context* 4, no. 2 (1991): 367–386.

中文译文 | G04-P006-007

18 丹尼尔·罗什，《文人共和国：18世纪的文化与光明人士》（巴黎，1998年）。关于文人共和国的本土性与世界性特征，可参考洛拉林·达斯顿所著的《启蒙时代文人共和国的理想与现实》，载于《科学背景》第4卷第2期（1991年），第367–386页。

G04-P006-008 | 英文原文（原段接续 / 分段）

[19] See http://republicofletters.stanford.edu/publications/locke/, and Willan's article "'John Locke Likes This': An Ego-Network Analysis of Locke's Letters" (under review).

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19 见http://republicofletters.stanford.edu/publications/locke/，以及威兰的文章"'约翰·洛克喜欢这个'：对洛克信件的自我网络分析"（正在审稿中）。

G04-P006-009 | 英文原文（原段接续 / 分段）

[20] See Candice Delisle, "Accessing Nature, Circulating Knowledge: Conrad Gessner's Correspondence Networks and His Medical and Naturalist Practices," *History of Universities* 23, no. 2 (2008): 35–58, here 53.

中文译文 | G04-P006-009

20 参见坎迪丝·德利尔，《接触自然，传播知识：康拉德·格斯纳的通信网络及其医学与自然学实践》，《大学历史》第23卷，第2期（2008年）：35–58页，此处为第53页。

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G04-P007-002 | 英文原文

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FIGURE 2: Network of Royal Society fellows (orange nodes) in John Locke's correspondence.

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图2：约翰·洛克通信中的皇家学会会员网络（橙色节点）

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letters claimed to be, and in what ways overseas colonies mattered in the increasingly far-flung imperial reach of the Europe-centered Republic of Letters.[21] Caroline Winterer and Claire Arcenas—another of our research teams—have explored these two questions in the network of a figure long renowned for his cosmopolitanism and colonial status, Benjamin Franklin (1706–1790).[22] Franklin may well have been the most famous North American of the eighteenth century, but his transatlantic letters initially did not include a significant number of international correspondents; only over decades spent in the imperial capital of London did he forge a letter network that included French, German, Italian, and Scottish correspondents. (See Figure 3.) Even at the close of his "London Decades" (1757–1775), and despite trips to Scotland, the Netherlands, and France, Franklin's correspondents were overwhelmingly found at ei-

中文译文 | G04-P007-004 | 原文注号：21、22

声称属于文人共和国的信件，以及海外殖民地在以欧洲为中心的文人共和国日益广泛的帝国统治范围内具有何种重要性。21 卡罗琳·温特勒和克莱尔·阿卡内斯——我们研究团队中的另一位成员——通过以博世派与殖民背景著称的本杰明·富兰克林（1706–1790）这一人物网络来探讨这两个问题。22 富兰克林或许是十八世纪最著名的北美人，但他最初跨大西洋的信件中并未包含大量国际通信者；直到他在帝国首都伦敦度过的几十年里，他才建立起一个包含法国、德国、意大利和苏格兰通信者的信件网络。（见图3。）即便在他的"伦敦十年"（1757–1775）时期结束之后，尽管他前往了苏格兰、荷兰和法国，富兰克林的通信者仍主要集中于这些地区。

G04-P007-005 | 英文原文

[21] Margaret C. Jacob, *Strangers Nowhere in the World: The Rise of Cosmopolitanism in Early Modern Europe* (Philadelphia, 2006); Caroline Winterer, "Where Is America in the Republic of Letters?," *Modern Intellectual History* 9, no. 3 (2012): 597–623; Winterer, *American Enlightenments: Pursuing Happiness in the Age of Reason* (New Haven, Conn., 2016).

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21 玛格丽特·C·雅各布，《世界上的陌生人：早期现代欧洲世界主义主义的兴起》（费城，2006）；卡罗琳·温特勒尔，"文人共和国中的美洲位于何方？"，《现代思想史》第9卷，第3期（2012）：597–623；温特勒尔，《美国启蒙：理性时代对幸福的追求》（康涅狄格州纽黑文，2016）。

G04-P007-006 | 英文原文（原段接续 / 分段）

[22] See http://republicofletters.stanford.edu/publications/franklin/; and Winterer and Arcenas's article "The Correspondence Network of Benjamin Franklin: The London Decades, 1757–1775" (under review).

中文译文 | G04-P007-006

22 参见http://republicofletters.stanford.edu/publications/franklin/；以及温特勒和阿卡内斯的文章《本尼迪克特·富兰克林的通信网络：1757–1775年的伦敦十年》（正在审稿中）。

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FIGURE 3: Benjamin Franklin's total correspondence, with locations sized by volume of letters.

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图3：本杰明·富兰克林的全部通信，其中位置的大小由信件数量表示

G04-P008-004 | 英文原文

ther end of the London-Philadelphia axis. His fame may have spread across Europe, but his letters circulated chiefly in the British imperial world.

中文译文 | G04-P008-004

伦敦-费城轴的终点。他的名声或许传播到了欧洲各地，但他的信件主要流通于英国帝国范围内

G04-P008-005 | 英文原文（原段接续 / 分段）

Our visualizations also provide new perspectives on murky problems of cultural transmission. As our data does not include letter content, it can tell us only so much about the circulation of ideas. But correspondence maps nonetheless raise questions that invite us to read letters and other documents with a fresh perspective. Starting with the observation that Voltaire (1694–1778) seems to have exchanged few letters with British correspondents, Dan Edelstein and Biliana Kassabova have argued in their work for Mapping the Republic of Letters that the standard narrative about the English origins of Voltaire's philosophy (and by extension of the French Enlightenment) must be revised in light of the fact that Voltaire's interest in, and admiration of, England was both qualified and chronologically restricted.[23] (See Figure 4.) England's period of cultural greatness, in his eyes, lay in the reign of Charles II, and was thus a thing of the past by the time he arrived in London. What is more, Voltaire credited the *siècle de Louis le grand* for many of the English intellectual exploits. The author of the bestselling and controversial *Letters Concerning the English Nation* had more than a few concerns, it turns out, about that nation.

中文译文 | G04-P008-005 | 原文注号：23

我们的可视化分析也为文化传承中的复杂问题提供了新的视角。由于我们的数据未包含信件内容，因此只能告诉我们关于思想传播的一些信息。不过，信件地图却引发了诸多问题，促使我们以全新的视角来阅读信件及其他文件。根据观察，伏尔泰（1694–1778）似乎与英国通信者丹·埃德尔斯坦和比利亚纳·卡萨博瓦几乎没有交换过信件。丹·埃德尔斯坦和比利亚纳·卡萨博瓦在《绘制文人共和国地图》一书中指出，关于伏尔泰哲学源自英国这一传统说法必须修正，因为伏尔泰对英格兰的兴趣与欣赏既有限又具有时间限制。（参见图4。）在他看来，英格兰文化发展的巅峰时期出现在查理二世统治时期，因此当他抵达伦敦时，这一时期已属于过去。此外，伏尔泰认为"路易大帝时代"是许多英国知识分子成就的起源。这位畅销且具有争议性的著作《关于英国人的信件》的作者，显然对那个国家有着诸多疑虑。

G04-P008-006 | 英文原文（原段接续 / 分段）

As the above case studies indicate, travel—whether Kircher's religious flight from war-torn Germany, Locke's exile from England, Franklin's transatlantic voyages, or Voltaire's timely crossing of the English Channel to mitigate scandal—played a crucial role in how many citizens of the Republic of Letters extended and built their networks. One of our groups, as detailed in the second article in this *AHR* Forum, has

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正如上述案例研究所表明的那样，旅行——无论是基尔彻从战争频发的德国逃往文人的共和国的宗教之旅，洛克从英格兰流亡的经历，富兰克林的跨大西洋航行，还是伏尔泰及时穿越英吉利海峡以化解丑闻——在许多文人共和国公民拓展和构建其关系网络的过程中发挥了至关重要的作用。我们中的一个小组，如《AHR Forum》第二篇文章中详细所述，

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[23] See http://republicofletters.stanford.edu/publications/voltaire/; and Edelstein and Kassabova's article "Where Are Voltaire's Letters Concerning the English Nation? Maps, Networks, and Intellectual History" (under review).

中文译文 | G04-P008-007

23 参见http://republicofletters.stanford.edu/publications/voltaire/；以及埃德尔斯坦和卡萨博娃的文章"沃勒尔的《关于英国人的信》在哪里？地图、网络与思想史"（正在审稿中）。

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FIGURE 4: Histogram of Voltaire's correspondence, by year, highlighting English correspondence.

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图4：伏尔泰书信的年度直方图，突出显示英文书信部分。

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been mapping Grand Tour travel, and the role it played in the emergence of architecture as a profession. The research process involved transforming John Ingamells's *A Dictionary of British and Irish Travellers in Italy, 1701–1800* into an interactive database filled with detailed information about the intersecting itineraries of the men and women who traveled to Italy, and creating a prosopography of this community.[24] (See Figure 5.) By studying British architects on the Grand Tour, Giovanna Ceserani, Giorgio Caviglia, Nicole Coleman, Thea De Armond, Sarah Murray, and Molly Taylor-Poleskey illuminate how the dynamic exploration of early modern travelers allows us to identify with greater precision the role of travel in relation to professional aspirations and cultural formation.[25] Architects who completed their education in Italy often met prospective clients on the Grand Tour, and presented their studies abroad as a compelling professional credential when they returned home.

中文译文 | G04-P009-004 | 原文注号：24、25

研究了大旅行团之旅，以及这种旅行在建筑业成为职业过程中所扮演的角色。研究过程包括将约翰·英加梅尔斯的《1701–1800年英国和爱尔兰旅者意大利志》转化为一个包含有关前往意大利的男女旅行者行程详细信息的互动数据库，并编制出该群体的人物志。（见图5。）通过研究大旅行团之旅中的英国建筑师，乔万娜·切塞拉尼、乔治奥·卡维利拉、妮可·科尔曼、泰亚·德·阿蒙德、莎拉·默里和莫莉·泰勒-波尔斯基揭示了早期现代旅行者的动态探索如何帮助我们更精准地理解旅行在职业追求与文化形成中的作用。在意大利完成教育的建筑师们通常在大旅行团之旅中遇到潜在客户，并在回国后将其海外学习经历作为有力的职业资历来展示。

G04-P009-005 | 英文原文（原段接续 / 分段）

When we have presented our work, a common complaint has been that for all our investment in digital tools, data editing, and data creation, the field has not been revolutionized as a result. This criticism can be voiced with optimism ("Show us you're the greatest thing since sliced bread," in the words of one enthusiastic reviewer) or with skepticism ("So you have nothing more to show for yourselves?"). But why *should* new methods produce radically different results? We are studying the same objects that scholars have been painstakingly exploring for hundreds of years. They naturally came up with penetrating insights about the Republic of Letters without the help of digital tools. But what these tools allow us to do is to sort through, refine, confirm, or refute the different insights these historians had, as well as to formulate new ones that emerge when we step back from the details of any particular subset of a correspondence to observe its contours. This is not to embrace a positivist, cliometric vision of computational supremacy; such a vision is ill-suited for historical archives that are so shot through with uncertainty and gaps. But where possible, basic calculations are still useful and can provide correctives to sheer speculation. They also force us to look closely at the information we have, resulting in numerous silent corrections to minor errors, assumptions that have solidified into facts, and other problems that arise when we take information for granted.

中文译文 | G04-P009-005

当我们展示我们的研究成果时，常见的批评是：尽管我们投入了大量资源用于数字工具、数据编辑和数据创建，该领域却并未因此发生革命性变化。这种批评可以用乐观的态度来表达（“用你们最棒的成果来证明吧”，正如一位热情的评论者所说），也可以用怀疑的态度来表达（“那么你们就没什么可展示的了？”）。但为何新方法能带来截然不同的结果呢？我们研究的是那些学者数百年来精心探索过的对象。它们自然能够让我们在不依赖数字工具的情况下获得深刻的见解，从而了解文人共和国的运作机制。而这些工具则使我们能够筛选、完善、验证或反驳这些历史学家所提出的各种见解，并提出新的见解——这些新见解是在我们从特定子集的细节中抽离出来，观察其整体特征时产生的。这并非意味着要接受一种实证主义、计量学的计算至上观点；因为历史档案中充满了不确定性与空白，这种观点并不适用。但在可能的情况下，基本计算仍然很有用，可以纠正纯粹的推测。它们还迫使我们仔细审视所拥有的信息，从而对小的错误、固化的假设以及其他被视为理所当然的问题进行许多无声的修正。

G04-P009-006 | 英文原文（原段接续 / 分段）

The data visualizations we rely on in our research and publications do not offer clear snapshots of the past, but rather fuzzy, blurry pictures. They reveal the general shape of things, orders of magnitude, and large-scale trends; they also draw our atten-

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我们在研究和出版中依赖的数据可视化图表并非能清晰呈现过去的情况，而是模糊不清的图像。它们揭示了事物的整体形态、数量级以及大规模趋势；同时也引导我们的注意力...

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[24] John Ingamells, comp., *A Dictionary of British and Irish Travellers in Italy, 1701–1800* (New Haven, Conn., 1997). For more on this project, see https://grandtour.stanford.edu/.

中文译文 | G04-P009-007

24 约翰·英加梅尔斯著，《意大利的英国与爱尔兰旅行者词典，1701–1800年》（康涅狄格州纽黑文，1997年）。关于该项目的更多信息，请参见https://grandtour.stanford.edu/。

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[25] See http://republicofletters.stanford.edu/publications/grandtour/; and Giovanna Ceserani, Giorgio Caviglia, Nicole Coleman, Thea De Armond, Sarah Murray, and Molly Taylor-Poleskey, “British Travelers in Eighteenth-Century Italy: The Grand Tour and the Profession of Architecture,” this issue.

中文译文 | G04-P009-008

25 参见 http://republicofletters.stanford.edu/publications/grandtour/；以及 Giovanna Ceserani、Giorgio Caviglia、Nicole Coleman、Thea De Armond、Sarah Murray 和 Molly Taylor-Poleskey 的论文《十八世纪意大利的英国旅行者：大旅行与建筑职业》，该文发表于本期。

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FIGURE 5: Timechart of British Grand Tour travelers in Italy in the eighteenth century.

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图5：18世纪前往意大利的英国长途旅行者的时间分布图。

G04-P010-004 | 英文原文

tion to microhistories that we might otherwise have missed. The results are never definitive, and always require further verification—an absence on a map might be due only to a gap in the data. But the critical point is that they open up new avenues of inquiry and are springboards for further research.

中文译文 | G04-P010-004

它们让我们能够发现那些原本可能忽略的微观历史。这些结果并非定论，仍需进一步验证——地图上缺失的部分可能仅因数据不足所致。但关键在于，它们为新的研究途径打开了大门，成为进一步研究的起点。

G04-P010-005 | 英文原文（原段接续 / 分段）

In some instances, the computational processes in question are not complex: one could count Kircher's global missionary correspondence, Voltaire's letters to England, or Franklin's international correspondents by hand. But computational methods are good for more than just performing difficult statistical operations (though one of our projects, on John Locke, does exactly that): they can also serve to reveal elements of surprise in the data. The point is not, therefore, whether you need computational tools to count the number of Franklin's French (or Voltaire's English, or Kircher's Jesuit) correspondents, but rather that computational methods provide novel insights into your sources. We designed and developed data visualizations not to do our work for us, but rather to point to where our work lies.

中文译文 | G04-P010-005

在某些情况下，所涉及的计算过程并不复杂：可以手动统计基尔彻的全球传教信件、伏尔泰给英国的信件，或富兰克林的国际通信。但计算方法不仅适用于执行复杂的统计操作（尽管我们的一个项目确实如此，该项目研究约翰·洛克）：它们还能揭示数据中的意外之处。因此，问题不在于是否需要计算工具来统计富兰克林的法国通信者、伏尔泰的英国通信者或基尔彻的耶稣会士的数量，而在于计算方法能为我们提供对资料的新见解。我们设计和开发数据可视化工具的目的不是代替我们完成工作，而是指出我们工作的方向。

G04-P010-006 | 英文原文

While our Mapping project came of age at a time when historians were just starting to focus on the promises and perils of big data, it also draws on a long historiography. Indeed, the Republic of Letters has never been a small or easily contained subject. Starting in the mid-fourteenth century, early humanists such as Petrarch wrote open letters as a means of communicating ideas and shaping opinion, and also as a process of intellectual self-definition. In 1417 the Venetian humanist Francesco Barbaro provided an indelible portrait of the nature of this emerging scholarly community when

中文译文 | G04-P010-006

尽管我们的研究项目在历史学家刚刚开始关注大数据的机遇与风险之际兴起，但它也借鉴了漫长的历史研究传统。事实上，文人共和国从来都不是一个微不足道或易于被涵盖的主题。从14世纪中叶起，像彼特拉克这样的早期人文主义者通过撰写公开信件来传播思想、塑造观点，同时也以此作为自我认知的途径。1417年，威尼斯人文主义者弗朗切斯科·巴尔巴罗对这一新兴学术社群的本质作出了不可磨灭的描绘。

G04-P010-007 | 英文原文

American Historical Review April 2017

中文译文 | G04-P010-007

美国历史评论 2017年4月

G04-P011-002 | 英文原文

he congratulated his Florentine friend Poggio Bracciolini for unearthing a treasure trove of ancient manuscripts in the monastic libraries of Northern Europe, acknowledging the utility of Poggio's rediscovery for "this Republic of letters" (*huic litteraria Reipublicae*).[26] It is hardly coincidental that the phrase *Respublica litteraria* first appeared in a letter. In the age of great letter writers such as the early-sixteenth-century Dutch humanist Desiderius Erasmus, who, to paraphrase Lisa Jardine, delighted in making absent friends present, the idea of conversation through correspondence became the norm rather than the exception.[27]

中文译文 | G04-P011-002 | 原文注号：26、27

他祝贺他的佛罗伦萨朋友波吉奥·布拉奇奥利尼在北欧的修道院图书馆中发现了大量古代写本，并肯定波吉奥的重新发现对"该文人共和国"的益处。26 恰好的是，"文人共和国"这一表述首次出现在一封信中。在像16世纪早期的荷兰人文主义者德西德里乌斯·伊拉斯谟这样的伟大作家时代，用丽莎·贾丁的话来说，他们乐于让不在场的朋友变得在场，通过通信进行交流的观念便成为常态而非例外。27

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G04-P011-004 | 英文原文

When Elizabeth Eisenstein, in her landmark study *The Printing Press as an Agent of Change* (1979), proclaimed printing to have been a transformative development in early modern Europe, she contrasted the limits of letters to the possibilities of books in disseminating knowledge while observing that we need to know a great deal more about the Republic of Letters.[28] More recent scholarship has reexamined this issue and arrived at a different conclusion, one that emphasizes the productive and continuous interactions between writing and publishing. A world of books and journals was also a world of readers, authors, and critics, whose conversations took place not only in academies, salons, and other sites of scholarly sociability, but also through long-distance travel and communication via letters. Such letters provide us with a historical roadmap from which we can reconstruct intellectual networks across time and space.

中文译文 | G04-P011-004 | 原文注号：28

当伊丽莎白·艾森斯坦在她的里程碑式研究《印刷机作为变革的媒介》（1979年）中宣称印刷术是近代欧洲的一场变革性发展时，她将文人的局限性与书籍在传播知识方面的潜力进行了对比，同时指出我们还需要更多了解文人共和国。28 近期的研究重新探讨了这一问题，得出了不同的结论，即强调写作与出版之间富有成效且持续的互动关系。书籍与期刊的世界也是读者、作者和评论家们的世界，他们的交流不仅发生在学术院、沙龙等学术社交场所，还通过长途旅行和书信往来实现。这些信件为我们提供了历史线索，使我们能够重建跨越时间与空间的知识网络。

G04-P011-005 | 英文原文（原段接续 / 分段）

In the 1970s, a number of scholars began to investigate the nature and meaning of the Republic of Letters. An international community of scholars has developed and defined this subject in the decades since, tracing the rise and decline of a rhetorical ideal and its connection to cultural and intellectual practices and scholarly communities.[29] But rhetoric, as Lorraine Daston cogently observed in an important essay, ulti-

中文译文 | G04-P011-005 | 原文注号：29

在20世纪70年代，许多学者开始研究文人共和国的本质与意义。此后数十年间，国际学者群体不断发展并界定了这一主题，探索了一种修辞理想的兴衰及其与文化和知识实践以及学术社群的关系。但正如洛雷恩·达斯顿在一篇重要论文中精辟指出的那样，修辞……

G04-P011-006 | 英文原文

[26] The standard English translation of this letter, dated July 6, 1417, can be found in Phyllis Walter Goodhart Gordan, ed. and trans., *Two Renaissance Book Hunters: The Letters of Poggius Bracciolini to Nicolaus de Niccolis* (New York, 1974), 196–203.

中文译文 | G04-P011-006

26 这封写于1417年7月6日的信件的标准英文译文可参见菲莉丝·沃尔特·古德哈特·戈丹编著的《两位文艺复兴时期的书籍猎人：波吉乌斯·布拉奇奥利尼致尼古拉乌斯·德·尼科利斯的信》（纽约，1974年），第196–203页。

G04-P011-007 | 英文原文（原段接续 / 分段）

[27] For an overview of these developments, see Marc Fumaroli, "The Republic of Letters," trans. R. Scott Walker, *Diogenes* 36, no. 143 (1988): 129–152; Hans Bots and Françoise Waquet, *La République des lettres* (Paris, 1997); Anthony Grafton, "A Sketch Map of a Lost Continent: The Republic of Letters," *Republics of Letters: A Journal for the Study of Knowledge, Politics, and the Arts* 1, no. 1 (2009), http://arcade.stanford.edu/rofl/sketch-map-lost-continent-republic-letters. On Erasmus, a good starting point is Lisa Jardine, *Erasmus, Man of Letters: The Construction of Charisma in Print* (Princeton, N.J., 1993), chap. 6; Constance M. Furey, *Erasmus, Contarini, and the Religious Republic of Letters* (Cambridge, 2008); Hanan Yoran, *Between Utopia and Dystopia: Erasmus, Thomas More, and the Humanist Republic of Letters* (Lanham, Md., 2010).

中文译文 | G04-P011-007

27 关于这些发展的概述，可参考 Marc Fumaroli 的《文人共和国》，由 R. Scott Walker 翻译，《Diogenes》36卷，第143期（1988年）：129–152页；Hans Bots 和 Franc¸oise Waquet 所著的《文人共和国》（巴黎，1997年）；Anthony Grafton 的《失落大陆的草图：文人共和国》，《文人共和国：知识、政治与艺术研究期刊》1卷，第1期（2009年），http://arcade.stanford.edu/rofl/sketch-map-lost-continent-republic-letters。关于伊拉斯谟，Lisa Jardine 的《伊拉斯谟：文人人物——印刷品中的魅力构建》（新泽西州普林斯顿，1993年）第6章；Constance M. Furey 所著的《伊拉斯谟、康塔里尼与宗教文人共和国》（剑桥，2008年）；Hanan Yoran 所著的《介于乌托邦与极权主义之间：伊拉斯谟、托马斯·莫尔与人文主义文人共和国》（马里兰州兰哈姆，2010年）可供参考。

G04-P011-008 | 英文原文（原段接续 / 分段）

[28] Elizabeth Eisenstein, *The Printing Press as an Agent of Change: Communications and Cultural Transformations in Early Modern Europe*, 2 vols. (Cambridge, 1979), 1: 136–137. The relationship of Eisenstein's work to the Republic of Letters is discussed by J. L. Pearl, "The Role of Personal Correspondence in the Exchange of Scientific Information in Early Modern France," *Renaissance and Reformation / Renaissance et Réforme*, New Series / Nouvelle Série, 8, no. 2 (1984): 106–113, here 107; and April G. Shelford, *Transforming the Republic of Letters: Pierre-Daniel Huet and European Intellectual Life, 1650–1720* (Rochester, N.Y., 2007), 3.

中文译文 | G04-P011-008

28 伊丽莎白·艾森斯坦，《印刷机作为变革的媒介：早期现代欧洲的传播与文化转型》，共2卷（剑桥，1979年），第1卷，第136–137页。J. L. 珀尔在《个人通信在早期现代法国科学信息交流中的作用》一文中探讨了艾森斯坦著作与文人共和国之间的关系，该文发表于《文艺复兴与宗教改革/文艺复兴与Reforme，新系列/新系列》，第8卷，第2期（1984年），第106–113页，此处引用为第107页；此外，阿普尔·G. 谢尔福德在《转变文人共和国：皮埃尔-丹尼尔·于埃与1650–1720年的欧洲知识分子生活》中也有相关论述，该书于（纽约州罗切斯特）出版，2007年，第3页。

G04-P011-009 | 英文原文（原段接续 / 分段）

[29] The earliest historical study of this concept may well be Annie Barnes, *Jean Le Clerc (1657–1736) et la République des lettres* (Paris, 1938). Readers interested in following the development of this subject since the 1970s should begin with Paul Dibon, "Les échanges épistolaires dans l'Europe savante du XVII[e] siècle," *Revue de synthèse* 81–82 (1976): 31–50; Dibon, "Communication in the Respublica litteraria of the 17th Century," *Respublica Litteraria: Studies in the Classical Tradition* 1 (1978): 43–55; Hans Bots, *Republiek der letteren: Ideaal en werkelijkheid: Rede uitgesproken bij aanvaarding van het ambt van buitengewoon hoogleraar aan de Katholieke universiteit te Nijmegen* (Amsterdam, 1977); Françoise Waquet, "Qu'est-ce que c'est la République des lettres? Essai de sémantique historique," *Bibliothèque de l'École des Chartres* 147 (1989): 473–502. The fundamental work of synthesis on this subject is Bots and Waquet, *La République des lettres*.

中文译文 | G04-P011-009

29 关于这一概念的最早历史研究，可能来自安妮·巴恩斯、让·勒克莱尔（1657–1736）所著的《文学共和国》（巴黎，1938年）。对于希望了解自1970年代以来该主题发展情况的读者而言，应首先阅读保罗·迪邦的《17世纪学术欧洲的书信交流》，《综合评论》第81–82期（1976年）：31–50页；迪邦的《17世纪文学共和国中的沟通》，《文学共和国：古典传统研究》第1卷（1978年）：43–55页；汉斯·博茨的《文学共和国：理想与现实——尼美根天主教大学特别教授就职演说》（阿姆斯特丹，1977年）；弗朗索瓦·瓦凯特的《什么是文学共和国？历史语义学尝试》，《沙特尔学院图书馆》第147期（1989年）：473–502页。关于这一主题的综合研究著作，主要是博茨和瓦凯特所著的《文学共和国》。

G04-P011-010 | 英文原文

中文译文 | G04-P011-010

美国历史评论，2017年4月

原PDF第12页（原刊第411页）

G04-P012-002 | 英文原文

mately inspired questions about the reality, or lived experience, of the ideal of the Republic of Letters.[30] Taking the case of Leibniz, Maarten Ultee envisioned a social history that would explore the concrete details of intellectual membership in this imagined community, including its geography, the volume and frequency of letters, and the social positions of its participants.[31] Such questions inspired the subsequent work of a talented cadre of scholars, most notably Dena Goodman, Anne Goldgar, and Daniel Roche, who increasingly brought the methods of cultural history and gender history to bear on the subject in the late 1980s and early 1990s.[32] Simultaneously, a rejuvenated intellectual history developed by scholars such as Joseph Levine, Anthony Grafton, Peter Miller, and Laurence Brockliss focused on key participants in the Republic of Letters as a means of better understanding the practices of intellectual life from the advent of Renaissance humanism to the rise of antiquarianism and ultimately the Age of Enlightenment.[33]

中文译文 | G04-P012-002 | 原文注号：30、31、32、33

关于文人共和国这一理想现实或实际体验的富有启发性的问题。以莱布尼茨为例，马腾·乌尔蒂设想了一种社会历史研究，旨在探讨这一想象性社群中知识成员资格的具体细节，包括其地理分布、信件的数量与频率以及参与者的社会地位。这类问题激发了一群才华横溢的学者——尤其是德娜·古德曼、安妮·戈尔德加和丹尼尔·罗什——在20世纪80年代末和90年代初运用文化史与性别史的研究方法来研究这一主题。与此同时，约瑟夫·莱文、安东尼·格拉夫顿、彼得·米勒和劳伦斯·布罗克利斯等学者所构建的复兴后的知识史研究，聚焦于文人共和国的关键参与者，以此作为理解从文艺复兴人文主义出现到古学兴起，最终直至启蒙时代期间知识生活实践的一种途径。

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G04-P012-004 | 英文原文

Initially there were a limited number of letters, but soon the volume of letters that individuals exchanged expanded astronomically. When Jardine wondered aloud what scholars should do with the "mass of 'data' which Erasmus's voluminous correspondence provides," she asked a pertinent question: 3,162 letters to and from Erasmus survive. We can juxtapose this number to the prolific letter-writing habits of Italian Renaissance humanists, but also of learned Protestant reformers: Martin Luther's surviving correspondence numbers 4,337 letters, and John Calvin's 4,271. Ignatius Loyola, the founder of the Society of Jesus, was a veritable epistolary machine who practiced what he preached in institutionalizing letter-writing as an essential expression of religious confraternity and bureaucracy; he left behind an impressive 9,178 letters (6,381 written by him) at the time of his death in 1556.[34] While not quantitatively large by contemporary standards of big data, such numbers are filled with hidden multipliers, because each letter typically contains a great deal of valuable information—about people, places, conversations, manuscripts, publications, projects, and institutions—and offers a trail of references that invariably leads the patient researcher to

中文译文 | G04-P012-004 | 原文注号：34

起初，交换的写本数量有限，但很快个人之间交流的写本数量却急剧增加。当贾丁大声询问学者们该如何处理埃拉斯莫那大量通信所包含的那"海量数据"时，她提出了一个重要问题：至今仍保存有3,162封来自或寄给埃拉斯莫的写本。我们可以将这一数字与意大利文艺复兴时期人文主义者的多产写信习惯以及新教改革者们的习惯相提并论：马丁·路德的现存通信中有4,337封写本，约翰·加尔文则有4,271封。耶稣会的创始人伊格纳修斯·洛约拉则是一位真正的书信制造者，他将写信行为制度化，使其成为宗教团体与官僚体系的重要表现方式；他在1556年去世时留下了令人印象深刻的9,178封写本（其中6,381封由他本人撰写）。虽然按当时的大数据标准来看这些数量并不庞大，但这些数字中蕴含着丰富的隐藏价值——每封写本通常包含大量有价值的信息，涉及人物、地点、对话、写本、出版物、项目及机构等——并且其中所包含的参考文献总能引导耐心的研究者找到线索。

G04-P012-005 | 英文原文

[30] Daston, "The Ideal and Reality of the Republic of Letters in the Enlightenment."

[31] Maarten Ultee, "The Republic of Letters: Learned Correspondence, 1680–1720," *Seventeenth Century* 2, no. 1 (1987): 95–112.

中文译文 | G04-P012-005

30 达斯顿，《启蒙时代文人共和国的理想与现实》。31 马尔滕·乌尔泰，《文人共和国：学术通信，1680–1720年》，《十七世纪》第2卷第1期（1987年）：95–112页。

G04-P012-006 | 英文原文（原段接续 / 分段）

[32] Dena Goodman, *The Republic of Letters: A Cultural History of the French Enlightenment* (Ithaca, N.Y., 1994); Anne Goldgar, *Impolite Learning: Conduct and Community in the Republic of Letters, 1680–1750* (New Haven, Conn., 1995); Roche, *Les républicains des lettres*. For subsequent efforts to build on Goodman's early insights on the importance of gender in the Republic of Letters, a good starting point is Susan Dalton, *Engendering the Republic of Letters: Reconnecting Public and Private Spheres in Eighteenth-Century Europe* (Montreal, 2003); Antoine Lilti, *Le monde des salons: Sociabilité et mondanité à Paris au XVIIIe siècle* (Paris, 2005); Dena Goodman, *Becoming a Woman in the Age of Letters* (Ithaca, N.Y., 2009).

中文译文 | G04-P012-006

32 德娜·古德曼，《文人共和国：法国启蒙运动的文化史》（纽约州伊萨卡，1994年）；安妮·戈尔加，《不雅的学习：1680–1750年文人共和国中的行为与社区》（康涅狄格州纽黑文，1995年）；罗什，《文人共和国的共和主义者》。对于后续研究古德曼关于文人共和国中性别重要性的早期见解的尝试，苏珊·道尔顿所著的《构建文人共和国：18世纪欧洲公共与私人领域的重新连接》（蒙特利尔，2003年）、安托万·利尔蒂所著的《沙龙世界：18世纪巴黎的社会交往与世俗生活》（巴黎，2005年），以及德娜·古德曼所著的《在文人时代成为女性》（纽约州伊萨卡，2009年）均为良好起点。

G04-P012-007 | 英文原文（原段接续 / 分段）

[33] Joseph M. Levine, *The Battle of the Books: History and Literature in the Augustan Age* (Ithaca, N.Y., 1994); Peter N. Miller, *Peiresc's Europe: Learning and Virtue in the Seventeenth Century* (New Haven, Conn., 2000); L. W. B. Brockliss, *Calvet's Web: Enlightenment and the Republic of Letters in Eighteenth-Century France* (Oxford, 2002). From Grafton's considerable corpus, one might follow the trail backward from his AHA Presidential Address, "The Republic of Letters in the American Colonies: Francis Daniel Pastorius Makes a Notebook," *American Historical Review* 117, no. 1 (February 2012): 1–39; and "A Sketch Map of a Lost Continent."

[34] Jardine, *Erasmus, Man of Letters*, 149. For a good overview of some of the metrics of early modern correspondence, see Francisco Bethencourt and Florike Egmond, eds., *Cultural Exchange in Early Modern Europe*, vol. 3: *Correspondence and Cultural Exchange in Europe, 1400–1700* (Cambridge, 2007), 10–12. For an introduction to religious letter-writing and Loyola's reputation as the most prolific letter writer of the sixteenth century, see Thomas M. Lucas, *Landmarking: City, Church and Jesuit Urban Strategy* (Chicago, 1997), 131.

中文译文 | G04-P012-007

33 约瑟夫·M·莱文，《文献总集中的战斗：奥古斯都时代的历史与文学》（纽约州伊萨卡，1994年）；彼得·N·米勒，《佩雷斯的欧洲：十七世纪的学习与美德》（康涅狄格州纽黑文，2000年）；L. W. B. 布罗克利斯，《卡尔维特的网络：十八世纪法国的启蒙与文人共和国》（牛津，2002年）。从格拉夫顿的丰富文献总集中，可以追溯其思路——从他的《美国历史评论》总统演讲"美国殖民地中的文人共和国：弗朗西斯·丹尼尔·帕斯托里乌斯撰写笔记"开始，该文发表于《美国历史评论》117卷第1期（2012年2月），第1–39页；以及"一个失落大陆的草图地图"。34 贾丁，《伊拉斯谟，文人》，第149页。关于早期现代通信的一些衡量标准，可参考弗朗西斯科·贝特恩库尔和弗洛丽克·埃格蒙德编著的《早期现代欧洲的文化交流》，第3卷：1400–1700年间的欧洲通信与文化交流（剑桥，2007年），第10–12页。关于宗教书信写作以及洛约拉作为十六世纪最活跃的写信人之一的声誉，可参考托马斯·M·卢卡斯所著的《地标化：城市、教会与耶稣会城市策略》（芝加哥，1997年），第131页。

G04-P012-008 | 英文原文

中文译文 | G04-P012-008

《美国历史评论》2017年4月

原PDF第13页（原刊第412页）

G04-P013-002 | 英文原文

other terrifyingly large masses of correspondence. This is far too much to absorb at a glance or to keep in one's head, which is why scholars have often worked with discrete portions of large correspondences to get to know specific content really well rather than attempting to consider the whole.

中文译文 | G04-P013-002

其他那些极其庞大的信件堆。这实在太多，无法一瞥之下或记住全部，因此学者们常常只处理这些庞大信件中的部分内容，以便更透彻地了解具体信息，而非试图考虑整体。

G04-P013-003 | 英文原文（原段接续 / 分段）

Such numbers pale in relation to the scale of mercantile correspondence, as the case of a late-fourteenth-century merchant in Prato, Francesco Datini, illustrates all too well: his surviving papers contain approximately 150,000 commercial and familial letters. While Petrarch carefully and deliberately identified the living and the dead, the real and the imaginary, to whom he might address a well-crafted Latin letter in imitation of Cicero, Quintilian, and other great Roman epistolarians, the demands of the marketplace required a constant stream of long-distance correspondence traveling in all directions to many different kinds of people who did business together.[35] The same can also be said of diplomatic correspondence. Take the example of Isabella d'Este, the fifteenth-century marchioness of Mantua, for whom some 25,000 letters (approximately 16,000 by her) survive in an archive containing about 100,000 letters related to the totality of her activities as one of the most powerful political and cultural brokers in Renaissance Italy.[36] Such examples serve as a stark reminder of just how fundamental and ubiquitous letter-writing was to a variety of human endeavors in the premodern world. While our project has focused on the intellectual and cultural dimensions of that particular subject, we hope that the techniques we have explored and experimented with may prove useful for historians with other interests.

中文译文 | G04-P013-003 | 原文注号：35、36

这些数字与商业通信的规模相比显得微不足道，14世纪末在普拉托的商人弗朗切斯科·达蒂尼的案例便充分说明了这一点：他保存下来的文件中包含约15万封商业和家庭信件。虽然彼特拉克精心地区分了生者与死者、现实与想象中的对象，他可能会模仿西塞罗、昆体良等伟大的罗马书信作家，用拉丁文撰写精美的信件，但商业需求要求不断有大量跨地区的通信往来，这些信件要寄给各种进行贸易的人士。35 外交信件也是如此。以15世纪的曼图亚侯爵伊莎贝拉·德斯特为例，她保存了约25,000封信件（她自己保存了约16,000封），这些信件被保存在一个档案中，该档案包含约100,000封与她作为文艺复兴时期意大利最强大的政治与文化调解人相关的信件。36 这些例子清楚地表明，在前现代世界中，书信写作对各种人类活动具有多么重要的地位。虽然我们的研究重点在于这一主题的学术与文化层面，但我们希望所探讨和尝试的技术方法能够对其他领域的历史学家也有帮助。

G04-P013-004 | 英文原文（原段接续 / 分段）

In the early sixteenth century, Erasmus wistfully envisioned a Christian Republic of Letters populated by scholars in communication across geographic, religious, and political boundaries. Increasingly, however, humanistic ideals of writing to someone as an expression of learned friendship intertwined with other priorities, as different cultures of letter-writing intersected and combined. Diplomats and agents made correspondence an instrument of politics, policy, and information; adventurers, missionaries, traders, and emigrants boarded mercantile vessels to parts unknown and sent letters home, waiting months, sometimes years, to receive responses from friends, family, and superiors back in Europe. Everywhere people went, letters accompanied them. Publishing letters became a lucrative editorial activity. Our project begins with this fundamental fact of the early modern world: it was a society whose patterns of increased mobility and communication were expressed in the numerical explosion of letters.

中文译文 | G04-P013-004

16世纪初，伊拉斯谟憧憬着一个文人共和国，那里由学者们组成，他们能够在地理、宗教和政治界限之外进行交流。然而，随着时间推移，人文主义理念——将文字作为学问友谊的表达——与其他优先事项交织在一起，不同书写文化也相互融合。外交官与代理人将通信视为政治、政策和信息的工具；冒险家、传教士、商人和移民则乘船前往未知之地，并寄出信件回家，等待数月甚至数年才能收到来自欧洲朋友、家人及上级的回复。无论人们前往何处，信件都伴随他们同行。出版信件也成为了一项盈利丰厚的编辑活动。我们的项目始于现代早期这一基本事实：这是一个流动性与交流能力增强的社会，其特征体现在信件的数量激增上。

G04-P013-005 | 英文原文（原段接续 / 分段）

Growing curiosity about experiences, ideas, languages, and artifacts that could be acquired only through long-distance travel multiplied the number of networks and communities of scholars in pursuit of knowledge. The superabundance of information

中文译文 | G04-P013-005

对只能通过长途旅行才能获得的经历、思想、语言和文物的强烈兴趣，使得追求知识的研究者网络与社群数量激增。信息的过剩

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G04-P013-007 | 英文原文

[35] The most recent description of the Datini correspondence by the Istituto Datini can be found at http://www.istitutodatini.it/schede/archivio/eng/arc-dat2.htm. On mercantile correspondence, see especially Francesca Trivellato, *The Familiarity of Strangers: The Sephardic Diaspora, Livorno, and Cross-Cultural Trade in the Early Modern Period* (New Haven, Conn., 2009); Trivellato, "Merchants' Letters across Geographical and Social Boundaries," in Bethencourt and Egmond, *Correspondence and Cultural Exchange in Europe*, 80–103; Trivellato, "A Republic of Merchants?," in Anthony Molho and Diogo Ramada Curto, eds., *Finding Europe: Discourses on Margins, Communities, Images, ca. 13th–ca. 18th Centuries* (Oxford, 2007), 133–158.

中文译文 | G04-P013-007

35 伊蒂斯蒂乌奥·达蒂尼机构对达蒂尼通信的最新描述可于 http://www.istitutodatini.it/schede/archivio/eng/arc-dat2.htm 查看。关于商业通信，可参考 Francesca Trivellato 的《陌生人的熟悉：塞法迪散居群体、利沃诺与早期现代时期的跨文化贸易》（新哈文，康涅狄格州，2009年）；Trivellato 在 Bethencourt 和 Egmond 所著的《欧洲的通信与文化交流》一书中撰写的"商人的信件跨越地理与社会边界"一文，第80–103页；以及 Trivellato 在 Anthony Molho 和 Diogo Ramada Curto 主编的《寻找欧洲：边缘、社区与图像相关的论述，约13世纪至18世纪》一书中撰写的"一个商人共和国？"一文，第133–158页。

4. 所有年份和阿拉伯数字必须原样保留，不得更改或替换。需保留的年份为：2007年、2009年。

G04-P013-008 | 英文原文（原段接续 / 分段）

[36] Deanna Shemek, "In Continuous Expectation: Isabella d'Este's Epistolary Desire," in Dennis Looney and Deanna Shemek, eds., *Phaethon's Children: The Este Court and Its Culture in Early Modern Italy* (Tempe, Ariz., 2005), 269–300.

中文译文 | G04-P013-008

36 迪安娜·谢梅克，《在持续的期待中：伊莎贝拉·德斯特的书信式渴望》，载于丹尼斯·卢尼与迪安娜·谢梅克编著的《法伊顿的子女：早期现代意大利德斯特宫廷及其文化》（亚利桑那州坦佩，2005年），第269–300页。

G04-P013-009 | 英文原文

中文译文 | G04-P013-009

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G04-P014-002 | 英文原文

so eloquently described in Ann Blair's 2010 book *Too Much to Know* was intimately connected to the increased scale and scope of scholarly correspondence in the late sixteenth and seventeenth centuries.[37] Where individuals had previously exchanged dozens or hundreds of letters, now their correspondences numbered in the thousands. Not surprisingly, the emergence of better postal systems in various regions accompanied this development.[38]

中文译文 | G04-P014-002 | 原文注号：37、38

正如安·布莱尔在2010年出版的《太多难以理解》一书中如此精妙地描述的那样，这一现象与16世纪末和17世纪学术通信的规模与范围扩大密切相关。37 过去人们互相交换数十或数百封信件，现在则多达数千封。毫不奇怪，各地区的邮政系统改进也伴随着这一发展。38

G04-P014-003 | 英文原文

But the exponential growth of information did not end there. Between the seventeenth and the eighteenth century, the profusion of new learned institutions, the invention of journals, and the attractions of cosmopolitan centers transformed the idea of scholarly community from a humanist model of Latin men of letters to a more socially diffuse model of learned and vernacular communities of men and women writing, traveling, reading, and publishing. The Republic of Letters in this era was defined not only by the intellectual currents of the Scientific Revolution and the Enlightenment, but also by global missionary and mercantile networks, the Grand Tour, and the emergence of urban centers in the British and European colonies whose scholars both corresponded and traveled across the Mediterranean, Atlantic, and Indian Oceans. Individual correspondences that had routinely numbered in the low thousands now ballooned to tens of thousands of letters. The Aix savant Nicolas-Claude Fabri de Peiresc's famous scholarly network of the early seventeenth century, which has left us somewhere between 10,000 and 14,000 letters, and Leibniz's approximately 15,000 surviving letters exchanged with numerous correspondents between 1663 and 1716 are well-known examples of what the world of prolific letter writers now was capable of producing.[39]

中文译文 | G04-P014-003 | 原文注号：39

但信息的指数级增长并未止步于此。在十七至十八世纪期间，大量新成立的学术机构、期刊的诞生以及世界性中心的兴起，使学术社群的概念从以拉丁语撰写的文人模式，转变为一种更加广泛的社会化模式——即由男女学者组成、通过写作、旅行、阅读和出版来互动的学术与民间社区。这一时期的文人共和国不仅受到科学革命与启蒙运动带来的思想潮流的影响，还受到全球传教与商业网络、大旅行以及英国和欧洲殖民地城市中心的发展所驱动。那些原本数量仅数千封的信件，如今则增至数万封。十七世纪初，艾克斯学者尼古拉-克劳德·法布里·德·佩雷斯克建立的著名学术网络留下了约10,000至14,000封信件；莱布尼茨在1663年至1716年间与众多通信者交换的约15,000封信件，都是当时多产信件作者所创造成果的典型例子。39

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G04-P014-005 | 英文原文

While many documents pertaining to the Republic of Letters have not survived, the quantity of extant letters from these centuries, even when restricted primarily to scholarly correspondence, offers historians of the early modern period a body of material containing far more data than any one scholar, or even a team of researchers, can assimilate through traditional research methods. At the same time, letters are invaluable resources when they are brought together in a traditional edited volume or

中文译文 | G04-P014-005

尽管许多与文人共和国相关的文献已无法保存，但这些世纪间的现存信件数量——即便主要局限于学术通信——为早期现代时期的历史学家提供了大量资料，其信息量远超任何一位学者或研究团队通过传统研究方法所能掌握的范围。同时，当这些信件被汇编成传统编辑书籍时，便成为极其宝贵的资源。

G04-P014-006 | 英文原文

[37] Blair, *Too Much to Know*.

[38] On postal systems, see Wolfgang Behringer, *Im Zeichen des Merkur: Reichspost und Kommunikationsrevolution in der Frühen Neuzeit* (Göttingen, 2003); Eric R. Dursteler, "Power and Information: The Venetian Postal System in the Eastern Mediterranean, 1573–1645," in Diogo Ramada Curto, Eric R. Dursteller, Julius Kirshner, and Francesca Trivellato, eds., *From Florence to the Mediterranean and Beyond: Essays in Honour of Anthony Molho*, 2 vols. (Florence, 2009), 2: 601–623; James Daybell, "Postal Conditions," in Daybell, *The Material Letter in Early Modern England: Manuscript Letters and the Culture and Practices of Letter-Writing, 1512–1635* (London, 2012), 109–147; Richard R. John, *Spreading the News: The American Postal System from Franklin to Morse* (Cambridge, Mass., 1995). For a GIS analysis of the French postal system, see Anne Bretagnolle and Nicolas Verdier, "Expanding the Network of Postal Routes in France, 1708–1833," in Muriel Le Roux, ed., *Post Offices of Europe, 18th–21st Century: A Comparative History* (Bern, 2014), 183–202.

中文译文 | G04-P014-006

37 布莱尔，《太多难以理解》。38 关于邮政系统，可参考沃尔夫冈·贝林格所著的《Mercur的印记：近代早期帝国邮政与通信革命》（哥廷根，2003年）；埃里克·R·杜尔斯泰勒所著的"权力与信息：东地中海地区的威尼斯邮政系统，1573–1645年"，载于迪奥戈·拉马达·库尔托、埃里克·R·杜尔斯泰勒、尤利乌斯·基什纳和弗朗西斯卡·特里韦拉托编著的《从佛罗伦萨到地中海及更远地区：为安东尼·莫洛而作的论文集》，共2卷（佛罗伦萨，2009年），第2卷，第601–623页；詹姆斯·戴贝尔所著的"邮政状况"，载于戴贝尔所著的《早期现代英格兰的物质信件：1512–1635年的写本信件与书写文化与实践》（伦敦，2012年），第109–147页；理查德·R·约翰所著的《传播新闻：从富兰克林到莫尔斯的美国邮政系统》（马萨诸塞州剑桥，1995年）。关于法国邮政系统的GIS分析，可参考安妮·布雷塔尼奥尔和尼古拉斯·韦尔迪埃所著的"1708–1833年间法国邮政线路网络的扩展"，载于穆里尔·勒鲁克斯编著的《18至21世纪的欧洲邮局：比较历史》（伯尔尼，2014年），第183–202页。

G04-P014-007 | 英文原文（原段接续 / 分段）

[39] On Peiresc's correspondence, a good starting point is Robert A. Hatch, "Peiresc as Correspondent: The Republic of Letters and the Geography of Ideas," in Brian P. Dolan, ed., *Science Unbound: Geography, Space and Discipline* (Umeå, 1998), 19–58; and recent essays by Peter Miller such as "Mapping Peiresc's Mediterranean: Geography and Astronomy, 1610–1636," in Dirk van Miert, ed., *Communicating Observations in Early Modern Letters (1500–1675): Epistolography and Epistemology in the Age of the Scientific Revolution* (London, 2013), 135–160. On Leibniz, see Ultee, "The Republic of Letters," 98; Paul Lodge, ed., *Leibniz and His Correspondents* (Cambridge, 2004); Nora Gädeke, "Gottfried Wilhelm Leibniz," in Christiane Berkvens-Stevelinck, Hans Bot, and Jens Häseler, eds., *Les grands intermédiaires culturels de la République des lettres: Études de réseaux de correspondances du XVIe au XVIIIe siècles* (Paris, 2005), 257–306.

中文译文 | G04-P014-007

39 关于皮尔雷斯的通信，一个好的研究起点是罗伯特·A·哈奇所写的《作为通信者的皮尔雷斯：文人共和国与思想地理》，载于布赖恩·P·多兰编著的《科学无界：地理、空间与学科》（乌梅亚，1998年），第19–58页；以及彼得·米勒的近期文章，如《绘制皮尔雷斯的地中海地图：1610–1636年的地理与天文学》，载于迪克·范·米尔特编著的《早期现代通信中的观测交流（1500–1675年）：科学革命时代的书信写作与认识论》（伦敦，2013年），第135–160页。关于莱布尼茨，参见乌尔蒂所写的《文人共和国》，第98页；保罗·洛奇编著的《莱布尼茨及其通信者》（剑桥，2004年）；诺拉·盖德克所写的《戈特弗里德·威廉·莱布尼茨》，载于克里斯蒂安·贝尔肯斯-斯特维林克、汉斯·博特和耶恩斯·赫塞勒编著的《文人共和国的重要文化中介：16至18世纪通信网络研究》（巴黎，2005年），第257–306页。

4. 所有年份和阿拉伯数字必须原样保留，不得更改或转换为汉字。需保留的年份包括：1500、1610、1636、1675、1998、2004、2005、2013。

G04-P014-008 | 英文原文

American Historical Review April 2017

中文译文 | G04-P014-008

《美国历史评论》2017年4月

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G04-P015-002 | 英文原文

increasingly as an online resource. Around the world, but particularly in Europe, the UK, and the United States, research projects are actively digitizing and cataloguing the surviving archives of the Republic of Letters.[40] While there is a considerable selection bias affecting whose records are being given new digital life, since most projects of this kind continue to focus on well-known figures, a host of forgotten characters are nonetheless emerging from this work: take Mlle Curton de Farges, who received a letter from Voltaire but is otherwise unknown to posterity.[41] Correspondence is diffuse and not entirely selective, after all; the act of writing a letter was possible for anyone possessing a certain degree of literacy or having access to someone who could write on his or her behalf. As such, it moves us beyond the usual criterion of selecting historical actors solely on the basis of their posthumous reputation. A great man or woman, after all, is only one node in an ego network. We have found ourselves repeatedly creating microhistories of people we did not expect to find and now wish to know.[42]

中文译文 | G04-P015-002 | 原文注号：40、41、42

作为在线资源，其重要性日益凸显。全球范围内，尤其是欧洲、英国和美国，研究项目正在积极将文人共和国的现存档案进行数字化和编目。尽管存在显著的选择偏差，导致某些人的记录获得新的数字形式，但由于大多数此类项目仍专注于知名人物，许多被遗忘的人物依然从这项工作中浮现出来——比如Mlle Curton de Farges，她收到了伏尔泰的信件，但后人对其几乎一无所知。毕竟，通信内容并不具有选择性；只要具备一定的识字能力或能够委托他人代笔，任何人都可以撰写信件。因此，这促使我们超越仅以死后声誉作为选择历史行动者的标准。毕竟，伟大的人物也只是自我网络中的一个节点而已。我们不断发现那些我们原本不期望见到、现在又想了解的人物的微观历史。

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In many respects, the early modern Republic of Letters has proved to be an ideal testing ground for developing new ways of thinking about historical data. Put a different way, it is a rich source of multifaceted, if incomplete, data that traverses time and space and connects people. The Republic of Letters fundamentally relied on the exchange of words and things between people who established, and in some instances maintained, bonds of family, friendship, and patronage through letters to realize intellectual, religious, and political projects. Travel, long-distance communication, writing culture, and the production and circulation of printed matter structured many of its core activities. The Republic of Letters was not only an ideal, an imagined community in Benedict Anderson's sense, but a series of practices conducted by people who formed relationships. Human brokers, agents, gatekeepers, and go-betweens emerge from the terrifying mounds of paper that were produced. If books and journals were the culminating products of the Republic of Letters, conversation was its lifeblood—and those conversations took place as much in lively social gatherings and in more intimate dialogues (whether with people who lived nearby or with people who were brought into proximity by their travels) as they did in a form at times tangibly preserved for us: letters.

中文译文 | G04-P015-004

在许多方面，早期现代文人共和国被证明是探索历史数据新思考方式的理想实验场。换句话说，它是一处丰富的资料来源，包含多种虽不完整的信息，这些信息跨越时间与空间，连接着人们。文人共和国主要依靠通过书信建立或维持家庭、友谊与庇护关系的人们之间的文字与物品交流，从而实现学术、宗教与政治项目。旅行、远程通信、书写文化以及印刷品的制作与传播构成了其许多核心活动。文人共和国不仅是本尼迪克特·安德森所定义的理想化社区，更是一系列由人之间建立关系所进行的实践行为。从那些大量产生的纸张中，便会出现人类的中介者、代理人、守门人及协调者。如果书籍与期刊是文人共和国的最终产物，那么对话则是其生命线——这些对话既发生在活跃的社会聚会中，也发生在更亲密的对话中（无论是与附近的人还是因旅行而相遇的人），同时也以有时被清晰保存下来的形式——书信——存在。

G04-P015-005 | 英文原文（原段接续 / 分段）

Even the earliest efforts to study the Republic of Letters confronted the difficulties of the data that this subject generates. Take Robert Mandrou's decision in 1973 to create a map of Erasmus's and Peiresc's correspondence and the diffusion of universities and Jesuit colleges in Western Europe. As Robert Hatch observed, Mandrou's Annaliste concern with the "geography of knowledge" led him to create static

中文译文 | G04-P015-005

甚至最初研究文人共和国的尝试也面临该课题所产生数据的困难。以罗伯特·曼德罗在1973年决定绘制埃拉斯姆斯与佩雷斯通信记录以及西方欧洲大学与耶稣会学院分布图为例。正如罗伯特·哈奇所指出的，曼德罗的《Annaliste》对"知识地理"的关注促使他创建了静态资料。

G04-P015-006 | 英文原文

[40] Indeed, one of our case studies began life as a digital archive more than ten years ago. For Paula Findlen, Suzanne Sutherland Duchacek, and Iva Lelková's use of the Athanasius Kircher Correspondence Project, jointly hosted by the Museo Galileo and Stanford University and originally created by Michael John Gorman and Nick Wilding at the European University Institute with assistance from Findlen, and most recently updated by Sutherland Duchacek and Lelková with technical support by Glen Worthey, see Athanasius Kircher at Stanford, "Correspondence," http://www.stanford.edu/group/kircher/cgi-bin/site/?page_id=7.

中文译文 | G04-P015-006

事实上，我们的一个案例研究在十多年前就以数字档案的形式开始发展。对于 Paula Findlen、Suzanne Sutherland Duchacek 和 Iva Lelkov□a 对阿塔纳修斯·基尔彻通信项目的使用而言，该项目由伽利略博物馆和斯坦福大学共同主办，最初由迈克尔·约翰·戈尔曼和尼克·怀德在欧洲大学研究所创建，并在 Findlen 的协助下完成，而最近则由 Sutherland Duchacek 和 Lelkov□a 在 Glen Worthey 的技术支持下进行更新。相关资料可查阅斯坦福大学的阿塔纳修斯·基尔彻项目："通信记录"，http://www.stanford.edu/group/kircher/cgi-bin/site/?page_id1⁄47。

G04-P015-007 | 英文原文（原段接续 / 分段）

[41] Letter of January 11, 1776, in Nicholas Cronk, ed., *Digital Correspondence of Voltaire*, http://dx.doi.org/10.13051/ee:doc/voltfrVF1260336a1c.

中文译文 | G04-P015-007

41 1776年1月11日的信件，出自Nicholas Cronk编著的《伏特瓦尔的数字书信集》，网址为http://dx.doi.org/10.13051/ee:doc/voltfrVF1260336a1c。

G04-P015-008 | 英文原文（原段接续 / 分段）

[42] Note also how in our Grand Tour case study, published in this issue of the *AHR*, a host of little-known and often forgotten architects are brought back into the fold of eighteenth-century British architecture.

中文译文 | G04-P015-008

42 另外请注意，在本期《AHR》杂志发表的我们的"大旅行"案例研究中，许多鲜为人知且常被遗忘的建筑师被重新纳入十八世纪英国建筑的范畴。

G04-P015-009 | 英文原文

AMERICAN HISTORICAL REVIEW APRIL 2017

中文译文 | G04-P015-009

《美国历史评论》2017年4月

原PDF第16页（原刊第415页）

G04-P016-002 | 英文原文

portraits of a number of correspondents in different locations, while Hatch subsequently articulated Peiresc's network around people rather than places.[43] Mandrou's project was a pioneering attempt to create a historical visualization of an intellectual network, but his idea did not really bear fruit for the next few decades. In 1987, Ultee reiterated the need for developing new ways to explore this kind of historical data while he completed his analysis of Leibniz's correspondence, acknowledging the limitations of his ability to convey what he had found in words when he concluded, "Eventually a graphic presentation will clarify the links in this network."[44]

中文译文 | G04-P016-002 | 原文注号：43、44

不同地点多位通信人的肖像，而哈奇随后将佩雷斯克的网络描述为以人而非地点为节点。43 曼德鲁的项目试图开创一种对知识网络的历史可视化方式，但其想法在接下来的几十年里并未真正取得成果。1987年，乌尔蒂重申了开发新方法探索此类历史数据的必要性，同时完成了他对莱布尼茨通信的分析，他承认自己无法用文字完整表达所发现的内容，最后指出："最终，图形展示将能阐明该网络中的联系。"44

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G04-P016-004 | 英文原文

The more historians began to consider correspondence in relationship to the reconstruction of historical networks, the more urgent these questions became. By 2001, David Kronick, one of the pioneering historians of the early modern journal, noted the preliminary results of new techniques of digitizing and analyzing early modern correspondence, especially citing Urs Boschung's electronic catalogue of the voluminous correspondence of the eighteenth-century Swiss physician, naturalist, and poet Albrecht von Haller—almost 17,000 letters from nearly 1,200 correspondents—which subsequently became a highly productive case study in network analysis and visualizations by a team of researchers.[45] Haller's correspondence is one of numerous projects that have emerged in recent years in which digitization has stimulated new forms of analysis and new ways of presenting the results of these investigations.[46]

中文译文 | G04-P016-004 | 原文注号：45、46

随着更多历史学家开始将通信文献与历史网络重建联系起来进行研究，这些问题变得更加紧迫。到2001年，作为早期现代期刊领域的先驱学者之一的大卫·克罗尼克指出，数字化与分析早期现代通信文献的新技术已取得了初步成果。他特别提到了乌尔斯·博斯洪所编制的18世纪瑞士医生、自然学家及诗人阿尔布雷希特·冯·哈勒的通信目录——该目录包含来自近1200位通信者的约17,000封信件——这一案例后来被一组研究人员作为网络分析与可视化研究的典型范例被广泛使用。哈勒的通信文献是近年来众多通过数字化技术推动新分析形式及结果呈现方式出现的项目之一。

G04-P016-005 | 英文原文

As we began our project in 2008, experiments with mapping the Republic of Letters proliferated. Robert Mayhew's account of early modern British geography demonstrates effectively how text-mining can help us understand the intellectual geography of the Republic of Letters and its understanding of physical geography.[47] Inspired by Steven J. Harris's early studies of mapping Jesuit science and Christopher Bayly's influential account of the relationship between information and empire, Simon Schaffer created a map of the "information order" of Isaac Newton's *Principia mathe-*

中文译文 | G04-P016-005 | 原文注号：47

2008年我们开始该项目时，对文人共和国的地图绘制实验日益增多。罗伯特·梅休对近代英国地理的论述有效地表明，文本挖掘有助于我们理解文人共和国的智力地理及其对物理地理的理解。受史蒂文·J·哈里斯早期对耶稣会科学地图绘制的研究，以及克里斯托弗·贝利关于信息与帝国关系的重要论述的启发，西蒙·谢弗绘制了艾萨克·牛顿《自然哲学原理》中的"信息秩序"地图。

G04-P016-006 | 英文原文

[43] Robert Mandrou, *From Humanism to Science, 1480–1700*, trans. Brian Pearce (New York, 1979), 309–313; Hatch, "Peiresc as Correspondent," 29–30.

中文译文 | G04-P016-006

43 罗伯特·曼德罗，《从人文主义到科学，1480–1700》，布赖恩·皮尔斯译（纽约，1979），309–313页；哈奇，"佩雷斯克作为通信者"，29–30页。

G04-P016-007 | 英文原文（原段接续 / 分段）

[44] Ultee, "The Republic of Letters," 103. Shelford responded to this idea by providing useful maps of Huet's Republic of Letters; *Transforming the Republic of Letters*, 9, 32, 128.

中文译文 | G04-P016-007

44 乌尔蒂，《文人共和国》，第103页。谢尔福德针对这一观点，提供了胡埃特的文人共和国的相关地图；见《转化文人共和国》，第9、32、128页。

G04-P016-008 | 英文原文（原段接续 / 分段）

[45] David A. Kronick, "The Commerce of Letters: Networks and 'Invisible Colleges' in Seventeenth- and Eighteenth-Century Europe," *Library Quarterly* 71 (2001): 28–43, here 40; Urs Boschung, *Repertorium zu Albrecht von Hallers Korrespondenz 1724–1777* (Basel, 2002), http://www.albrecht-von-haller.ch/medien/pdf/repertorium.pdf. For a network analysis of Haller's correspondence, see Martin Stuber, Stefan Hächler, and Luc Lienhard, eds., *Hallers Netz: Ein europäischer Gelehrtenbriefwechsel zur Zeit der Aufklärung* (Basel, 2005); Florence Catherine, "La pratique et les réseaux savants d'Albrecht von Haller (1708–1777), vecteurs de transfert culturel entre les espaces français et germaniques au XVIIIe siècle" (doctoral thesis, Université Nancy 2, 2009). A full bibliography of the research emerging from this collaborative project can be found at http://www.albrecht-von-haller.ch/e/correspondence.php.

中文译文 | G04-P016-008

45 大卫·A·克罗尼克，《书信交流：十七与十八世纪欧洲的网络与"无形交流"》，《图书馆季刊》第71期（2001年）：28–43页，此处为40页；乌尔斯·博斯洪，《阿尔布雷希特·冯·哈勒通信集目录，1724–1777年》（巴塞尔，2002年），http://www.albrecht-von-haller.ch/medien/pdf/repertorium.pdf。关于哈勒通信网络的分析，可参考马丁·施图伯、斯特凡·哈赫勒和卢克·利恩哈德编著的《哈勒的网络：启蒙时代欧洲学者间的通信》（巴塞尔，2005年）；弗洛伦斯·凯瑟琳，《阿尔布雷希特·冯·哈勒（1708–1777）的学术实践与网络：十八世纪法语与德语地区文化传播的载体》（南锡第二大学博士论文，2009年）。该合作项目所产出的研究文献完整目录可查阅http://www.albrecht-von-haller.ch/e/correspondence.php。

G04-P016-009 | 英文原文（原段接续 / 分段）

[46] To pick another rich example that has produced a considerable body of recent scholarship engaged with questions of correspondence networks and the nature of epistolary commerce, see the Clusius Project, sponsored by the Scaliger Institute of the University of Leiden, http://www.library.leiden.edu/special-collections/scaliger-institute/projects/clusius-project.html, which has digitized the correspondence of the Dutch naturalist Carolus Clusius. See Florike Egmond, Paul Hoftijzer, and Robert Visser, eds., *Carolus Clusius: Towards a Cultural History of a Renaissance Naturalist* (Amsterdam, 2008); Florike Egmond, *The World of Carolus Clusius: Natural History in the Making, 1550–1610* (London, 2010). This archive has recently been integrated into another Dutch project co-sponsored by Huygens ING whose goal is to create a complete digital archive of manuscript facsimiles, transcriptions, and biographical information. Digital Edition of the Clusius Correspondence, http://www.dwc.knaw.nl/biografie/clusius/digital-edition-of-the-clusius-correspondence/.

中文译文 | G04-P016-009

46 另一个丰富的例子表明，近期有大量研究致力于探讨通信网络与书信贸易的本质问题。可参阅由莱顿大学斯卡尔吉格研究所支持的Clusius项目，网址为：http://www.library.leiden.edu cnecial-collections/scaliger-institute/projects/clusius-project.html。该项目已将荷兰自然学家卡洛卢斯·克鲁修斯的通信资料数字化。参见弗洛利克·埃格蒙德、保罗·霍夫蒂泽尔和罗伯特·维瑟编著的《卡洛卢斯·克鲁修斯：一位文艺复兴时期自然学家的文化史》（阿姆斯特丹，2008年）；以及弗洛利克·埃格蒙德所著的《卡洛卢斯·克鲁修斯的世界：1550–1610年的自然历史》（伦敦，2010年）。这个档案库最近被整合到另一个由胡伊根斯ING资助的荷兰项目之中，该项目的目标是建立完整的数字化档案，包含写本副本、转录文本及人物传记信息。克鲁修斯通信书的数字版可访问网址：http://www.dwc.knaw.nl/biografie/clusius/digital-edition-ofthe-clusius-correspondence/.

G04-P016-010 | 英文原文（原段接续 / 分段）

[47] Robert Mayhew, "British Geography's Republic of Letters: Mapping an Imagined Community, 1600–1800," *Journal of the History of Ideas* 65, no. 2 (2004): 251–276.

中文译文 | G04-P016-010

47 Robert Mayhew，"英国地理学的文人共和国：1600–1800年想象社区的映射"，《思想史杂志》第65卷，第2期（2004年）：251–276页。

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G04-P017-002 | 英文原文

matica (1687) to debunk the notion that Newton was a solitary scholar untouched by a changing world.[48] Similarly, Yves Gingras explored the uses of citation and co-citation analysis, taking advantage of the digitization of Royal Society secretary Henry Oldenburg's correspondence by the Electronic Enlightenment Project at Oxford University and JSTOR's fully searchable version of the *Philosophical Transactions* to reconstruct the Scientific Revolution as a "visible map of the intellectual relations between people," arguing for the importance of combining "the usual micro-analysis of the specific content of the letters" with an ability to consider it instead as "a global corpus of data."[49] A group of French early modern historians, collaborating under the acronym CITERE, have also combined quantitative data and maps to analyze such variables as the speed and size of learned correspondence in eighteenth-century Europe.[50]

中文译文 | G04-P017-002 | 原文注号：48、49、50

Matica (1687)试图打破"牛顿是一位不受时代变迁影响的孤独学者"这一观念。48 同样，Yves Gingras探讨了引用与共引用分析的应用，利用牛津大学Electronic Enlightenment项目对皇家学会秘书Henry Oldenburg通信的数字化成果，以及JSTOR可完全搜索的《哲学交易录》版本，他将科学革命视为"人们之间知识关系的可视化图谱"，主张将"对信件具体内容的常规微观分析"与将其视为"全球数据文献总集"的能力相结合的重要性。49 一群法国早期现代史学家以CITERE为代号开展合作，他们结合定量数据与地图，分析了18世纪欧洲学术通信的速度与规模等变量。50

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G04-P017-004 | 英文原文

Many research questions underpinning scholarship on the Republic of Letters ultimately rest on data issues.[51] While recognizing that not everyone likes the idea of presenting early modern letters as filled with "data," worrying that it runs the risk of flattening the analysis or losing sight of the fact that these are historical documents whose creation is context-specific, we nonetheless believe that the information they contain deserves greater attention. Our experience in the Mapping the Republic of Letters project addresses a number of these questions to demonstrate how a digital and visual approach can indeed facilitate this sort of analysis. Building upon the rich historiography we have outlined above, we began with the insights gleaned from this considerable body of research. The inhabitants of the early modern Republic of Letters had elaborate visions of what their intellectual community resembled—most viewed it as a cosmopolitan, egalitarian place, and proudly highlighted their correspondents in far-flung places. Montesquieu famously satirized this self-perception in his *Persian Letters* (1721), where an astronomer boasts about corresponding with "a man in Stockholm, another in Leipzig, and another in London, whom I have never seen, and no doubt shall never see." Such descriptions parody the actual practices of the great secretaries of the Republic of Letters such as Oldenburg, who reportedly "never read a Letter before he had Pen, Ink, and Paper ready to answer it forthwith."[52] But could these descriptions really be trusted? Or was the reality of scholarly

中文译文 | G04-P017-004 | 原文注号：51、52

许多支撑文人共和国研究工作的学术问题，最终都依赖于数据问题。51 虽然认识到并非所有人都喜欢将早期现代信件视为充满"数据"的载体，且担心这种处理方式会使分析变得扁平化，或忽视这些文件属于具有特定背景的历史文献这一事实，但我们仍认为其中所包含的信息值得给予更多关注。我们在"绘制文人共和国地图"项目中的经验解决了许多这些问题，展示了数字与视觉方法如何真正促进此类分析。基于上述丰富的历史研究，我们从这一大量研究成果中获得的见解开始着手。早期现代文人共和国的居民对他们的知识共同体有着复杂的认知——大多数人认为这是一个世界性的、平等的场所，并自豪地展示他们在遥远地区的通信对象。孟德斯鸠在《波斯信札》（1721年）中以讽刺的方式描述了这种自我认知，其中一位天文学家夸耀与"斯德哥尔摩的人、莱比锡的人以及伦敦的人通信，而我一直未见到他们，无疑也永远见不到。"这类描述模仿了文人共和国伟大秘书们的实际行为，例如奥尔登堡据说"在准备好笔、墨水和纸张以立即回复之前，从不阅读任何信件"。52 但这些描述真的可信吗？还是说现实情况并非如此？

G04-P017-005 | 英文原文

[48] Simon Schaffer, *The Information Order of Isaac Newton's Principia Mathematica* (Uppsala, 2008). The history of science has actively incorporated certain insights from sociologist John Law's and sociologist and anthropologist Bruno Latour's contributions to Actor-Network Theory (ANT) to develop a rich account of the role of human and non-human actors in social networks. See, for example, Steven J. Harris, "Mapping Jesuit Science: The Role of Travel in the Geography of Knowledge," in John W. O'Malley, Gauvin Alexander Bailey, Steven J. Harris, and T. Frank Kennedy, eds., *The Jesuits: Cultures, Sciences, and the Arts, 1540–1773* (Toronto, 1999), 212–240.

中文译文 | G04-P017-005

48 西蒙·沙弗，《艾萨克·牛顿《自然哲学的数学原理》中的信息秩序》（乌普萨拉，2008）。科学史积极吸收了社会学家约翰·劳与社会学家、人类学家布鲁诺·拉图尔对"行动者-网络理论"的贡献，以全面阐述人类与非人类行动者在社交网络中的作用。例如，参见史蒂文·J·哈里斯的论文《绘制耶稣会科学图谱：旅行在知识地理中的作用》，载于约翰·W·奥马利、高文·亚历山大·贝利、史蒂文·J·哈里斯和T·弗兰克·肯尼迪编著的《耶稣会：文化、科学与艺术，1540–1773年》（多伦多，1999），第212–240页。

G04-P017-006 | 英文原文（原段接续 / 分段）

[49] Yves Gingras, "Mapping the Structure of the Intellectual Field Using Citation and Co-Citation Analysis of Correspondences," *History of European Ideas* 36, no. 3 (2010): 330–339, here 330 and 339; and Yves Gingras and Alexandre Guay, "The Use of Analogies in Seventeenth and Eighteenth Century Science," *Perspectives on Science* 19, no. 2 (2011): 154–191.

中文译文 | G04-P017-006

49 伊夫·金格拉斯，《通过引用与共引分析来绘制思想领域的结构图谱》，《欧洲思想史》第36卷，第3期（2010年）：330–339页，此处引用330和339页；以及伊夫·金格拉斯与亚历山大·盖伊，《类比在十七和十八世纪科学中的应用》，《科学视角》第19卷，第2期（2011年）：154–191页。

G04-P017-007 | 英文原文（原段接续 / 分段）

[50] See Pierre-Yves Beaurepaire, ed., *La communication en Europe de l'âge classique au siècle de Lumières* (Paris, 2014). The English name for this project is Communicating Europe: Early Modern Circulations, Territories and Networks.

中文译文 | G04-P017-007

50 参见皮埃尔-伊夫·博尔佩尔编，《从古典时代到启蒙时代的欧洲传播》（巴黎，2014年）。该项目的英文名称为《欧洲传播：早期现代传播、领域与网络》。

G04-P017-008 | 英文原文（原段接续 / 分段）

[51] For example, see the approach taken in many of the articles in Berkvens-Stevelinck, Bots, and Häseler, *Les grands intermédiaires culturels de la République des lettres*.

中文译文 | G04-P017-008

51 例如，可参考Berkvens-Stevelinck、Bots和H€aseler所著的《文学共和国的重要文化中介》一书中许多文章所采用的方法。

G04-P017-009 | 英文原文（原段接续 / 分段）

[52] Montesquieu, *Persian Letters*, trans. Margaret Mauldon (Oxford, 2008), 221; Marie Boas Hall, *Henry Oldenburg: Shaping the Royal Society* (Oxford, 2002), 129, quoting Martin Lister's recollection of Oldenburg in 1698.

中文译文 | G04-P017-009

52 蒙泰斯奎，《波斯信札》，玛格丽特·莫尔登译（牛津，2008），221页；玛丽·博阿斯·霍尔，《亨利·奥尔德堡：对皇家学会的影响》（牛津，2002），129页，引自马丁·利斯特对奥尔德堡在1698年的回忆。

G04-P017-010 | 英文原文

中文译文 | G04-P017-010

《美国历史评论》2017年4月

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G04-P018-002 | 英文原文

communication in the early modern age far more prosaic? Was the Republic of Letters genuinely cosmopolitan, or was it more regional, national, and confessional in scope? What did membership in this republic actually look like, and how might we bring different versions of this imagined community into focus by exploring well-known, concrete examples simultaneously? And how did it evolve over time and space? These were the questions that our research group set out to explore, initially without the aid of data or visualizations. But like so many scholars who came before us, we inevitably found ourselves grappling with what to do with the growing mountains of historical information about people, places, and time that lay at the center of our enterprise.

中文译文 | G04-P018-002

早期现代时期的交流是否更为平凡？文人共和国究竟具有真正的世界性特征，还是更偏向地区性、国家性或宗教性范畴？加入该共和国的成员资格究竟是什么样，我们又如何通过探索著名的、具体的实例来凸显这种想象中的社群的不同版本？它又是如何随着时间和空间的发展而演变的？这些正是我们的研究小组试图探讨的问题，最初并未借助数据或可视化工具。但如同许多之前的学者一样，我们不得不面对那些关于人物、地点和时代的历史信息堆积如山，这些信息正是我们研究的核心。

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G04-P018-004 | 英文原文

Is a big historical data approach the future or an illusion? It is easy to understand why some commentators have been predicting a "big data revolution," even in the humanities.[53] Not only are increasing numbers of texts and archives being digitized, but their accompanying metadata can now be validated and enriched via cultural heritage resources. For instance, we can begin to identify published authors across different datasets thanks to the adoption of authority files (such as VIAF), which link library records and other data sources through unique identifiers.[54] More and more libraries, including the OCLC, the Bibliothèque nationale de France, and Europeana, are also adopting Linked Open Data (LOD) models, making structured and curated data currently contained in library catalogues available for research.[55] The bibliographer's domain is fast becoming terrain for scholars to plumb and mine.

中文译文 | G04-P018-004 | 原文注号：53、54、55

这是一个重要的历史数据方法，还是一种幻想？很容易理解为何一些评论人士预测会出现"大数据革命"，甚至是在人文学领域也是如此。53 不仅越来越多的文本和档案被数字化，其相关的元数据现在也可以通过文化遗产资源进行验证和丰富。例如，通过采用权威文件（如VIAF），我们可以在不同数据集中识别出已发表的作者，这些文件通过唯一的标识符将图书馆记录和其他数据源连接在一起。54 越来越多的图书馆，包括OCLC、法国国家图书馆和Europeana，都在采用链接开放数据模型，使得目前存储在图书馆目录中的结构化、经过整理的数据能够用于研究。55 图书学领域正迅速成为学者探索和利用的领域。

G04-P018-005 | 英文原文（原段接续 / 分段）

Much attention in the past fifteen years has been directed toward *text* digitization, but the production of vast stores of data—in particular, of metadata—is an equally exciting story. Anyone who has spent a few hours on Google Books can probably imagine a future in which all print material has been digitized; but imagining the future of data collection is more challenging. Ideally, if data models become reasonably standardized and well implemented, we might be able to look forward to a world in which datasets connect quickly and seamlessly, combining a wealth of discrete and useful information available to be reorganized into complex combinatorial patterns. If the print era was defined by the revolutionary technology of movable type, this coming

中文译文 | G04-P018-005

过去十五年来，人们大量关注文本数字化工作，但海量数据的生成——尤其是元数据——同样是一个令人兴奋的领域。任何在谷歌图书上花费几小时浏览过的人，或许都能想象出所有印刷资料都被数字化的未来；但设想数据收集的未来则更具挑战性。理想情况下，如果数据模型能够得到合理标准化并有效实施，我们或许能期待一个数据集能够快速、无缝连接的世界，将大量独立且有用的信息整合成复杂的组合模式。如果印刷时代由活字印刷技术所定义，那么即将到来的时代将会如何呢？

G04-P018-006 | 英文原文

[53] See Mayer-Schönberger and Cukier, *Big Data*; Erez Aiden and Jean-Baptiste Michel, *Uncharted: Big Data as a Lens on Human Culture* (New York, 2013). For a similar endorsement by historians, see Guldi and Armitage, *The History Manifesto*.

中文译文 | G04-P018-006

53 参见 Mayer-Schoenberg 和 Cukier 的《大数据》；Erez Aiden 和 Jean-Baptiste Michel 的《未知领域：大数据作为人类文化的视角》（纽约，2013年）。关于历史学家对此观点的支持，可参考 Guldi 和 Armitage 的《历史宣言》。

G04-P018-007 | 英文原文（原段接续 / 分段）

[54] VIAF stands for "Virtual International Authority File" (http://viaf.org/). As the website states, this service "combines multiple name authority files [from different databases, mainly libraries] into a single OCLC-hosted name authority service. The goal of the service is to . . . increase the utility of library authority files by matching and linking widely-used authority files and making that information available on the Web."

[55] On Linked Open Data, see notably Tom Heath and Christian Bizer, *Linked Data: Evolving the Web into a Global Data Space* (San Rafael, Calif., 2011), http://www.morganclaypool.com/doi/abs/10.2200/S00334ED1V01Y201102WBE001; see also Eero Hyvönen, *Publishing and Using Cultural Heritage Linked Data on the Semantic Web* (San Rafael, Calif., 2012), http://www.morganclaypool.com/doi/abs/10.2200/S00452ED1V01Y201210WBE003. Our thanks to Glauco Mantegari for these references. The open data portal of the Bibliothèque nationale can be found at http://data.bnf.fr; for Europeana, see http://pro.europeana.eu/linked-open-data. Information about the OCLC's linked data plans (through WorldCat) can be found at http://www.oclc.org/data.en.html. Unfortunately, the process for converting MARC library data into LOD (in BIBFRAME format) is likely to take many, many years.

中文译文 | G04-P018-007

54 VIAF 代表"虚拟国际权威文件"（http://viaf.org/）。正如该网站所述，这项服务"将来自不同数据库（主要是图书馆）的多个名称权威文件整合到由 OCLC 托管的单一名称权威服务中。该服务的目标是……通过匹配和链接常用权威文件，使图书馆权威文件的实用性提升，并将这些信息在互联网上公开。"55 关于链接开放数据，可参考 Tom Heath 和 Christian Bizer 所著的《链接数据：将网络转化为全球数据空间》（加利福尼亚州圣拉斐尔，2011年），http://www.morganclaypool.com/doi/abs/10.2200/S00334ED1V01Y201102WBE001；另可参考 Eero Hyvoenen 所著的《在语义网中发布和使用文化遗产链接数据》（加利福尼亚州圣拉斐尔，2012年），http://www.morganclaypool.com/doi/abs/10.2200/S00452ED1V01Y201210WBE003。我们感谢 Glauco Mantegari 提供这些参考文献。法国国家图书馆的开放数据门户可访问于 http://data.bnf.fr；关于 Europeana，可访问 http://pro.europeana.eu/linked-open-data。有关 OCLC 链接数据计划的信息（通过 World- Cat 平台）可访问于 http://www.oclc.org/data.en.html。遗憾的是，将 MARC 图书馆数据转换为 LOD 格式（BIBFRAME 格式）的过程可能需要很长时间。

G04-P018-008 | 英文原文

American Historical Review April 2017

中文译文 | G04-P018-008

《美国历史评论》2017年4月

原PDF第19页（原刊第418页）

G04-P019-002 | 英文原文

era could well be defined by the equally revolutionary practice of "movable data." Once set in print and locked in books, catalogues, or tables, data can now be pried loose from their original settings and made available for new arrangements. Therein lies perhaps the greatest promise of data for historians: until now, most of us have been beholden to search forms for querying library catalogues, digital corpora, and other databases. We are constrained by what the library or the archive permits us to ask. Linked data models have the potential to make data more nimble and malleable, and subject to a much wider variety of uses.

中文译文 | G04-P019-002

“可移动数据”这种同样具有革命性的实践方式，或许可以定义这一时代。一旦被记录在印刷品中并保存在书籍、目录或表格中，数据就可以从原有的环境中被释放出来，用于新的用途。这或许就是数据对历史学家最具潜力的价值：直到现在，我们大多只能依赖搜索表单来查询图书馆目录、数字语料库及其他数据库。我们的提问受到图书馆或档案馆允许的范围所限制。链接式数据模型则有可能使数据更加灵活且可塑，从而适用于更多不同的用途。

G04-P019-003 | 英文原文（原段接续 / 分段）

But this prospect still lies in the future, and may yet prove to be a mirage. For one thing, when considering the promise of linking data across archives, we must consider the original state of the underlying data. The metadata for the historical records used in many of our case studies were constructed for data retrieval within particular online systems, not for a research inquiry.[56] The “place of publication” field in the Bibliothèque nationale contains a number of values similar enough for us to consider, but with subtle variations: Paris appears as “Paris,” “(Paris),” “[Paris],” “(Paris,)” etc. To use these data effectively, we need to carefully decode the meanings in the parentheses, brackets, and commas that may or may not be documented (for instance, when is Paris the stated but false place of publication vs. when is it the unstated but real place?). The same can be said for digitized modern print editions: content structured for publishing is effectively unstructured for the purposes of data analysis. The scholarly parsing of existing data often leads us to create our own new columns of data that serve our research agenda.

中文译文 | G04-P019-003 | 原文注号：56

然而，这一前景仍有待未来实现，也可能最终只是海市蜃楼。首先，在考虑跨档案关联数据的潜力时，我们必须考察底层数据的原始状态。在我们许多个案中，历史记录的元数据最初是为特定在线系统内部的数据检索而建立，并非为某项研究问题而设计。〔56〕法国国家图书馆的“出版地”字段中，有一些值彼此足够相似，值得一同考察，却又存在细微差别：巴黎会写成“Paris”“(Paris)”“[Paris]”“(Paris,)”等。要有效使用这些数据，就必须仔细解读圆括号、方括号和逗号所代表的含义，而这些含义未必有说明文件。例如，什么时候巴黎是书上标明却虚假的出版地，什么时候又是未标明但真实的出版地？经过数字化的现代印刷版本也有同样的问题：为出版而组织的内容，对于数据分析而言，实际上仍是非结构化的。学者对既有数据的解析，往往会促使我们建立服务于自身研究目标的新数据列。

G04-P019-004 | 英文原文（原段接续 / 分段）

Secondly, there is also the problem of incompleteness, often of unknown proportions. To borrow an example from our research, we were fortunate to receive at an early stage of our project a large dataset from the Electronic Enlightenment Project (EEP) at Oxford University.[57] The records contained the metadata for about 50,000 letters to and from leading seventeenth- and eighteenth-century authors, including Thomas Hobbes, John Locke, Voltaire, Rousseau, David Hume, Claude Adrien Helvétius, Adam Smith, and Jeremy Bentham. Along with these metadata, we also benefited from the painstaking curatorial work of the researchers at EEP, who had included information about nationality, occupation, gender, and dates for each individual in the collection. But the data were filled with gaps and uncertainties. This was in no way the fault of EEP, which had produced as fine a database as possible (in addition to their remarkable full-text correspondence database). Rather, the data were incomplete because the historical record was incomplete. Many letters had been passed down without any indication of their source or destination locations; some were also undated. Others had vanished off the face of the earth.

中文译文 | G04-P019-004 | 原文注号：57

其次，还存在数据不完整的问题，而且不完整的程度通常无法确定。以我们的研究为例：项目早期，我们有幸从牛津大学的“电子启蒙运动项目”（EEP）获得一个大型数据集。〔57〕其中包含约50,000封信的元数据，涉及17至18世纪的重要作者，包括Thomas Hobbes、John Locke、伏尔泰、卢梭、David Hume、Claude Adrien Helvétius、Adam Smith和Jeremy Bentham等，既有他们发出的信，也有他们收到的信。除了这些元数据，我们还受益于EEP研究者细致的资料整理工作：他们为收藏中的每个人补充了国籍、职业、性别和生卒日期等信息。但这些数据仍有大量空缺与不确定性。这绝不是EEP的过错；除建立出色的书信全文数据库外，他们已尽可能建成优良的数据库。数据之所以不完整，是因为历史记录本身不完整。许多传世书信没有注明发信地或收信地，有的也没有日期，另一些则已彻底亡佚。

G04-P019-005 | 英文原文（原段接续 / 分段）

This critical reading and assessment of the state of the data is common to each of our case studies in Mapping the Republic of Letters. This distinction points to a third significant methodological difference in our approach to the objects of our study. If our historical subjects had been communicating digitally, and we could have captured their exchanges and movements from a real-time live data stream, we might be able to

中文译文 | G04-P019-005

对数据状况的这种批判性解读与评估，在我们所有关于文人共和国的测绘案例研究中均存在。这一差异表明，我们在研究对象处理上存在第三个重要的方法论区别。如果我们的历史主体通过数字方式进行交流，且我们能够从实时直播数据中获取他们的交流与动态，或许我们就能……

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G04-P019-007 | 英文原文

[56] This is why scholarly cataloguing projects such as Early Modern Letters Online (EMLO), developed by the Cultures of Knowledge Project at Oxford University, are so important for work on early modern correspondence. See http://emlo.bodleian.ox.ac.uk/.

中文译文 | G04-P019-007

56 正因如此，牛津大学知识文化项目所开发的“早期现代信件在线”（EMLO）等学术编目项目，对研究早期现代通信文献至关重要。详见 http://emlo.bodleian.ox.ac.uk/。

G04-P019-008 | 英文原文（原段接续 / 分段）

[57] See http://www.e-enlightenment.com/.

中文译文 | G04-P019-008

57 参见 http://www.e-enlightenment.com/。

G04-P019-009 | 英文原文

中文译文 | G04-P019-009

原PDF第20页（原刊第419页）

G04-P020-002 | 英文原文

adopt the statistical methods favored by data scientists. But historical information does not stream forth like a Twitter feed. The datasets that we have explored, all of which were manually scrutinized, edited, and enriched by our research teams, came to us already structured, and sometimes designed for purposes contrary to our own. Most of the metadata we rely upon from these digitized archives have been framed and reframed for various uses over decades. We need to unmask their underlying assumptions in order to get down to the level of historical evidence. The role of the scholar thus becomes one of finding meaning in the mass, which in the case of historical data is often also a mess.

中文译文 | G04-P020-002

采用数据科学家偏爱的统计方法。但历史信息并不像推特消息流那样不断涌出。我们考察的所有数据集，都经过研究团队手工审查、编辑和补充；它们到达我们手中之前，已经具有既定结构，有时其设计目的甚至与我们的目的相悖。我们从这些数字档案中使用的大多数元数据，在数十年间曾为了不同用途被反复组织和重新组织。必须揭示其背后的假设，才能深入到历史证据的层面。因此，学者的职责变成了从大量资料中寻找意义；而对于历史数据，这种“大量资料”往往也是一团乱麻。

G04-P020-003 | 英文原文（原段接续 / 分段）

Our final problem is more pragmatic and has greatly preoccupied our research project: Where are the tools to support humanistic inquiry into multidimensional, heterogeneous, and incomplete datasets? To take maximum advantage of this coming data deluge, we will need new technologies to help us construct meaningful queries and then explore, filter, combine, and analyze the results.[58] We need to see the past from many different vantage points in order to understand it, and that cannot happen in front of an infinite tabular spreadsheet.[59] Visualization has proven to be an extremely effective tool for representing data and navigating data across many disciplines.[60] ArcGIS, Tableau, R, and Gephi are some of the most powerful desktop applications and programming languages for data visualization used in humanities research today. And yet the quantitative bias that defines much of the functionality of those tools does little to support the kind of qualitative analysis practiced in our case studies. We require different tools that can accommodate ambiguities, paradoxes, and contingencies to help us process historical and more generally humanistic data effectively.

中文译文 | G04-P020-003 | 原文注号：58、59、60

最后一个问题更为实际，也一直占据我们研究项目的大量精力：支持人文学者研究多维、异质且不完整数据集的工具，究竟在哪里？要充分利用即将到来的数据洪流，我们需要新技术，帮助提出有意义的查询，再探索、筛选、组合和分析结果。〔58〕为了理解过去，我们需要从许多不同视角观察它；坐在一张无穷无尽的电子表格前，无法做到这一点。〔59〕可视化已被证明是跨学科表现与浏览数据的极为有效的工具。〔60〕ArcGIS、Tableau、R和Gephi，是当今人文研究中用于数据可视化的强大桌面应用或编程语言。不过，这些工具的许多功能以定量取向为基础，对于我们在个案中实践的定性分析帮助有限。我们需要能够容纳歧义、悖论和偶然性的不同工具，以便有效处理历史数据乃至更广泛的人文数据。

G04-P020-004 | 英文原文（原段接续 / 分段）

Designing digital tools is a long-term, expensive, complicated process, one that typically involves working with grant officers, data scientists, programmers, and interface designers. Needless to say, we had no idea what we were getting into when we set off down this path. This phase of our collaboration arose from the growing realization that we simply could not begin to explore, to any serious degree, the questions that we wanted to pose without experimenting with different approaches to visualizing historical data. We could borrow or build a database or two, but what did we want to do with it? This became the crucial question that led us in the direction that ultimately resulted in the creation of a web-based data visualization platform, Palladio.[61]

中文译文 | G04-P020-004 | 原文注号：61

设计数字工具是一个长期、昂贵且复杂的过程，通常需要与资助官员、数据科学家、程序员和界面设计师合作。不言而喻，当我们开始这一进程时，我们完全不知道自己将面临什么挑战。我们的合作阶段源于一个逐渐清晰的认识：如果我们不尝试不同的数据可视化方法，就根本无法认真探讨我们想要提出的问题。我们可以借用或构建一两个数据库，但究竟要利用它们做些什么呢？这成为了一个关键问题，最终引导我们创造了基于网络的数据可视化平台——Palladio。61

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G04-P020-006 | 英文原文

[58] For a similar point, see Guldi and Armitage, *The History Manifesto*, chap. 4.

[59] For a fascinating historical reflection on this problem that has inspired aspects of our own historical visualizations, see Daniel Rosenberg and Anthony Grafton, *Cartographies of Time: A History of the Timeline* (New York, 2010).

中文译文 | G04-P020-006

58 关于类似观点，参见Guldi和Armitage所著的《历史宣言》第4章。59 关于对这一问题的有趣历史分析，该分析影响了我们自身历史呈现的方式，可参阅Daniel Rosenberg和Anthony Grafton所著的《时间的地图：时间线的历史》（纽约，2010年）。

G04-P020-007 | 英文原文（原段接续 / 分段）

[60] Evidence of this can be found in the development of the commercial visualization software packages mentioned below, as well as in the popularity of recent books and websites on the topic from the fields of statistics, design, business analytics, and computer science. For some prominent examples, see Edward Tufte, http://www.edwardtufte.com; Manuel Lima, http://www.visualcomplexity.com/vc/; Stephen Few, http://perceptualedge.com; Alberto Cairo, http://www.thefunctionalart.com; David McCandless, http://www.informationisbeautiful.net.

中文译文 | G04-P020-007

60. 这一证据可体现在下面提到的商业可视化软件包的发展中，也可体现在统计学、设计、商业分析及计算机科学领域关于该主题的书籍和网站的热度上。一些典型的例子包括：爱德华·塔夫特，http://www.edwardtufte.com；曼努埃尔·利马，http://www.visualcomplexity.com/vc/；斯蒂芬·费乌，http://perceptualedge.com；阿尔贝托·卡伊罗，http://www.thefunctionalart.com；大卫·麦克坎德尔斯，http://www.informationisbeautiful.net。

G04-P020-008 | 英文原文（原段接续 / 分段）

[61] Palladio is a data-visualization platform that allows users to upload and analyze their data in the browser; funds for its development came from the National Endowment for the Humanities and Stanford University. It brings together much of what we learned in the design and development of prototype tools over the past six years as members of the Mapping the Republic of Letters project at Stanford University, primarily in collaboration with the Density Design research lab at the Politecnico di Milano, and in conversation with allied initiatives such as the Cultures of Knowledge project at Oxford University, as well as the Circulation of Knowledge project jointly sponsored by Huygens ING in The Hague and the Des-

中文译文 | G04-P020-008

61. Palladio 是一个数据可视化平台，允许用户在浏览器中上传并分析其数据；其开发资金来自国家人文基金会和斯坦福大学。该平台整合了过去六年间作为斯坦福大学“文人共和国”项目成员在设计与开发原型工具过程中所学到的诸多经验，主要与米兰理工大学的密度设计研究实验室合作，并与牛津大学知识文化项目以及海牙Huygens ING与Des联合赞助的知识传播项目等相关项目进行交流。

G04-P020-009 | 英文原文

中文译文 | G04-P020-009

G04-P021-002 | 英文原文

Since our project has always had humanistic pursuits at heart, it is no surprise that our underlying philosophy of tool development would seek to enhance, not replace, traditional historical methods. Big data are not useful only to scholars who adopt statistical methods. The large scale also allows us to pursue traditional research in a much broader space of possibilities. When we construct visual models with data, we consider those visualizations to be powerful heuristics, not proofs. We wield our data with the assumption that they are incomplete, ambiguous, and uncertain. We visualize them to help guide us, to uncover patterns and provoke questions that we otherwise might not ever have considered. Our publications include online access precisely so that readers can explore and manipulate the data of our case studies. These use cases of Palladio allow direct insight into the process of heuristic visualizations, and on these same web pages readers can download the data to pursue their own research questions as well.[62] But we do not believe that the data in and of themselves provide answers. The interpretive work is done outside of the computer, in the decisions that go into building our datasets, in the design of the instruments we use to visualize the data, and finally in making sense of the very imprecise maps, graphs, and diagrams we produce. We still need to read those shelf-loads of books and boxes of manuscripts—in the archive or online—to make sense of our findings.

中文译文 | G04-P021-002 | 原文注号：62

由于我们的项目始终以人文追求为核心理念，因此我们的工具开发理念致力于提升传统历史研究方法，而非取代它们，这并不令人意外。大数据并非仅对采用统计方法的学者有用。大规模数据还使我们能够在更广阔的领域开展传统研究。当我们用数据构建可视化模型时，我们认为这些可视化结果属于强大的启发式工具，而非证明。我们以数据不完整、模糊且不确定为前提来使用这些数据。通过可视化方式，我们可以获得指导，发现模式，并引发那些原本可能无法思考的问题。我们的出版物包含在线访问功能，以便读者能够探索和操作我们的案例研究数据。这些Palladio应用案例使我们能够直接了解启发式可视化的过程，而在这些网页上，读者还可以下载数据以自行研究。但我们认为，数据本身并不提供答案。解释性工作是在计算机之外完成的：体现在构建数据集的决策中，体现在用于可视化的工具设计上，最终也体现在对我们所生成的不精确地图、图表和图形的解读中。我们仍需阅读那些成堆的书籍和写本——无论是在档案中还是在线——才能理解我们的发现。

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G04-P021-004 | 英文原文

As a result of dealing with data visualizations, we found ourselves thinking anew about space. Many of the patterns that interest us, and that our visualizations help to uncover, concern geographical relations. This puts our project in dialogue with spatial history, an area in which digital research has had a great impact. But our material and questions have inspired a distinct methodology for our research. Many spatial history projects analyze the relationships between different categories of space—for instance, between spatial practice and represented space, to borrow Henri Lefebvre's terms.[63]

中文译文 | G04-P021-004 | 原文注号：63

由于处理数据可视化工作，我们重新思考了空间的概念。许多我们感兴趣的模式，以及可视化技术能够揭示的模式，都与地理关系有关。这使得我们的项目与空间历史领域产生联系，而数字研究在这一领域具有重大影响。不过，我们的材料与问题也启发了我们一种独特的研究方法。许多空间历史项目都分析不同空间类别之间的关系——例如，用亨利·勒菲弗的话来说，就是分析空间实践与表征空间之间的关系。63

G04-P021-005 | 英文原文

This approach often involves comparing *layers* of space—quite literally in the case of the HyperCities project, for example.[64] By contrast, our project has been much more concerned with the relationships between different areas within a single layer of space. In fleeing Germany, Kircher was cast adrift from his native land, and his correspondence captures beautifully the essence of Jesuit mobility as well as how people maintained a strong sense of identity at a distance. Locke's exile in France belonged to one of those moments when a portion of the "English" world was on the Continent, as would later occur on a much larger scale when the Grand Tour created a world in

中文译文 | G04-P021-005 | 原文注号：64

这种方法通常涉及比较不同空间层之间的关系——例如 HyperCities 项目就是如此。64 相比之下，我们的项目更关注单一空间层内不同区域之间的关系。基尔彻在逃离德国后，便与故乡分离，他的信件生动展现了耶稣会士的流动性，以及人们在远方如何保持强烈的身份认同。洛克被流放到法国，也是"英国"世界的一部分出现在欧洲大陆的时刻之一，后来随着大旅行运动的开展，这一情况又以更大的规模发生了。

G04-P021-006 | 英文原文

cartes Centre at the University of Utrecht. See http://hdlab.stanford.edu/projects/palladio/. We count it as a success of our project that Palladio has now been used widely in a variety of projects well beyond our own within Mapping the Republic of Letters. For a recent assessment, see Graham, Milligan, and Weingart, *Exploring Big Historical Data*, 112, 268–269.

中文译文 | G04-P021-006

乌得勒支大学内的 Cartes 中心。请访问 http://hdlab.stanford.edu/projects/palladio/。我们认为，Palladio 如今已在《文人共和国》项目之外的多种项目中得到广泛应用，这标志着我们项目的成功。关于最近的评估结果，请参阅 Graham、Milligan 和 Weingart 所著的《探索大型历史数据》第 112 页，第 268–269 行。

G04-P021-007 | 英文原文（原段接续 / 分段）

[62] See http://republicofletters.stanford.edu/publications, which hosts the online interactive visualizations, databases, and data schemas for our case studies. Note that the datasets for each case study are also stored in SDR (the Stanford Digital Repository), https://sdr.stanford.edu/collections/druid:zn653qj0117, to ensure their longevity in what might otherwise at times seem the quite ephemeral afterlife of digital projects, and where they can be accessed openly by the scholarly community.

中文译文 | G04-P021-007

62 参见 http://republicofletters.stanford.edu/publications，该网站提供了我们案例研究的在线互动可视化图表、数据库和数据模式。请注意，每个案例研究的数据集也存储在 SDR（斯坦福数字存储库）中，网址为 https://sdr.stanford.edu/collections/druid: zn653qj0117，这样这些数据集就能在数字项目可能短暂消失之后仍然保存下来，并且学术界也可以自由访问这些数据。

G04-P021-008 | 英文原文（原段接续 / 分段）

[63] See Richard White, "What Is Spatial History?," Spatial History Lab: Working Paper, submitted February 1, 2010, http://web.stanford.edu/group/spatialhistory/cgi-bin/site/pub.php?id=29. White draws on Henri Lefebvre, *The Production of Space*, trans. Donald Nicholson-Smith (Cambridge, Mass., 1991).

中文译文 | G04-P021-008

63 参见理查德·怀特《什么是空间历史？》，《空间历史实验室：工作论文》，2010年2月1日提交，http://web.stanford.edu/group/spatialhistory/cgi-bin/site/pub.php?id1⁄429。怀特引用了亨利·勒菲弗的《空间的生产》，由唐纳德·尼科尔森-史密斯翻译（马萨诸塞州剑桥，1991年）。

G04-P021-009 | 英文原文（原段接续 / 分段）

[64] HyperCities: Thick Mapping in the Digital Humanities, http://hypercities.com/.

中文译文 | G04-P021-009

64 HyperCities：数字人文领域的深度映射，http://hypercities.com/。

G04-P021-010 | 英文原文

中文译文 | G04-P021-010

《美国历史评论》2017 年 4 月刊

原PDF第22页（原刊第421页）

G04-P022-002 | 英文原文

between England and Italy for multiple generations of Britons. Benjamin Franklin, remembered today as the archetypal American, in fact spent much of the last third of his life largely in England and the Continent, often doing business in which the lines between what was "British" and what was "American" were blurred to the point of meaninglessness. When we reflect on the connections among England, France, and North America in the eighteenth century, for example, we are essentially asking how an "English space" was situated in relation to a "French space," or an American one within a British world. Instead of superimposing layers, we look at a patchwork of spaces, all connected on the same plane by the human interactions that constituted the early modern Republic of Letters.[65]

中文译文 | G04-P022-002 | 原文注号：65

在多个世代的英国人看来，这是英格兰与意大利之间的关系。本杰明·富兰克林作为典型的美国人被后人铭记，但实际上他生命中的最后三分之一时间大多在英格兰和欧洲度过，他经常从事那些"英国"与"美国"界限模糊到毫无意义的事务。例如，当我们思考18世纪英格兰、法国与北美之间的关系时，实际上是在探讨"英国空间"如何与"法国空间"或"美国空间"在英国世界中相互关联。我们并非叠加不同层次，而是将这些空间视为一个整体，它们通过人类互动在同一个层面上相连，而这些互动构成了早期现代文人共和国的格局。65

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G04-P022-004 | 英文原文

The kind of spaces we study might best be described as “cultural zones.”[66] Such zones exist, almost by definition, in the plural: to a large degree, they are defined relative to and in distinction from one another. Our project requires us to identify these zones spatially and conceptually, but also to reflect on their interactions.

中文译文 | G04-P022-004 | 原文注号：66

我们研究的这种空间，或许最好称为‘文化区域’。〔66〕此类区域几乎按其定义就不是单一存在的：在很大程度上，它们是通过彼此相对、彼此区分来界定的。我们的项目要求我们从空间和概念两个层面识别这些区域，同时也思考它们之间的互动。

G04-P022-005 | 英文原文（原段接续 / 分段）

At the geographical level, we do not place much importance on exact boundaries or even precise locations; for this reason, we have not used GIS (geographic information systems) in our visualizations, and we tend to prefer abstract maps over historical ones. We take as a given that cultural zones have fuzzy borders and can overlap: eighteenth-century Geneva, for instance, could be seen as belonging to both a Swiss Protestant and a French cultural zone. We use the term “cultural zone” rather than “cultural space” in order to stress its imprecise, often amorphous, edges. A zone also covers a large area, which is why we chose this term over “place.” It can indeed capture connections between places in a given period.

中文译文 | G04-P022-005

在地理层面，我们并不太关注精确的边界或位置；因此，我们在可视化中未使用地理信息系统，而是更倾向于使用抽象地图而非历史地图。我们认为文化区域具有模糊的边界且可能相互重叠：例如，18世纪的日内瓦可以被视为属于瑞士新教和法国文化区域两者所涵盖的范围。我们使用“文化区域”这一术语而非“文化空间”，是为了强调其不精确、往往无定形的边界。一个区域涵盖较大的面积，正因如此，我们选择使用这一术语而非“地点”。它确实能够体现特定时期中各地点之间的关联。

G04-P022-006 | 英文原文（原段接续 / 分段）

Much of spatial theory, from Georg Simmel and Henri Lefebvre to David Harvey, has roots in urban studies.[67] Cities are also central to our thinking about cultural zones, but in our case we look *extra muros*. Indeed, a cultural zone can often be defined as the area affected by the dominance of a city (usually a capital). The French cultural zone radiates around Paris; in the eighteenth century, its reach could be felt as far away as St. Petersburg. The English cultural zone extended around London, and reached as far as Philadelphia and the Indian subcontinent. Rome’s reputation as *caput mundi* was certainly enhanced by its role as the nerve center of overlapping global missionary networks whose reach extended to every corner of the world. Of course, capitals did not always play this role: not every region created cultural zones

中文译文 | G04-P022-006 | 原文注号：67

许多空间理论，从格奥尔格·西梅尔和亨利·勒菲弗到大卫·哈维，其根源都在于城市研究。城市也是我们思考文化区概念的核心，但在我们的研究中，我们更关注“城外”地区。实际上，文化区通常可以被定义为受某座城市（通常是首都）影响的区域。法国的文化区围绕巴黎展开；在18世纪，其影响甚至波及到圣彼得堡。英国的文化区则围绕伦敦扩展，延伸至费城和印度次大陆。罗马作为“世界之都”的声誉，无疑得益于其作为全球传教网络中心的作用，这些网络的影响范围覆盖世界的每一个角落。当然，首都并非总是扮演这种角色：并非每个地区都能形成文化区。

G04-P022-007 | 英文原文

[65] On place vs. space, see Edward S. Casey, *The Fate of Place: A Philosophical History* (Berkeley, Calif., 1997); Philip J. Ethington, "Placing the Past: 'Groundwork' for a Spatial Theory of History," *Rethinking History* 11, no. 4 (2007): 465–493.

中文译文 | G04-P022-007

65 关于"地点"与"空间"，参见爱德华·S·凯西所著的《地点的命运：哲学史》（伯克利，加州，1997年）；菲利普·J·伊辛顿的"定位过去：历史空间理论的基础"，载于《重新思考历史》第11卷第4期（2007年），第465–493页。

G04-P022-008 | 英文原文（原段接续 / 分段）

[66] The concept of "zone," to the extent that it is used in spatial theory, tends to be deployed by literary scholars in relation either to cities (see, for instance, David Bell, Jon Binnie, Ruth Holliday, Robyn Longhurst, and Robin Peace, *Pleasure Zones: Bodies, Cities, Spaces* [Syracuse, N.Y., 2001]) or to climatic "zones" (see Felicity Nussbaum, *Torrid Zones: Maternity, Sexuality, and Empire in Eighteenth-Century English Narratives* [Baltimore, 1995]). There has also been some reflection on this spatial category in postcolonial scholarship; see Imre Szeman, *Zones of Instability: Literature, Postcolonialism, and the Nation* (Baltimore, 2003). Among historians, Peter Sahlins considers zones in relation to frontiers in *Boundaries: The Making of France and Spain in the Pyrenees* (Berkeley, Calif., 1991). We use the term in the sense provided by the *Oxford English Dictionary*: "A definite region or area of the earth, or of any place or space, distinguished from adjacent regions by some special quality or condition (indicated by a defining word or phrase)."

[67] See Georg Simmel, "The Metropolis and Mental Life" (1903), in Gary Bridge and Sophie Watson, eds., *The Blackwell City Reader* (Oxford, 2002), 11–19; Lefebvre, *The Urban Revolution*, trans. Robert Bononno (Minneapolis, 2003); David Harvey, *Social Justice and the City*, revised ed. (Athens, Ga., 2009).

中文译文 | G04-P022-008

66 "区域"这一概念，在空间理论中常被文学学者用于与城市相关（例如，参见大卫·贝尔、乔恩·宾尼、露丝·霍利迪、罗宾·朗赫斯特和罗宾·皮斯所著的《愉悦区域：身体、城市与空间》[纽约州锡拉库扎，2001年]），或用于气候"区域"的概念（参见费莉西蒂·努斯鲍姆所著的《炎热区域：十八世纪英国叙事中的生育、性爱与帝国》[巴尔的摩，1995年]）。后殖民学术领域也有对这一空间范畴的探讨；参见伊姆雷·塞曼所著的《不稳定区域：文学、后殖民主义与国家}[巴尔的摩，2003年]。在历史学家中，彼得·萨林斯在《边界：法国和西班牙在比利牛斯山的形成》[加州伯克利，1991年]中将区域与边界联系起来。我们按照《牛津英语词典》的定义使用这一术语："地球或任何地点、空间的特定区域，因某种特殊性质或条件而与相邻区域区分开来（由定义词或短语表示）"。67 参见格奥尔格·齐美尔在加里·布里奇和索菲·沃森编著的《布莱克威尔城市读本》[牛津，2002年]中的文章《大都市与精神生活》（1903年），第11–19页；罗贝尔·勒菲布尔的《城市革命》，罗伯特·博诺诺译[明尼阿波利斯，2003年]；大卫·哈维所著的《社会正义与城市》，修订版[佐治亚州雅典，2009年]。

G04-P022-009 | 英文原文

中文译文 | G04-P022-009

原PDF第23页（原刊第422页）

G04-P023-002 | 英文原文

around a single capital city. But in these instances, cultural zones still formed primarily around a constellation of cities.

中文译文 | G04-P023-002

围绕单一首都城市。但在这些情况下，文化区域仍主要由一系列城市构成。

G04-P023-003 | 英文原文（原段接续 / 分段）

Our general focus on capitals is not meant to deny the importance of other centers in the circulation or creation of knowledge and goods.[68] For example, Rome's centrality in Grand Tour travels gains meaning only in relation to other Italian destinations—and in any case travelers could not even reach Rome without journeying through other cities and regions. But for definitional purposes, the periphery of a cultural zone is extremely hard to identify spatially (i.e., to map). The edges of a cultural zone tend to be unstable, and can vary depending on the contingencies of war, empire-building, and governmental policy. The Jesuit diaspora was constantly expanding and contracting as new opportunities emerged and others became unstable or vanished entirely. For instance, pockets of a French cultural zone emerged in London and the Dutch Republic after the revocation of the Edict of Nantes in 1685. Cultural zones are rarely demarcated by a wall, a line, or a river. At the edges, they just fade away, as other cultural zones take their place. Cities, by contrast, are much more easily localized. They anchor the cultural zone in cartographic space.

中文译文 | G04-P023-003 | 原文注号：68

我们关注首都的立场并非旨在否定其他中心在知识与商品流通或创造中的重要性。68 例如，罗马在大旅行中的中心性意义仅与其他意大利目的地相关联——无论如何，旅行者若不经过其他城市和地区，根本无法抵达罗马。但从定义的角度来看，文化区的边缘区域在空间上极难确定（即难以绘制地图）。文化区的边界往往不稳定，可能因战争、帝国建设及政府政策的变化而发生变化。耶稣会士的流散群体也不断扩大和缩小，因为新的机会出现，而一些群体则变得不稳定或完全消失。例如，1685年《南特敕令》被废除后，法国文化区在伦敦和荷兰共和国出现了一些区域。文化区很少被墙、线或河流所划分。在边缘处，它们便逐渐消失，被其他文化区取代。相比之下，城市则更容易被定位。它们将文化区固定在地图空间中。

G04-P023-004 | 英文原文（原段接续 / 分段）

At the conceptual level, what makes these zones cultural? Here we can only point to a wide array of features, most of which have, in themselves, little to do with space. A language, a religion, a system of social hierarchy, literary and artistic tastes, fashions, food, constitutional theories, and so forth: all these features, and others, contribute to the production of a cultural zone, and each zone will be more or less determined by different features. That said, these practices and ideas are nonetheless spatial in that they prevail in a certain zone, and beyond that zone, different practices and ideas prevail. Political institutions are obviously important for establishing differences between cultural zones, but they are not sufficient: the American states remained in an English cultural zone well after independence. In other words, a cultural zone and a political sphere are not necessarily the same thing.

中文译文 | G04-P023-004

在概念层面，这些区域为何属于文化区域呢？此处只能指出一系列特征，其中大多数本身与空间并无直接关系。语言、宗教、社会等级制度、文学与艺术风格、时尚、食物、宪法理论等等：所有这些及其他特征共同构成了文化区域，而每个区域的特征也因不同而有所差异。不过，这些实践与观念终究具有空间性，因为它们在某一特定区域内存在，而在该区域之外则存在不同的实践与观念。政治制度显然对确立不同文化区域之间的差异至关重要，但单独依靠它并不足够：美国各州在独立后仍处于英语文化区域之中。换句话说，文化区域与政治领域未必是同一回事。

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G04-P023-006 | 英文原文

THE ARGUMENTS ADVANCED HERE are mainly intended to answer questions about our methodology; they are not meant to serve as a manifesto, and there are of course slight but telling differences in how each of our case studies maps an episode in the Republic of Letters. Since every database is different, and since the needs of different scholarly agendas are best met by different technologies, we do not think it possible or even desirable to dictate a unique methodological approach for all experiments in digital history. One objective of these reflections is to encourage other scholars to venture out into these still largely uncharted waters, using what we have learned along the way as either a guide or a foil.

中文译文 | G04-P023-006

此处提出的论点主要旨在解答关于我们研究方法的疑问；它们并非旨在形成一份宣言，当然，我们的各个案例研究在描述文人共和国中的某一事件时也存在细微但重要的差异。由于每个数据库都不同，且不同学术课题的需求最适合通过不同技术来实现，我们认为不可能甚至不值得为数字历史中的所有实验制定一种统一的研究方法。这些思考的一个目的就是鼓励其他学者探索这些仍未被充分探索的领域，利用我们一路上学到的知识作为指导或参照。

G04-P023-007 | 英文原文（原段接续 / 分段）

But we also have our own pipe dreams about the future of digital history, particularly where intellectual and cultural history are concerned. Given that our own collection of metadata was pooled from different sources, we experienced firsthand the advantages of studying historical networks from multiple angles.[69] This multiplicity can

中文译文 | G04-P023-007 | 原文注号：69

但我们对数字历史未来的前景也抱有自己的幻想，尤其是涉及思想与文化历史方面。由于我们的元数据集合来自不同来源，我们亲身体验了从多个角度研究历史网络的优势。这种多样性可以

G04-P023-008 | 英文原文

[68] See, e.g., Londa Schiebinger, *Plants and Empire: Colonial Bioprospecting in the Atlantic World* (Cambridge, Mass., 2004).

中文译文 | G04-P023-008

68 参见例如 Londa Schiebinger 的《植物与帝国：大西洋世界的殖民生物勘探》（马萨诸塞州剑桥，2004年）。

G04-P023-009 | 英文原文（原段接续 / 分段）

[69] Our largest source of metadata was the EEP; the Humanities Packard Institute gave us the metadata for the Benjamin Franklin correspondence; Stanford Libraries, for the Kircher correspondence; the

中文译文 | G04-P023-009

69 我们最大的元数据来源是EEP；人文帕卡德研究所提供了本杰明·富兰克林书信的元数据；斯坦福图书馆提供了基尔彻书信的元数据；

G04-P023-010 | 英文原文

中文译文 | G04-P023-010

G04-P024-002 | 英文原文

uncover networks that exist at one or two (if not six) degrees of separation. Attempting to make our databases converge has not been an explicit goal of the project, since it is not yet clear that doing so would yield any strong insights. Franklin and Voltaire never corresponded directly (and we know of only one meeting in person), but because we are in possession of the metadata for both their correspondences, we can identify everyone with whom they both corresponded.[70] Kircher and Locke had even less reason to meet, and in fact never did meet, and they certainly never corresponded, so perhaps there is nothing particularly revealing to be gained by exploring the degree to which their correspondence intersects, since they shared so little. Instead, each illuminates a largely distinctive dimension of seventeenth-century intellectual life and its political, religious, and commercial entanglements. By contrast, the Grand Tour Project thrives on identifying points of intersection that emerge as people meet in Rome, Florence, Venice, and Naples, among other places, form temporary communities forged by travel, and potentially renew their acquaintance around an architectural commission that reminds a patron and an architect that they once shared the experience of Italy together. These examples highlight the potential benefits of having overlapping metadata: it allows us to explore possible connections by following the trail.

中文译文 | G04-P024-002 | 原文注号：70

揭示那些通过一或两层（甚至六层）联系而存在的网络。让我们的数据库趋于一致并非该项目的明确目标，因为目前还不清楚这样做能否带来重要的发现。富兰克林和伏尔泰从未直接通信（我们只知道他们有过一次面对面的会面），但由于我们拥有他们的通信元数据，因此可以识别出他们共同通信的每一个人。基尔彻和洛克更没有机会见面，实际上他们从未会面，也肯定没有通信，因此探索他们的通信在何种程度上有交集可能并无特别重要的发现，因为他们交流甚少。相反，每一方都揭示了十七世纪知识生活及其政治、宗教和商业互动中独特的一面。相比之下，“大旅行项目”则专注于识别人们在罗马、佛罗伦萨、威尼斯和那不勒斯等地方相遇后形成的临时社区，这些社区由旅行形成，并可能围绕一项建筑项目重新建立联系，使赞助人和建筑师想起他们曾经一起体验过意大利的生活。这些例子凸显了共享元数据带来的潜在好处：它让我们能够沿着线索探索可能的联系。

G04-P024-003 | 英文原文（原段接续 / 分段）

The metadata in our own databases is heterogeneous and somewhat arbitrary; it reflects our varied interests and the availability of such information, so we have been opportunistic in selecting projects where it was already possible to do this kind of work. Its limitations also showcase, *a contrario*, what a more systematic approach could yield. Imagine a database even larger in scale than the one currently under construction by Oxford's Cultures of Knowledge project—Early Modern Letters Online (EMLO), which contains the entire Kircher database, among many others—in which all metadata relating to scholarly communication could be pooled.[71] Each person would have a unique identifier, connecting all of his or her interactions, and embedded in a much larger pool of information that spanned two hundred, three hundred, or four hundred years of intellectual activity. Add to this the possible bibliographic dimensions, with each author linked to the relevant holdings in major libraries; and throw in other assortments of metadata, such as membership in learned academies, participation in salons or clubs, university education, occupation, travels, and so on. This would indeed be a multifaceted archive for all of us to explore.

中文译文 | G04-P024-003 | 原文注号：71

我们自身数据库中的元数据具有多样性且有些随意；它反映了我们的不同兴趣以及此类信息的可获得性，因此我们在选择项目时总是抓住机会，优先选择那些已经能够开展此类工作的项目。然而，其局限性也恰恰揭示了更系统化的方法能带来什么。想象一个规模比牛津大学“知识文化”项目正在建设的数据库——《早期现代文献在线》（EMLO）——还要大得多的数据库，该数据库包含整个基尔彻数据库以及许多其他数据库——在这个数据库中，所有与学术交流相关的元数据都可以集中起来。每个人都会有一个独特的标识符，连接起其所有的互动记录，并嵌入到一个涵盖两百年、三百年或四百年智力活动的更庞大的信息库中。再加上可能的文献学维度，每位作者都与主要图书馆中的相关藏书相关联；此外还有各种其他元数据，如加入学术学会的情况、参与沙龙或俱乐部的情况、大学教育经历、职业背景、旅行记录等。这确实会成为一个多层面的档案库，供我们所有人探索。

G04-P024-004 | 英文原文（原段接续 / 分段）

No doubt such a database is a fantasy, and we are not suggesting that it could ever exist in complete form, notably for the reasons advanced above. But a partial realization of this encyclopedia of metadata is not beyond our abilities. A pan-European re-

中文译文 | G04-P024-004

毫无疑问，这样的数据库只是幻想而已，我们并不认为它能够以完整形式存在，尤其是基于上述原因。但部分实现这种元数据百科全书并非超出我们的能力范围。泛欧洲范围内的……

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G04-P024-006 | 英文原文

data for the Grand Tour came from Ingamells, *A Dictionary of British and Irish Travellers in Italy*. For an interesting reflection on the advantages of pooling data, see Gingras, "Mapping the Structure of the Intellectual Field Using Citation and Co-Citation Analysis of Correspondences." We also warmly thank Robert Hatch at the University of Florida for generously sharing his experience constructing and working simultaneously with correspondence databases for his research on the relationship between the Republic of Letters and the Scientific Revolution.

中文译文 | G04-P024-006

大旅行相关的数据来自英加梅尔斯所著的《英国与爱尔兰旅者词典》。关于数据整合优势的有趣思考，可参阅金格拉斯的文章《利用引文与共引分析绘制知识领域结构图》。我们还要向佛罗里达大学的罗伯特·哈奇表示衷心的感谢，他慷慨分享了他在构建及维护通信数据库方面的经验，这些工作与他关于文人共和国与科学革命之间关系的研究密切相关。

G04-P024-007 | 英文原文（原段接续 / 分段）

[70] Franklin met Voltaire when the latter was initiated into a Masonic Lodge; see Nicholas Hans, "UNESCO of the Eighteenth Century: *La Loge des Neuf Sœurs* and Its Venerable Master, Benjamin Franklin," *Proceedings of the American Philosophical Society* 97, no. 5 (1953): 513–524. The list of their shared correspondents includes Jean Le Rond d'Alembert, Pierre-Jean-Georges Cabanis, Octavie Durey de Mesnières, Mme Helvétius, David Hume, André Morellet, and Anne-Robert-Jacques Turgot.

中文译文 | G04-P024-007

70. 当伏尔泰加入一个共济会分会时，富兰克林与他相遇了；参见尼古拉斯·汉斯所写《十八世纪的联合国教科文组织：九姐妹分会及其尊贵领袖本杰明·富兰克林》，载于《美国哲学学会会刊》第97卷第5期（1953年）：513–524页。他们共同通信的名单中包括让·勒朗·达兰贝尔、皮埃尔-让-乔治·卡巴尼斯、奥克塔维·杜雷·德·梅斯涅雷、赫尔维蒂乌斯夫人、大卫·休谟、安德烈·莫雷莱以及安妮-罗贝尔-雅克·图尔戈。

G04-P024-008 | 英文原文（原段接续 / 分段）

[71] The EMLO catalogue at http://emlo.bodleian.ox.ac.uk/ is a promising step in this direction.

中文译文 | G04-P024-008

71 位于http://emlo.bodleian.ox.ac.uk/的EMLO目录是朝此方向迈出的一步，具有积极意义。

G04-P024-009 | 英文原文

中文译文 | G04-P024-009

G04-P025-002 | 英文原文

search network, Reassembling the Republic of Letters, has just set out with a similar vision, and in fact, we can draw inspiration from the remarkable work undertaken by classicists to create a “cyberinfrastructure” that enables and encourages interoperability between digital projects.[72] However, we also need tools, developed for and by humanists, to explore and analyze these data, to make them meaningful. And even in this ideal scenario, the time will invariably come when it will be necessary to turn away from the screen, pick up a book, and try to figure out what these piles of historical information mean. That is the exercise that all of our publications seek to do. We have not attempted to map the Republic of Letters without forgetting why we began this project: because the figures who populate our project are interesting for many reasons and left behind a rich trail of documentation whose content we need to read with care if we hope that we will ever be able to understand them.

中文译文 | G04-P025-002 | 原文注号：72

“搜索网络——重新构建文人共和国”——也秉持着类似的愿景开始运作。事实上，我们可以从经典学者所开展的杰出工作中获得灵感，他们创建了“网络基础设施”，从而推动并促进数字项目之间的互操作性。72然而，我们还需要由人文主义者开发的工具来探索和分析这些数据，使其具有实际意义。即便处于这种理想情境下，总会有那么一刻需要离开屏幕，拿起书籍，试图理解这些历史信息背后的含义。正是这种探索，我们的所有出版物也力求实现。我们并未试图绘制文人共和国的地图，却忘记了开展此项目的初衷：因为参与该项目的人员因多种原因而具有吸引力，他们留下的文献资料丰富，如果我们希望最终能够理解他们，就必须谨慎地阅读这些资料。

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G04-P025-004 | 英文原文

[72] See http://www.republicofletters.net/ for more information on the Reassembling the Republic of Letters project, with which Mapping the Republic of Letters is now affiliated, and which counts collaborators from more than thirty European countries, as well as from Canada and Australia. For the "cyberinfrastructure," see *Changing the Center of Gravity: Transforming Classical Studies through Cyberinfrastructure*, ed. Gregory Crane and Melissa Terras, Special Issue, *Digital Humanities Quarterly* 3, no. 1 (2009).

中文译文 | G04-P025-004

72 有关文人共和国重组项目更多信息，请访问 http://www.republicofletters.net/。该项目与《绘制文人共和国地图》项目目前隶属于同一机构，该项目拥有来自三十多个欧洲国家以及加拿大和澳大利亚的合作者。关于"网络基础设施"，请参阅 Gregory Crane 和 Melissa Terras 编著的《改变重心：通过网络基础设施转变古典研究》，《数字人文季刊》第3卷第1期（2009年）。

G04-P025-005 | 英文原文

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中文译文 | G04-P025-005

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G04-P025-006 | 英文原文

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中文译文 | G04-P025-006

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G04-P025-007 | 英文原文

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中文译文 | G04-P025-007

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G04-P025-008 | 英文原文

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中文译文 | G04-P025-008

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G04-P025-009 | 英文原文

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中文译文 | G04-P025-009

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G04-P025-010 | 英文原文

American Historical Review April 2017

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美国历史评论 2017年4月

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AHR Forum

Historical Research in a Digital Age: Reflections from the Mapping the Republic of Letters Project

DAN EDELSTEIN, PAULA FINDLEN, GIOVANNA CESERANI, CAROLINE WINTERER, AND NICOLE COLEMAN

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ABOUT EIGHT YEARS AGO, WE rather serendipitously found ourselves in the early stages of a group research project that seemed to have potential as an experiment with how to do digital history. The result was Mapping the Republic of Letters, a collaborative endeavor based at Stanford University. The project consists of a growing number of wide-ranging case studies, including "British Architects on the Grand Tour in Eighteenth-Century Italy," the subject of the second article in this *AHR* Forum. The case studies engage in multiple ways with the early modern Republic of Letters, and, as each is based on a different source of information, they pose a variety of problems for data-handling and unique challenges for visualizing it. All of them are available on our common website.[1]

At one time, archival material remained more or less in the archives. Historians inclined to use such materials routinely performed acts of intellectual pilgrimage, as many of us still do today, to be initiated into this essential rite of historical apprenticeship. They traveled, learned how to use their archives onsite, and extracted hard-earned information like Forty-Niners panning for gold, returning home with pages of notes, perhaps even a bit of microfilm to read on a machine or print out on that nauseatingly smelly paper whose pungent chemical odor was anything but the nirvana of

Research for this project was initially made possible by a Stanford Presidential Fund for Innovation in the Humanities Grant, and subsequently by funding from the Stanford Humanities Center, the Dean of Research, the Vice Provost of Graduate Education, the Vice Provost of Undergraduate Education, and the Vice Provost of Online Learning. Additional funding came from a National Endowment for the Humanities (NEH) "Digging into Data" Challenge Grant (HJ-50056, 2009–2011), as well as an NEH Digital Humanities Implementation Grant (HK-50087, 2013–2016). Important collaborators include the Electronic Enlightenment Project (Oxford University), the Packard Humanities Institute, DensityDesign Research Lab (Politecnico di Milano), Stanford University Libraries, the Cultures of Knowledge Project (Oxford University), and the Circulation of Knowledge and Learned Practices Project (Huygens Instiuut). The "Mapping the Republic of Letters" Project is part of CESTA, Stanford's Center for Spatial and Textual Analysis. The authors also wish to thank Keith Baker, Giorgio Caviglia, Zephyr Frank, Anthony Grafton, Glauco Mantegari, Robert McNamee, and the wonderful teams of graduate and undergraduate students who have worked with us on this project.

[1] See http://republicofletters.stanford.edu/publications/.

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the past.[2] A much smaller subset of the scholarly community, whose delight in documents long predated the rise of the professional historian of the mid-nineteenth century, ambitiously undertook the laborious project of transcription, creating critical editions of key documents that form the backbone of many histories of famous figures, defining moments, and other noteworthy events.[3] This tiny portion of the infinite documentation of the past found its way into print, while the majority remained in the archive.

In the 1960s, social science historians began to consider how computers facilitated quantitative analysis. An entire generation or two of historians discovered the punch card and began to code. During the 1970s, a steady stream of scholarship emerged from these early experiments with historical datasets. This approach appealed especially to historians who wished to migrate away from traditional political and intellectual history, writing the kind of historical sociology and family, labor, and economic history whose grandest manifestos might be Emmanuel Le Roy Ladurie's 1974 essay "History That Stands Still," which summarized well the goals and ambitions of the early Annales School, and historical sociologist Charles Tilly's evocatively titled *Big Structures, Large Processes, Huge Comparisons* a decade later.[4] The foundations of the new social history of the 1960s through the 1980s rest upon these impressive early experiments with big historical data and its insistence on the value of cliometrics.

This was the old era of big history with big data. We are now in the early stages of a new age of historical data that offers new possibilities as well as challenges. The past fifteen years have seen a gradual but perceptible shift in where the archive is located as an ever greater portion of the historical record migrates piecemeal from boxes and books to PDFs and JPGs, and a vast sea of digitized text. A growing number of manuscripts can be viewed online, and hefty multivolume document collections are now digitized, indexed, and easily searchable. Historians increasingly find themselves utilizing digital databases as the idea of the searchable document and the virtual archive reorganize how libraries, research institutes, teams of scholars, and even individual researchers present and share interesting sources. To take one example, bibliometrics once belonged primarily to librarians and archivists and now seems to interest a broad, interdisciplinary array of scholars as an analytic research tool. For all these reasons, it is not surprising that the history of information has become an important topic.[5] To paraphrase Raymond Williams, "information" is one of the keywords of the

[2] For an enjoyable meditation on the origins of these historical practices, see Carolyn Steedman, *Dust: The Archive and Cultural History* (New Brunswick, N.J., 2002). The early modern archive in particular is beautifully evoked in Arlette Farge, *The Allure of the Archives*, trans. Thomas Scott-Railton (New Haven, Conn., 2013).

[3] On this subject, a classic point of departure is Anthony Grafton, *The Footnote: A Curious History* (Cambridge, Mass., 1997).

[4] Emmanuel Le Roy Ladurie, "History That Stands Still," in Ladurie, *The Mind and Method of the Historian*, trans. Siân Reynolds and Ben Reynolds (Chicago, 1981), 1–27; Charles Tilly, *Big Structures, Large Processes, Huge Comparisons* (New York, 1984). The most recent overview of the Annales School is André Burguière, *The Annales School: An Intellectual History*, trans. Jane Marie Todd (Ithaca, N.Y., 2009). For early methodological reflections on quantitative history, see William O. Aydelotte, "Quantification in History," *American Historical Review* 71, no. 3 (April 1966): 803–825; William O. Aydelotte, Robert William Fogel, and Allan G. Bogue, eds., *The Dimensions of Quantitative Research in History* (Princeton, N.J., 1972).

[5] See, for instance, Paul N. Edwards, Lisa Gitelman, Gabrielle Hecht, Adrian Johns, Brian Larkin, and Neil Safier, "Historical Perspectives on the Circulation of Information," *AHR* Conversation, *American Historical Review* 116, no. 5 (December 2011): 1393–1435.

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early twenty-first century, and so too is "data."[6] The question, however, is not how to write the history of information but what to do with all of it, as historians in an era when we have a burgeoning array of techniques and technologies available to help us scrape, mine, curate, analyze, and visualize data.

Too much information, it turns out, might be a good thing. Perhaps the first-year graduate student, overwhelmed by shelves full of books in his or her field, may not immediately agree. "Information overload" is a common reaction—now as in the past—to the vast collections of knowledge that have accumulated in our libraries, and are increasingly migrating online.[7] But sometimes, to paraphrase Jean-Jacques Rousseau, the cure lies in the poison.[8] "Big data" is the name commonly given to the exponential increase in information, whether it results from the digitization of analog data stores or is collected in real time from remote sensors, the web, or mobile devices.[9] The difference between these datasets and that multitude of books is the potential for discovery and analysis. "Big data" refers not to size alone, as the term suggests, but to the interconnectedness of resources. For our purposes, big data is what you get when those shelves of books have been digitized, structured, and interlinked in such a way that the information contained within them can be filtered, plotted, measured, parsed, and visualized. The scholar starting out in a new field still has to ingest and process the literature the old-fashioned way—by reading it. But we can put computational tools to work to help with the ingesting and processing and to inspire other questions we might ask with this material as we look across, as well as within, our sources.

Few humanists would argue that reading books and manuscripts is a bad thing. Indeed, the newfound availability of large datasets in the humanities at times conjures up the worrisome specter of digitally savvy sorcerer's apprentices who, thanks to new databases, "cite anything and construe nothing."[10] Searchable, sortable data can make even beginning scholars look erudite and clever. An eager researcher can accumulate

[6] Raymond Williams, *Keywords: A Vocabulary of Culture and Society* (New York, 1976). On the history of information, see especially C. A. Bayly, *Empire and Information: Intelligence Gathering and Social Communication in India, 1780–1870* (Cambridge, 1996); Daniel R. Headrick, *When Information Came of Age: Technologies of Knowledge in the Age of Reason and Revolution, 1700–1850* (Oxford, 2001); and Jacob Soll, *The Information Master: Jean-Baptiste Colbert's Secret State Intelligence System* (Ann Arbor, Mich., 2009). Daniel Rosenberg has a project underway on the history of data, with related work published as "Data before the Fact," in Lisa Gitelman, ed., *"Raw Data" Is an Oxymoron* (Cambridge, Mass., 2013), 15–40; "Enter Data," in Katherine Behar, ed., *Data's Entry* (Istanbul, 2016), 14–33; and "An Archive of Words," in Lorraine Daston, ed., *Sciences of the Archive* (Chicago, forthcoming 2017).

[7] Daniel Rosenberg, "Early Modern Information Overload," *Journal of the History of Ideas* 64, no. 1 (2003): 1–9; Ann M. Blair, *Too Much To Know: Managing Scholarly Information before the Modern Age* (New Haven, Conn., 2010). More generally, see Peter Burke, *A Social History of Knowledge: From Gutenberg to Diderot* (Cambridge, 2000).

[8] Jean Starobinski, *Le remède dans le mal: Critique et légitimation de l'artifice à l'âge des Lumières* (Paris, 1989).

[9] See Danah Boyd and Kate Crawford, "Six Provocations for Big Data," paper presented at "A Decade in Internet Time: Symposium on the Dynamics of the Internet and Society," University of Oxford, September 21, 2011, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1926431; Viktor Mayer-Schönberger and Kenneth Cukier, *Big Data: A Revolution That Will Transform How We Live, Work, and Think* (Boston, 2013). For a humanistic take on this debate, see Gitelman, *"Raw Data" Is an Oxymoron*; Lev Manovich, "Trending: The Promises and the Challenges of Big Social Data," in Matthew K. Gold, ed., *Debates in the Digital Humanities* (Minneapolis, 2012), 460–475; Christof Schöch, "Big? Smart? Clean? Messy? Data in the Humanities," *Journal of Digital Humanities* 2, no. 3 (2013), http://journalofdigitalhumanities.org/2–3/big-smart-clean-messy-data-in-the-humanities/.

[10] Jonathan Barnes, quoted by Anthony Grafton in *Worlds Made by Words: Scholarship and Community in the Modern West* (Cambridge, Mass., 2009), 322.

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historical mass very quickly. Superficial engagement with this material can also lead to gross misinterpretations when the context for interpreting data is missing and when one has read only what is available online. Stopping there, however, would be premature, indeed shortsighted. These perils notwithstanding, the digitization and datafication of historical documents can open up promising methodological avenues and opportunities for new insights.

We began our project, Mapping the Republic of Letters, with these opportunities in mind and with an awareness of the growing number of projects that are digitizing letters and other key documents of scholarly, cultural, political, and religious life from the sixteenth through eighteenth centuries. Such projects generate tremendous amounts of valuable metadata about social networks, travel, institutional affiliations, publication history, and so forth. In the age of Edward Snowden, most people are now familiar with the concept of metadata, such as information about who communicated (or traveled) with whom, when, and where. Typically, metadata is everything except the "what" that was communicated. Mapping the Republic of Letters deals primarily with the metadata about letters or travels rather than with their actual content. Or rather, we use the metadata to produce maps, charts, and other data visualizations, and then refer back to the content to elucidate or complement what we see. Similarly (yes) to the NSA, we engage in "traffic analysis" to explore the breadth, shape, and hubs of intellectual networks.[11]

Since we began our project, the increase in available online data has been accompanied by a growing conversation about its impact and meaning for historical research. In the last couple of years alone, a string of new handbooks and manifestos have tackled "big data," laying out its promises for historians while introducing the various methodologies by which digital historians approach it.[12] In the pages of the *AHR*, Lara Putnam has recently shown how digital search—the now-widespread scholarly use of Google, JSTOR, and WorldCat, among others—is bringing many of the issues associated with big-data historical research to the doorsteps even of historians who have not explicitly adopted digital tools meant to quantify and visualize data.[13] Our reflections here, particularly about our experience with digital historical databases and how this has affected our work in early modern intellectual history, aim to contribute to this conversation. The publications emerging from our project all employ computational technologies and visualization techniques to make discoveries

[11] For the U.S. government's practice of traffic analysis, see Vera R. Filby, *A Collection of Writings on Traffic Analysis* (Fort Meade, Md., 1993), http://www.governmentattic.org/8docs/NSA-TrafficAnalysis Monograph_1993.pdf.

[12] See Jo Guldi and David Armitage, *The History Manifesto* (Cambridge, 2014), chap. 4, for their take on big data's promise for historians. Among the recent proliferation of companions and handbooks for digital humanities, and digital history in particular, note Shawn Graham, Ian Milligan, and Scott Weingart, *Exploring Big Historical Data: The Historian's Macroscope* (London, 2016), which most explicitly addresses digital history as big data history. For a methodological reflection on big data history, see also Cameron Blevins, "The Perpetual Sunrise of Methodology," paper prepared for AHA Session 158, "Authoring Digital Scholarship for History: Challenges and Opportunities," 129th Annual Meeting of the American Historical Association, New York City, January 4, 2015, http://www.cameronblevins.org/posts/perpetual-sunrise-methodology/.

[13] Lara Putnam, "The Transnational and the Text-Searchable: Digitized Sources and the Shadows They Cast," *American Historical Review* 121, no. 2 (April 2016): 377–402, with rich references on the recent conversation on history in the digital age. See also Ted Underwood, "Theorizing Research Practices We Forgot to Theorize Twenty Years Ago," *Representations* 127, no. 1 (2014): 64–72.

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about the past that would have been difficult, if not impossible, to reach by analog means. These scholarly outcomes have emerged in—indeed, could not exist outside of—a dialogue with a longstanding tradition of early modern intellectual historiography. They also grew out of research practices that our group established along the way, such as digital visualizations, tool development, publication format, and co-authored scholarship, which speak to how history might be done in a digital age.

Mapping the Republic of Letters is a research project with many heads. It contains two seventeenth-century (Athanasius Kircher and John Locke) and three eighteenth-century (Voltaire, Benjamin Franklin, and the Grand Tour) case studies.[14] Their geography primarily traverses Great Britain, Europe, and British North America, though each intellectual network makes strategic excursions that lie beyond these boundaries, raising questions about not only the nature of early modern cosmopolitanism and travel, but also the nature of global connections and communications. Historians of the Republic of Letters have also long debated how it evolved over time. Did the erudite *Respublica litteraria* of humanist scholars differ significantly from the worldly *République des lettres*? What can we learn by examining not only the most prolific correspondents who represent the strongest ties in a given network, but also the "long tail" of letter writers, often the vast majority in most instances, who sent no more than a single letter or a handful of letters? As David Lux and Harold Cook observe in an important essay inspired by the work of sociologist Mark Granovetter, understanding how people expand the reach of their network by "widening the network of weak ties" is an essential component of network analysis.[15]

One of our groups has focused on the German Jesuit polymath Athanasius Kircher (1602–1680). Paula Findlen, Suzanne Sutherland, and Iva Lelková have explored why Kircher's mid-seventeenth-century network was more cosmopolitan, in terms of geographic breadth, than those of his Enlightenment successors.[16] As a member of the Society of Jesus, Kircher could plug into an already established global missionary network while also making himself a useful participant in the scholarly Republic of Letters, and inserting himself successfully in Habsburg networks in Central Europe and Italian networks emanating to and from his adoptive city, Rome. (See Figure 1.)[17] A fine-grained analysis of the community that corresponded with him shows how his publishing projects grew and evolved with the expanding reach of the Jesuit missionary network into many different parts of the world. This research has been able to map the multiple ways in which Kircher self-consciously shaped and reshaped his religious, political, and intellectual networks to bolster his reputation as the man to know in the Eternal City, someone whose books and reputation circum-

[14] Interactive visualizations, datasets, and links to research articles related to these case studies can be found at http://republicofletters.stanford.edu/publications/.

[15] David S. Lux and Harold J. Cook, "Closed Circles or Open Networks? Communicating at a Distance during the Scientific Revolution," *History of Science* 36, no. 2 (1998): 179–211, here 190; Mark S. Granovetter, "The Strength of Weak Ties," *American Journal of Sociology* 78, no. 6 (1973): 1360–1380. For a recent study that uses network analysis to explore the correspondence of imprisoned Protestants during the English wars of religion, see Ruth Ahnert and Sebastian E. Ahnert, "Protestant Letter Networks in the Reign of Mary I: A Quantitative Approach," *ELH* 82, no. 1 (2015): 1–33.

[16] See http://republicofletters.stanford.edu/publications/kircher/. The result of this research will be published in a series of articles by Paula Findlen, Suzanne Sutherland, and Iva Lelková, culminating in a book provisionally titled *The Baroque Postmaster: Athanasius Kircher between Rome and the World*.

[17] Those who are reading this article in the print issue can view this and the other figures in color either in the online issue or at http://republicofletters.stanford.edu/publications/.

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FIGURE 1: Source locations for correspondence to Athanasius Kircher, sized by numbers of letters sent.

navigated the globe. While his network was less socially diverse than those of, say, Locke, Franklin, and Voltaire, who corresponded with broader swaths of society (and most notably, by the eighteenth century, with far more women), his letters traveled much farther.

Skeptics have challenged the rhetoric of the "republicans of letters," to invoke Daniel Roche's phrase, and questioned whether there really was *a* republic, or whether this illusory place masked a balkanized hodgepodge of more or less connected mini-republics.[18] In the context of Mapping the Republic of Letters, Claude Willan's work on John Locke (1632–1704) has used graphs and other visualizations to reveal just how disconnected the different social and national subgroups in his correspondence network were from one another.[19] (See Figure 2.) The republic that emerges from his study is not a unified network that any newcomer could join wholesale, but a patchwork of isolated communities that Locke fused together. His findings provide an empirical demonstration of the scholarly suspicion that there were really "a multitude of communities within, or rather underneath, the surface of the Republic of Letters."[20]

Mapping the correspondence networks of major Enlightenment figures underscores the limits of their geographic horizons, raising questions about what it meant to be a self-professed "citizen of the world," as many enlightened men and women of

[18] Daniel Roche, *Les républicains des lettres: Gens de culture et Lumières au XVIIIe siècle* (Paris, 1998). On the national vs. cosmopolitan makeup of the Republic of Letters, see especially Lorraine Daston, "The Ideal and Reality of the Republic of Letters in the Enlightenment," *Science in Context* 4, no. 2 (1991): 367–386.

[19] See http://republicofletters.stanford.edu/publications/locke/, and Willan's article "'John Locke Likes This': An Ego-Network Analysis of Locke's Letters" (under review).

[20] See Candice Delisle, "Accessing Nature, Circulating Knowledge: Conrad Gessner's Correspondence Networks and His Medical and Naturalist Practices," *History of Universities* 23, no. 2 (2008): 35–58, here 53.

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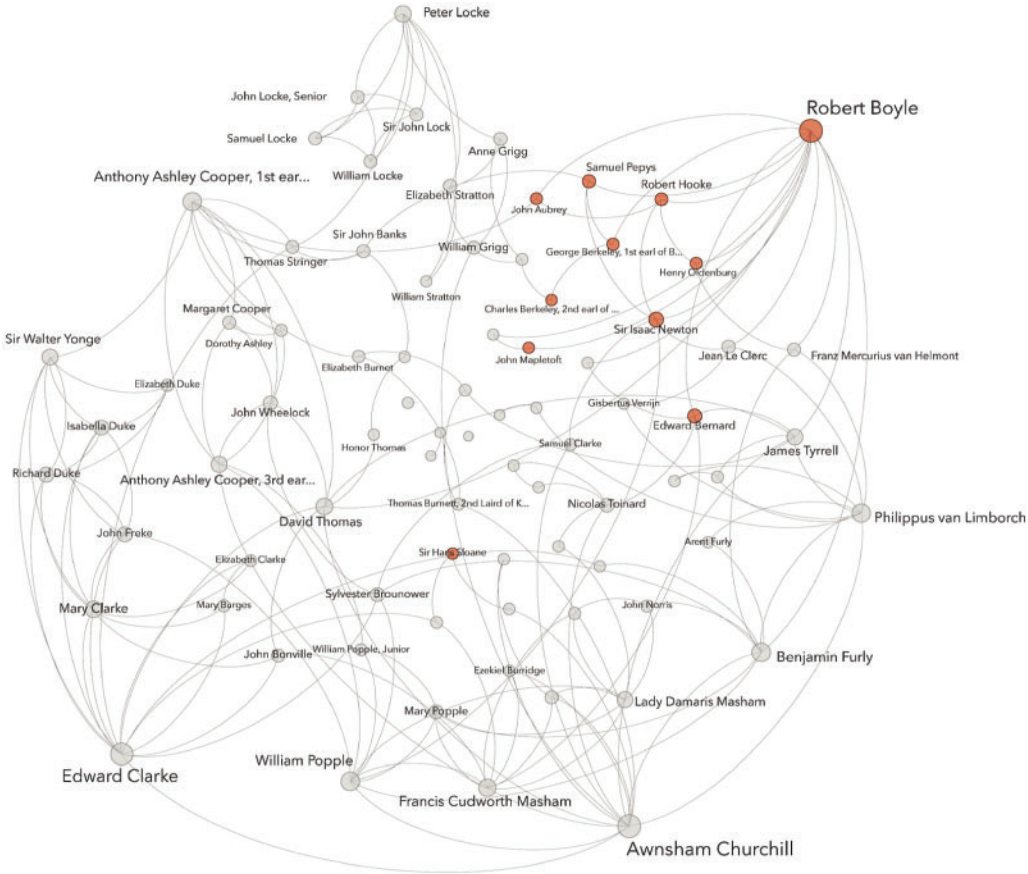


FIGURE 2: Network of Royal Society fellows (orange nodes) in John Locke's correspondence.

letters claimed to be, and in what ways overseas colonies mattered in the increasingly far-flung imperial reach of the Europe-centered Republic of Letters.[21] Caroline Winterer and Claire Arcenas—another of our research teams—have explored these two questions in the network of a figure long renowned for his cosmopolitanism and colonial status, Benjamin Franklin (1706–1790).[22] Franklin may well have been the most famous North American of the eighteenth century, but his transatlantic letters initially did not include a significant number of international correspondents; only over decades spent in the imperial capital of London did he forge a letter network that included French, German, Italian, and Scottish correspondents. (See Figure 3.) Even at the close of his "London Decades" (1757–1775), and despite trips to Scotland, the Netherlands, and France, Franklin's correspondents were overwhelmingly found at ei-

[21] Margaret C. Jacob, *Strangers Nowhere in the World: The Rise of Cosmopolitanism in Early Modern Europe* (Philadelphia, 2006); Caroline Winterer, "Where Is America in the Republic of Letters?," *Modern Intellectual History* 9, no. 3 (2012): 597–623; Winterer, *American Enlightenments: Pursuing Happiness in the Age of Reason* (New Haven, Conn., 2016).

[22] See http://republicofletters.stanford.edu/publications/franklin/; and Winterer and Arcenas's article "The Correspondence Network of Benjamin Franklin: The London Decades, 1757–1775" (under review).

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FIGURE 3: Benjamin Franklin's total correspondence, with locations sized by volume of letters.

ther end of the London-Philadelphia axis. His fame may have spread across Europe, but his letters circulated chiefly in the British imperial world.

Our visualizations also provide new perspectives on murky problems of cultural transmission. As our data does not include letter content, it can tell us only so much about the circulation of ideas. But correspondence maps nonetheless raise questions that invite us to read letters and other documents with a fresh perspective. Starting with the observation that Voltaire (1694–1778) seems to have exchanged few letters with British correspondents, Dan Edelstein and Biliana Kassabova have argued in their work for Mapping the Republic of Letters that the standard narrative about the English origins of Voltaire's philosophy (and by extension of the French Enlightenment) must be revised in light of the fact that Voltaire's interest in, and admiration of, England was both qualified and chronologically restricted.[23] (See Figure 4.) England's period of cultural greatness, in his eyes, lay in the reign of Charles II, and was thus a thing of the past by the time he arrived in London. What is more, Voltaire credited the *siècle de Louis le grand* for many of the English intellectual exploits. The author of the bestselling and controversial *Letters Concerning the English Nation* had more than a few concerns, it turns out, about that nation.

As the above case studies indicate, travel—whether Kircher's religious flight from war-torn Germany, Locke's exile from England, Franklin's transatlantic voyages, or Voltaire's timely crossing of the English Channel to mitigate scandal—played a crucial role in how many citizens of the Republic of Letters extended and built their networks. One of our groups, as detailed in the second article in this *AHR* Forum, has

[23] See http://republicofletters.stanford.edu/publications/voltaire/; and Edelstein and Kassabova's article "Where Are Voltaire's Letters Concerning the English Nation? Maps, Networks, and Intellectual History" (under review).

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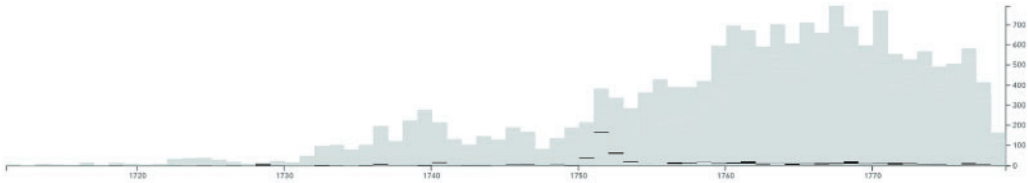


FIGURE 4: Histogram of Voltaire's correspondence, by year, highlighting English correspondence.

been mapping Grand Tour travel, and the role it played in the emergence of architecture as a profession. The research process involved transforming John Ingamells's *A Dictionary of British and Irish Travellers in Italy, 1701–1800* into an interactive database filled with detailed information about the intersecting itineraries of the men and women who traveled to Italy, and creating a prosopography of this community.[24] (See Figure 5.) By studying British architects on the Grand Tour, Giovanna Ceserani, Giorgio Caviglia, Nicole Coleman, Thea De Armond, Sarah Murray, and Molly Taylor-Poleskey illuminate how the dynamic exploration of early modern travelers allows us to identify with greater precision the role of travel in relation to professional aspirations and cultural formation.[25] Architects who completed their education in Italy often met prospective clients on the Grand Tour, and presented their studies abroad as a compelling professional credential when they returned home.

When we have presented our work, a common complaint has been that for all our investment in digital tools, data editing, and data creation, the field has not been revolutionized as a result. This criticism can be voiced with optimism ("Show us you're the greatest thing since sliced bread," in the words of one enthusiastic reviewer) or with skepticism ("So you have nothing more to show for yourselves?"). But why *should* new methods produce radically different results? We are studying the same objects that scholars have been painstakingly exploring for hundreds of years. They naturally came up with penetrating insights about the Republic of Letters without the help of digital tools. But what these tools allow us to do is to sort through, refine, confirm, or refute the different insights these historians had, as well as to formulate new ones that emerge when we step back from the details of any particular subset of a correspondence to observe its contours. This is not to embrace a positivist, cliometric vision of computational supremacy; such a vision is ill-suited for historical archives that are so shot through with uncertainty and gaps. But where possible, basic calculations are still useful and can provide correctives to sheer speculation. They also force us to look closely at the information we have, resulting in numerous silent corrections to minor errors, assumptions that have solidified into facts, and other problems that arise when we take information for granted.

The data visualizations we rely on in our research and publications do not offer clear snapshots of the past, but rather fuzzy, blurry pictures. They reveal the general shape of things, orders of magnitude, and large-scale trends; they also draw our atten-

[24] John Ingamells, comp., *A Dictionary of British and Irish Travellers in Italy, 1701–1800* (New Haven, Conn., 1997). For more on this project, see https://grandtour.stanford.edu/.

[25] See http://republicofletters.stanford.edu/publications/grandtour/; and Giovanna Ceserani, Giorgio Caviglia, Nicole Coleman, Thea De Armond, Sarah Murray, and Molly Taylor-Poleskey, "British Travelers in Eighteenth-Century Italy: The Grand Tour and the Profession of Architecture," this issue.

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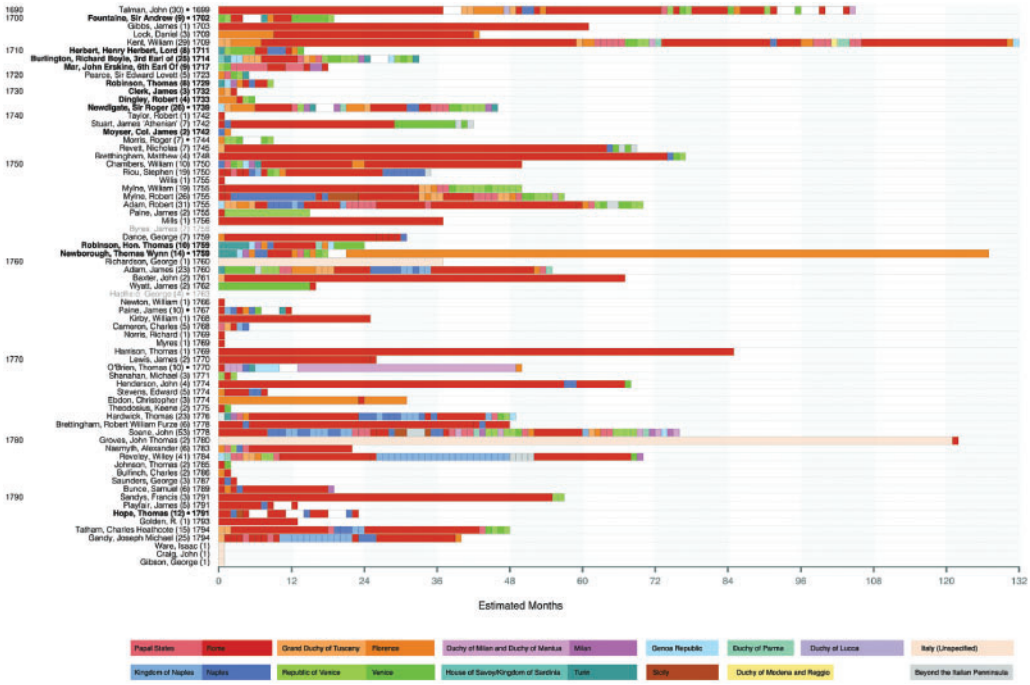


FIGURE 5: Timechart of British Grand Tour travelers in Italy in the eighteenth century.

tion to microhistories that we might otherwise have missed. The results are never definitive, and always require further verification—an absence on a map might be due only to a gap in the data. But the critical point is that they open up new avenues of inquiry and are springboards for further research.

In some instances, the computational processes in question are not complex: one could count Kircher's global missionary correspondence, Voltaire's letters to England, or Franklin's international correspondents by hand. But computational methods are good for more than just performing difficult statistical operations (though one of our projects, on John Locke, does exactly that): they can also serve to reveal elements of surprise in the data. The point is not, therefore, whether you need computational tools to count the number of Franklin's French (or Voltaire's English, or Kircher's Jesuit) correspondents, but rather that computational methods provide novel insights into your sources. We designed and developed data visualizations not to do our work for us, but rather to point to where our work lies.

WHILE OUR MAPPING PROJECT came of age at a time when historians were just starting to focus on the promises and perils of big data, it also draws on a long historiography. Indeed, the Republic of Letters has never been a small or easily contained subject. Starting in the mid-fourteenth century, early humanists such as Petrarch wrote open letters as a means of communicating ideas and shaping opinion, and also as a process of intellectual self-definition. In 1417 the Venetian humanist Francesco Barbaro provided an indelible portrait of the nature of this emerging scholarly community when

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he congratulated his Florentine friend Poggio Bracciolini for unearthing a treasure trove of ancient manuscripts in the monastic libraries of Northern Europe, acknowledging the utility of Poggio's rediscovery for "this Republic of letters" (*huic litteraria Reipublicae*).[26] It is hardly coincidental that the phrase *Respublica litteraria* first appeared in a letter. In the age of great letter writers such as the early-sixteenth-century Dutch humanist Desiderius Erasmus, who, to paraphrase Lisa Jardine, delighted in making absent friends present, the idea of conversation through correspondence became the norm rather than the exception.[27]

When Elizabeth Eisenstein, in her landmark study *The Printing Press as an Agent of Change* (1979), proclaimed printing to have been a transformative development in early modern Europe, she contrasted the limits of letters to the possibilities of books in disseminating knowledge while observing that we need to know a great deal more about the Republic of Letters.[28] More recent scholarship has reexamined this issue and arrived at a different conclusion, one that emphasizes the productive and continuous interactions between writing and publishing. A world of books and journals was also a world of readers, authors, and critics, whose conversations took place not only in academies, salons, and other sites of scholarly sociability, but also through long-distance travel and communication via letters. Such letters provide us with a historical roadmap from which we can reconstruct intellectual networks across time and space.

In the 1970s, a number of scholars began to investigate the nature and meaning of the Republic of Letters. An international community of scholars has developed and defined this subject in the decades since, tracing the rise and decline of a rhetorical ideal and its connection to cultural and intellectual practices and scholarly communities.[29] But rhetoric, as Lorraine Daston cogently observed in an important essay, ulti-

[26] The standard English translation of this letter, dated July 6, 1417, can be found in Phyllis Walter Goodhart Gordan, ed. and trans., *Two Renaissance Book Hunters: The Letters of Poggius Bracciolini to Nicolaus de Niccolis* (New York, 1974), 196–203.

[27] For an overview of these developments, see Marc Fumaroli, "The Republic of Letters," trans. R. Scott Walker, *Diogenes* 36, no. 143 (1988): 129–152; Hans Bots and Françoise Waquet, *La République des lettres* (Paris, 1997); Anthony Grafton, "A Sketch Map of a Lost Continent: The Republic of Letters," *Republics of Letters: A Journal for the Study of Knowledge, Politics, and the Arts* 1, no. 1 (2009), http://arcade.stanford.edu/rofl/sketch-map-lost-continent-republic-letters. On Erasmus, a good starting point is Lisa Jardine, *Erasmus, Man of Letters: The Construction of Charisma in Print* (Princeton, N.J., 1993), chap. 6; Constance M. Furey, *Erasmus, Contarini, and the Religious Republic of Letters* (Cambridge, 2008); Hanan Yoran, *Between Utopia and Dystopia: Erasmus, Thomas More, and the Humanist Republic of Letters* (Lanham, Md., 2010).

[28] Elizabeth Eisenstein, *The Printing Press as an Agent of Change: Communications and Cultural Transformations in Early Modern Europe*, 2 vols. (Cambridge, 1979), 1: 136–137. The relationship of Eisenstein's work to the Republic of Letters is discussed by J. L. Pearl, "The Role of Personal Correspondence in the Exchange of Scientific Information in Early Modern France," *Renaissance and Reformation / Renaissance et Réforme*, New Series / Nouvelle Série, 8, no. 2 (1984): 106–113, here 107; and April G. Shelford, *Transforming the Republic of Letters: Pierre-Daniel Huet and European Intellectual Life, 1650–1720* (Rochester, N.Y., 2007), 3.

[29] The earliest historical study of this concept may well be Annie Barnes, *Jean Le Clerc (1657–1736) et la République des lettres* (Paris, 1938). Readers interested in following the development of this subject since the 1970s should begin with Paul Dibon, "Les échanges épistolaires dans l'Europe savante du XVIIe siècle," *Revue de synthèse* 81–82 (1976): 31–50; Dibon, "Communication in the Respublica litteraria of the 17th Century," *Respublica Litteraria: Studies in the Classical Tradition* 1 (1978): 43–55; Hans Bots, *Republiek der letteren: Ideaal en werkelijkheid: Rede uitgesproken bij aanvaarding van het ambt van buitengewoon hoogleraar aan de Katholieke universiteit te Nijmegen* (Amsterdam, 1977); Françoise Waquet, "Qu'est-ce que c'est la République des lettres? Essai de sémantique historique," *Bibliothèque de l'École des Chartres* 147 (1989): 473–502. The fundamental work of synthesis on this subject is Bots and Waquet, *La République des lettres*.

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mately inspired questions about the reality, or lived experience, of the ideal of the Republic of Letters.[30] Taking the case of Leibniz, Maarten Ultee envisioned a social history that would explore the concrete details of intellectual membership in this imagined community, including its geography, the volume and frequency of letters, and the social positions of its participants.[31] Such questions inspired the subsequent work of a talented cadre of scholars, most notably Dena Goodman, Anne Goldgar, and Daniel Roche, who increasingly brought the methods of cultural history and gender history to bear on the subject in the late 1980s and early 1990s.[32] Simultaneously, a rejuvenated intellectual history developed by scholars such as Joseph Levine, Anthony Grafton, Peter Miller, and Laurence Brockliss focused on key participants in the Republic of Letters as a means of better understanding the practices of intellectual life from the advent of Renaissance humanism to the rise of antiquarianism and ultimately the Age of Enlightenment.[33]

Initially there were a limited number of letters, but soon the volume of letters that individuals exchanged expanded astronomically. When Jardine wondered aloud what scholars should do with the "mass of 'data' which Erasmus's voluminous correspondence provides," she asked a pertinent question: 3,162 letters to and from Erasmus survive. We can juxtapose this number to the prolific letter-writing habits of Italian Renaissance humanists, but also of learned Protestant reformers: Martin Luther's surviving correspondence numbers 4,337 letters, and John Calvin's 4,271. Ignatius Loyola, the founder of the Society of Jesus, was a veritable epistolary machine who practiced what he preached in institutionalizing letter-writing as an essential expression of religious confraternity and bureaucracy; he left behind an impressive 9,178 letters (6,381 written by him) at the time of his death in 1556.[34] While not quantitatively large by contemporary standards of big data, such numbers are filled with hidden multipliers, because each letter typically contains a great deal of valuable information—about people, places, conversations, manuscripts, publications, projects, and institutions—and offers a trail of references that invariably leads the patient researcher to

[30] Daston, "The Ideal and Reality of the Republic of Letters in the Enlightenment."

[31] Maarten Ultee, "The Republic of Letters: Learned Correspondence, 1680–1720," *Seventeenth Century* 2, no. 1 (1987): 95–112.

[32] Dena Goodman, *The Republic of Letters: A Cultural History of the French Enlightenment* (Ithaca, N.Y., 1994); Anne Goldgar, *Impolite Learning: Conduct and Community in the Republic of Letters, 1680–1750* (New Haven, Conn., 1995); Roche, *Les républicains des lettres*. For subsequent efforts to build on Goodman's early insights on the importance of gender in the Republic of Letters, a good starting point is Susan Dalton, *Engendering the Republic of Letters: Reconnecting Public and Private Spheres in Eighteenth-Century Europe* (Montreal, 2003); Antoine Lilti, *Le monde des salons: Sociabilité et mondanité à Paris au XVIIIe siècle* (Paris, 2005); Dena Goodman, *Becoming a Woman in the Age of Letters* (Ithaca, N.Y., 2009).

[33] Joseph M. Levine, *The Battle of the Books: History and Literature in the Augustan Age* (Ithaca, N.Y., 1994); Peter N. Miller, *Peiresc's Europe: Learning and Virtue in the Seventeenth Century* (New Haven, Conn., 2000); L. W. B. Brockliss, *Calvet's Web: Enlightenment and the Republic of Letters in Eighteenth-Century France* (Oxford, 2002). From Grafton's considerable corpus, one might follow the trail backward from his AHA Presidential Address, "The Republic of Letters in the American Colonies: Francis Daniel Pastorius Makes a Notebook," *American Historical Review* 117, no. 1 (February 2012): 1–39; and "A Sketch Map of a Lost Continent."

[34] Jardine, *Erasmus, Man of Letters*, 149. For a good overview of some of the metrics of early modern correspondence, see Francisco Bethencourt and Florike Egmond, eds., *Cultural Exchange in Early Modern Europe*, vol. 3: *Correspondence and Cultural Exchange in Europe, 1400–1700* (Cambridge, 2007), 10–12. For an introduction to religious letter-writing and Loyola's reputation as the most prolific letter writer of the sixteenth century, see Thomas M. Lucas, *Landmarking: City, Church and Jesuit Urban Strategy* (Chicago, 1997), 131.

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other terrifyingly large masses of correspondence. This is far too much to absorb at a glance or to keep in one's head, which is why scholars have often worked with discrete portions of large correspondences to get to know specific content really well rather than attempting to consider the whole.

Such numbers pale in relation to the scale of mercantile correspondence, as the case of a late-fourteenth-century merchant in Prato, Francesco Datini, illustrates all too well: his surviving papers contain approximately 150,000 commercial and familial letters. While Petrarch carefully and deliberately identified the living and the dead, the real and the imaginary, to whom he might address a well-crafted Latin letter in imitation of Cicero, Quintilian, and other great Roman epistolarians, the demands of the marketplace required a constant stream of long-distance correspondence traveling in all directions to many different kinds of people who did business together.[35] The same can also be said of diplomatic correspondence. Take the example of Isabella d'Este, the fifteenth-century marchioness of Mantua, for whom some 25,000 letters (approximately 16,000 by her) survive in an archive containing about 100,000 letters related to the totality of her activities as one of the most powerful political and cultural brokers in Renaissance Italy.[36] Such examples serve as a stark reminder of just how fundamental and ubiquitous letter-writing was to a variety of human endeavors in the premodern world. While our project has focused on the intellectual and cultural dimensions of that particular subject, we hope that the techniques we have explored and experimented with may prove useful for historians with other interests.

In the early sixteenth century, Erasmus wistfully envisioned a Christian Republic of Letters populated by scholars in communication across geographic, religious, and political boundaries. Increasingly, however, humanistic ideals of writing to someone as an expression of learned friendship intertwined with other priorities, as different cultures of letter-writing intersected and combined. Diplomats and agents made correspondence an instrument of politics, policy, and information; adventurers, missionaries, traders, and emigrants boarded mercantile vessels to parts unknown and sent letters home, waiting months, sometimes years, to receive responses from friends, family, and superiors back in Europe. Everywhere people went, letters accompanied them. Publishing letters became a lucrative editorial activity. Our project begins with this fundamental fact of the early modern world: it was a society whose patterns of increased mobility and communication were expressed in the numerical explosion of letters.

Growing curiosity about experiences, ideas, languages, and artifacts that could be acquired only through long-distance travel multiplied the number of networks and communities of scholars in pursuit of knowledge. The superabundance of information

[35] The most recent description of the Datini correspondence by the Istituto Datini can be found at http://www.istitutodatini.it/schede/archivio/eng/arc-dat2.htm. On mercantile correspondence, see especially Francesca Trivellato, *The Familiarity of Strangers: The Sephardic Diaspora, Livorno, and Cross-Cultural Trade in the Early Modern Period* (New Haven, Conn., 2009); Trivellato, "Merchants' Letters across Geographical and Social Boundaries," in Bethencourt and Egmond, *Correspondence and Cultural Exchange in Europe*, 80–103; Trivellato, "A Republic of Merchants?," in Anthony Molho and Diogo Ramada Curto, eds., *Finding Europe: Discourses on Margins, Communities, Images, ca. 13th–ca. 18th Centuries* (Oxford, 2007), 133–158.

[36] Deanna Shemek, "In Continuous Expectation: Isabella d'Este's Epistolary Desire," in Dennis Looney and Deanna Shemek, eds., *Phaethon's Children: The Este Court and Its Culture in Early Modern Italy* (Tempe, Ariz., 2005), 269–300.

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so eloquently described in Ann Blair's 2010 book *Too Much to Know* was intimately connected to the increased scale and scope of scholarly correspondence in the late sixteenth and seventeenth centuries.[37] Where individuals had previously exchanged dozens or hundreds of letters, now their correspondences numbered in the thousands. Not surprisingly, the emergence of better postal systems in various regions accompanied this development.[38]

But the exponential growth of information did not end there. Between the seventeenth and the eighteenth century, the profusion of new learned institutions, the invention of journals, and the attractions of cosmopolitan centers transformed the idea of scholarly community from a humanist model of Latin men of letters to a more socially diffuse model of learned and vernacular communities of men and women writing, traveling, reading, and publishing. The Republic of Letters in this era was defined not only by the intellectual currents of the Scientific Revolution and the Enlightenment, but also by global missionary and mercantile networks, the Grand Tour, and the emergence of urban centers in the British and European colonies whose scholars both corresponded and traveled across the Mediterranean, Atlantic, and Indian Oceans. Individual correspondences that had routinely numbered in the low thousands now ballooned to tens of thousands of letters. The Aix savant Nicolas-Claude Fabri de Peiresc's famous scholarly network of the early seventeenth century, which has left us somewhere between 10,000 and 14,000 letters, and Leibniz's approximately 15,000 surviving letters exchanged with numerous correspondents between 1663 and 1716 are well-known examples of what the world of prolific letter writers now was capable of producing.[39]

While many documents pertaining to the Republic of Letters have not survived, the quantity of extant letters from these centuries, even when restricted primarily to scholarly correspondence, offers historians of the early modern period a body of material containing far more data than any one scholar, or even a team of researchers, can assimilate through traditional research methods. At the same time, letters are invaluable resources when they are brought together in a traditional edited volume or

[37] Blair, *Too Much to Know*.

[38] On postal systems, see Wolfgang Behringer, *Im Zeichen des Merkur: Reichspost und Kommunikationsrevolution in der Frühen Neuzeit* (Göttingen, 2003); Eric R. Dursteler, "Power and Information: The Venetian Postal System in the Eastern Mediterranean, 1573–1645," in Diogo Ramada Curto, Eric R. Dursteller, Julius Kirshner, and Francesca Trivellato, eds., *From Florence to the Mediterranean and Beyond: Essays in Honour of Anthony Molho*, 2 vols. (Florence, 2009), 2: 601–623; James Daybell, "Postal Conditions," in Daybell, *The Material Letter in Early Modern England: Manuscript Letters and the Culture and Practices of Letter-Writing, 1512–1635* (London, 2012), 109–147; Richard R. John, *Spreading the News: The American Postal System from Franklin to Morse* (Cambridge, Mass., 1995). For a GIS analysis of the French postal system, see Anne Bretagnolle and Nicolas Verdier, "Expanding the Network of Postal Routes in France, 1708–1833," in Muriel Le Roux, ed., *Post Offices of Europe, 18th–21st Century: A Comparative History* (Bern, 2014), 183–202.

[39] On Peiresc's correspondence, a good starting point is Robert A. Hatch, "Peiresc as Correspondent: The Republic of Letters and the Geography of Ideas," in Brian P. Dolan, ed., *Science Unbound: Geography, Space and Discipline* (Umeå, 1998), 19–58; and recent essays by Peter Miller such as "Mapping Peiresc's Mediterranean: Geography and Astronomy, 1610–1636," in Dirk van Miert, ed., *Communicating Observations in Early Modern Letters (1500–1675): Epistolography and Epistemology in the Age of the Scientific Revolution* (London, 2013), 135–160. On Leibniz, see Ultee, "The Republic of Letters," 98; Paul Lodge, ed., *Leibniz and His Correspondents* (Cambridge, 2004); Nora Gädeke, "Gottfried Wilhelm Leibniz," in Christiane Berkvens-Stevelinck, Hans Bot, and Jens Häseler, eds., *Les grands intermédiaires culturels de la République des lettres: Études de réseaux de correspondances du XVIe au XVIIIe siècles* (Paris, 2005), 257–306.

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increasingly as an online resource. Around the world, but particularly in Europe, the UK, and the United States, research projects are actively digitizing and cataloguing the surviving archives of the Republic of Letters.[40] While there is a considerable selection bias affecting whose records are being given new digital life, since most projects of this kind continue to focus on well-known figures, a host of forgotten characters are nonetheless emerging from this work: take Mlle Curton de Farges, who received a letter from Voltaire but is otherwise unknown to posterity.[41] Correspondence is diffuse and not entirely selective, after all; the act of writing a letter was possible for anyone possessing a certain degree of literacy or having access to someone who could write on his or her behalf. As such, it moves us beyond the usual criterion of selecting historical actors solely on the basis of their posthumous reputation. A great man or woman, after all, is only one node in an ego network. We have found ourselves repeatedly creating microhistories of people we did not expect to find and now wish to know.[42]

In many respects, the early modern Republic of Letters has proved to be an ideal testing ground for developing new ways of thinking about historical data. Put a different way, it is a rich source of multifaceted, if incomplete, data that traverses time and space and connects people. The Republic of Letters fundamentally relied on the exchange of words and things between people who established, and in some instances maintained, bonds of family, friendship, and patronage through letters to realize intellectual, religious, and political projects. Travel, long-distance communication, writing culture, and the production and circulation of printed matter structured many of its core activities. The Republic of Letters was not only an ideal, an imagined community in Benedict Anderson's sense, but a series of practices conducted by people who formed relationships. Human brokers, agents, gatekeepers, and go-betweens emerge from the terrifying mounds of paper that were produced. If books and journals were the culminating products of the Republic of Letters, conversation was its lifeblood—and those conversations took place as much in lively social gatherings and in more intimate dialogues (whether with people who lived nearby or with people who were brought into proximity by their travels) as they did in a form at times tangibly preserved for us: letters.

Even the earliest efforts to study the Republic of Letters confronted the difficulties of the data that this subject generates. Take Robert Mandrou's decision in 1973 to create a map of Erasmus's and Peiresc's correspondence and the diffusion of universities and Jesuit colleges in Western Europe. As Robert Hatch observed, Mandrou's Annaliste concern with the "geography of knowledge" led him to create static

[40] Indeed, one of our case studies began life as a digital archive more than ten years ago. For Paula Findlen, Suzanne Sutherland Duchacek, and Iva Lelková's use of the Athanasius Kircher Correspondence Project, jointly hosted by the Museo Galileo and Stanford University and originally created by Michael John Gorman and Nick Wilding at the European University Institute with assistance from Findlen, and most recently updated by Sutherland Duchacek and Lelková with technical support by Glen Worthey, see Athanasius Kircher at Stanford, "Correspondence," http://www.stanford.edu/group/kircher/cgi-bin/site/?page_id=7.

[41] Letter of January 11, 1776, in Nicholas Cronk, ed., *Digital Correspondence of Voltaire*, http://dx.doi.org/10.13051/ee:doc/voltfrVF1260336a1c.

[42] Note also how in our Grand Tour case study, published in this issue of the *AHR*, a host of little-known and often forgotten architects are brought back into the fold of eighteenth-century British architecture.

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portraits of a number of correspondents in different locations, while Hatch subsequently articulated Peiresc's network around people rather than places.[43] Mandrou's project was a pioneering attempt to create a historical visualization of an intellectual network, but his idea did not really bear fruit for the next few decades. In 1987, Ultee reiterated the need for developing new ways to explore this kind of historical data while he completed his analysis of Leibniz's correspondence, acknowledging the limitations of his ability to convey what he had found in words when he concluded, "Eventually a graphic presentation will clarify the links in this network."[44]

The more historians began to consider correspondence in relationship to the reconstruction of historical networks, the more urgent these questions became. By 2001, David Kronick, one of the pioneering historians of the early modern journal, noted the preliminary results of new techniques of digitizing and analyzing early modern correspondence, especially citing Urs Boschung's electronic catalogue of the voluminous correspondence of the eighteenth-century Swiss physician, naturalist, and poet Albrecht von Haller—almost 17,000 letters from nearly 1,200 correspondents—which subsequently became a highly productive case study in network analysis and visualizations by a team of researchers.[45] Haller's correspondence is one of numerous projects that have emerged in recent years in which digitization has stimulated new forms of analysis and new ways of presenting the results of these investigations.[46]

As we began our project in 2008, experiments with mapping the Republic of Letters proliferated. Robert Mayhew's account of early modern British geography demonstrates effectively how text-mining can help us understand the intellectual geography of the Republic of Letters and its understanding of physical geography.[47] Inspired by Steven J. Harris's early studies of mapping Jesuit science and Christopher Bayly's influential account of the relationship between information and empire, Simon Schaffer created a map of the "information order" of Isaac Newton's *Principia mathe-*

[43] Robert Mandrou, *From Humanism to Science, 1480–1700*, trans. Brian Pearce (New York, 1979), 309–313; Hatch, "Peiresc as Correspondent," 29–30.

[44] Ultee, "The Republic of Letters," 103. Shelford responded to this idea by providing useful maps of Huet's Republic of Letters; *Transforming the Republic of Letters*, 9, 32, 128.

[45] David A. Kronick, "The Commerce of Letters: Networks and 'Invisible Colleges' in Seventeenth- and Eighteenth-Century Europe," *Library Quarterly* 71 (2001): 28–43, here 40; Urs Boschung, *Repertorium zu Albrecht von Hallers Korrespondenz 1724–1777* (Basel, 2002), http://www.albrecht-von-haller.ch/medien/pdf/repertorium.pdf. For a network analysis of Haller's correspondence, see Martin Stuber, Stefan Hächler, and Luc Lienhard, eds., *Hallers Netz: Ein europäischer Gelehrtenbriefwechsel zur Zeit der Aufklärung* (Basel, 2005); Florence Catherine, "La pratique et les réseaux savants d'Albrecht von Haller (1708–1777), vecteurs de transfert culturel entre les espaces français et germaniques au XVIIIe siècle" (doctoral thesis, Université Nancy 2, 2009). A full bibliography of the research emerging from this collaborative project can be found at http://www.albrecht-von-haller.ch/e/correspondence.php.

[46] To pick another rich example that has produced a considerable body of recent scholarship engaged with questions of correspondence networks and the nature of epistolary commerce, see the Clusius Project, sponsored by the Scaliger Institute of the University of Leiden, http://www.library.leiden.edu/special-collections/scaliger-institute/projects/clusius-project.html, which has digitized the correspondence of the Dutch naturalist Carolus Clusius. See Florike Egmond, Paul Hoftijzer, and Robert Visser, eds., *Carolus Clusius: Towards a Cultural History of a Renaissance Naturalist* (Amsterdam, 2008); Florike Egmond, *The World of Carolus Clusius: Natural History in the Making, 1550–1610* (London, 2010). This archive has recently been integrated into another Dutch project co-sponsored by Huygens ING whose goal is to create a complete digital archive of manuscript facsimiles, transcriptions, and biographical information. Digital Edition of the Clusius Correspondence, http://www.dwc.knaw.nl/biografie/clusius/digital-edition-of-the-clusius-correspondence/.

[47] Robert Mayhew, "British Geography's Republic of Letters: Mapping an Imagined Community, 1600–1800," *Journal of the History of Ideas* 65, no. 2 (2004): 251–276.

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matica (1687) to debunk the notion that Newton was a solitary scholar untouched by a changing world.[48] Similarly, Yves Gingras explored the uses of citation and co-citation analysis, taking advantage of the digitization of Royal Society secretary Henry Oldenburg's correspondence by the Electronic Enlightenment Project at Oxford University and JSTOR's fully searchable version of the *Philosophical Transactions* to reconstruct the Scientific Revolution as a "visible map of the intellectual relations between people," arguing for the importance of combining "the usual micro-analysis of the specific content of the letters" with an ability to consider it instead as "a global corpus of data."[49] A group of French early modern historians, collaborating under the acronym CITERE, have also combined quantitative data and maps to analyze such variables as the speed and size of learned correspondence in eighteenth-century Europe.[50]

Many research questions underpinning scholarship on the Republic of Letters ultimately rest on data issues.[51] While recognizing that not everyone likes the idea of presenting early modern letters as filled with "data," worrying that it runs the risk of flattening the analysis or losing sight of the fact that these are historical documents whose creation is context-specific, we nonetheless believe that the information they contain deserves greater attention. Our experience in the Mapping the Republic of Letters project addresses a number of these questions to demonstrate how a digital and visual approach can indeed facilitate this sort of analysis. Building upon the rich historiography we have outlined above, we began with the insights gleaned from this considerable body of research. The inhabitants of the early modern Republic of Letters had elaborate visions of what their intellectual community resembled—most viewed it as a cosmopolitan, egalitarian place, and proudly highlighted their correspondents in far-flung places. Montesquieu famously satirized this self-perception in his *Persian Letters* (1721), where an astronomer boasts about corresponding with "a man in Stockholm, another in Leipzig, and another in London, whom I have never seen, and no doubt shall never see." Such descriptions parody the actual practices of the great secretaries of the Republic of Letters such as Oldenburg, who reportedly "never read a Letter before he had Pen, Ink, and Paper ready to answer it forthwith."[52] But could these descriptions really be trusted? Or was the reality of scholarly

[48] Simon Schaffer, *The Information Order of Isaac Newton's Principia Mathematica* (Uppsala, 2008). The history of science has actively incorporated certain insights from sociologist John Law's and sociologist and anthropologist Bruno Latour's contributions to Actor-Network Theory (ANT) to develop a rich account of the role of human and non-human actors in social networks. See, for example, Steven J. Harris, "Mapping Jesuit Science: The Role of Travel in the Geography of Knowledge," in John W. O'Malley, Gauvin Alexander Bailey, Steven J. Harris, and T. Frank Kennedy, eds., *The Jesuits: Cultures, Sciences, and the Arts, 1540–1773* (Toronto, 1999), 212–240.

[49] Yves Gingras, "Mapping the Structure of the Intellectual Field Using Citation and Co-Citation Analysis of Correspondences," *History of European Ideas* 36, no. 3 (2010): 330–339, here 330 and 339; and Yves Gingras and Alexandre Guay, "The Use of Analogies in Seventeenth and Eighteenth Century Science," *Perspectives on Science* 19, no. 2 (2011): 154–191.

[50] See Pierre-Yves Beaurepaire, ed., *La communication en Europe de l'âge classique au siècle de Lumières* (Paris, 2014). The English name for this project is Communicating Europe: Early Modern Circulations, Territories and Networks.

[51] For example, see the approach taken in many of the articles in Berkvens-Stevelinck, Bots, and Häseler, *Les grands intermédiaires culturels de la République des lettres*.

[52] Montesquieu, *Persian Letters*, trans. Margaret Mauldon (Oxford, 2008), 221; Marie Boas Hall, *Henry Oldenburg: Shaping the Royal Society* (Oxford, 2002), 129, quoting Martin Lister's recollection of Oldenburg in 1698.

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communication in the early modern age far more prosaic? Was the Republic of Letters genuinely cosmopolitan, or was it more regional, national, and confessional in scope? What did membership in this republic actually look like, and how might we bring different versions of this imagined community into focus by exploring well-known, concrete examples simultaneously? And how did it evolve over time and space? These were the questions that our research group set out to explore, initially without the aid of data or visualizations. But like so many scholars who came before us, we inevitably found ourselves grappling with what to do with the growing mountains of historical information about people, places, and time that lay at the center of our enterprise.

Is a big historical data approach the future or an illusion? It is easy to understand why some commentators have been predicting a "big data revolution," even in the humanities.[53] Not only are increasing numbers of texts and archives being digitized, but their accompanying metadata can now be validated and enriched via cultural heritage resources. For instance, we can begin to identify published authors across different datasets thanks to the adoption of authority files (such as VIAF), which link library records and other data sources through unique identifiers.[54] More and more libraries, including the OCLC, the Bibliothèque nationale de France, and Europeana, are also adopting Linked Open Data (LOD) models, making structured and curated data currently contained in library catalogues available for research.[55] The bibliographer's domain is fast becoming terrain for scholars to plumb and mine.

Much attention in the past fifteen years has been directed toward *text* digitization, but the production of vast stores of data—in particular, of metadata—is an equally exciting story. Anyone who has spent a few hours on Google Books can probably imagine a future in which all print material has been digitized; but imagining the future of data collection is more challenging. Ideally, if data models become reasonably standardized and well implemented, we might be able to look forward to a world in which datasets connect quickly and seamlessly, combining a wealth of discrete and useful information available to be reorganized into complex combinatorial patterns. If the print era was defined by the revolutionary technology of movable type, this coming

[53] See Mayer-Schönberger and Cukier, *Big Data*; Erez Aiden and Jean-Baptiste Michel, *Uncharted: Big Data as a Lens on Human Culture* (New York, 2013). For a similar endorsement by historians, see Guldi and Armitage, *The History Manifesto*.

[54] VIAF stands for "Virtual International Authority File" (http://viaf.org/). As the website states, this service "combines multiple name authority files [from different databases, mainly libraries] into a single OCLC-hosted name authority service. The goal of the service is to . . . increase the utility of library authority files by matching and linking widely-used authority files and making that information available on the Web."

[55] On Linked Open Data, see notably Tom Heath and Christian Bizer, *Linked Data: Evolving the Web into a Global Data Space* (San Rafael, Calif., 2011), http://www.morganclaypool.com/doi/abs/10.2200/S00334ED1V01Y201102WBE001; see also Eero Hyvönen, *Publishing and Using Cultural Heritage Linked Data on the Semantic Web* (San Rafael, Calif., 2012), http://www.morganclaypool.com/doi/abs/10.2200/S00452ED1V01Y201210WBE003. Our thanks to Glauco Mantegari for these references. The open data portal of the Bibliothèque nationale can be found at http://data.bnf.fr; for Europeana, see http://pro.europeana.eu/linked-open-data. Information about the OCLC's linked data plans (through WorldCat) can be found at http://www.oclc.org/data.en.html. Unfortunately, the process for converting MARC library data into LOD (in BIBFRAME format) is likely to take many, many years.

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era could well be defined by the equally revolutionary practice of "movable data." Once set in print and locked in books, catalogues, or tables, data can now be pried loose from their original settings and made available for new arrangements. Therein lies perhaps the greatest promise of data for historians: until now, most of us have been beholden to search forms for querying library catalogues, digital corpora, and other databases. We are constrained by what the library or the archive permits us to ask. Linked data models have the potential to make data more nimble and malleable, and subject to a much wider variety of uses.

But this prospect still lies in the future, and may yet prove to be a mirage. For one thing, when considering the promise of linking data across archives, we must consider the original state of the underlying data. The metadata for the historical records used in many of our case studies were constructed for data retrieval within particular online systems, not for a research inquiry.[56] The "place of publication" field in the Bibliothèque nationale contains a number of values similar enough for us to consider, but with subtle variations: Paris appears as "Paris," "(Paris)," "[Paris]," "(Paris,)" etc. To use these data effectively, we need to carefully decode the meanings in the parentheses, brackets, and commas that may or may not be documented (for instance, when is Paris the stated but false place of publication vs. when is it the unstated but real place?). The same can be said for digitized modern print editions: content structured for publishing is effectively unstructured for the purposes of data analysis. The scholarly parsing of existing data often leads us to create our own new columns of data that serve our research agenda.

Secondly, there is also the problem of incompleteness, often of unknown proportions. To borrow an example from our research, we were fortunate to receive at an early stage of our project a large dataset from the Electronic Enlightenment Project (EEP) at Oxford University.[57] The records contained the metadata for about 50,000 letters to and from leading seventeenth- and eighteenth-century authors, including Thomas Hobbes, John Locke, Voltaire, Rousseau, David Hume, Claude Adrien Helvétius, Adam Smith, and Jeremy Bentham. Along with these metadata, we also benefited from the painstaking curatorial work of the researchers at EEP, who had included information about nationality, occupation, gender, and dates for each individual in the collection. But the data were filled with gaps and uncertainties. This was in no way the fault of EEP, which had produced as fine a database as possible (in addition to their remarkable full-text correspondence database). Rather, the data were incomplete because the historical record was incomplete. Many letters had been passed down without any indication of their source or destination locations; some were also undated. Others had vanished off the face of the earth.

This critical reading and assessment of the state of the data is common to each of our case studies in Mapping the Republic of Letters. This distinction points to a third significant methodological difference in our approach to the objects of our study. If our historical subjects had been communicating digitally, and we could have captured their exchanges and movements from a real-time live data stream, we might be able to

[56] This is why scholarly cataloguing projects such as Early Modern Letters Online (EMLO), developed by the Cultures of Knowledge Project at Oxford University, are so important for work on early modern correspondence. See http://emlo.bodleian.ox.ac.uk/.

[57] See http://www.e-enlightenment.com/.

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adopt the statistical methods favored by data scientists. But historical information does not stream forth like a Twitter feed. The datasets that we have explored, all of which were manually scrutinized, edited, and enriched by our research teams, came to us already structured, and sometimes designed for purposes contrary to our own. Most of the metadata we rely upon from these digitized archives have been framed and reframed for various uses over decades. We need to unmask their underlying assumptions in order to get down to the level of historical evidence. The role of the scholar thus becomes one of finding meaning in the mass, which in the case of historical data is often also a mess.

Our final problem is more pragmatic and has greatly preoccupied our research project: Where are the tools to support humanistic inquiry into multidimensional, heterogeneous, and incomplete datasets? To take maximum advantage of this coming data deluge, we will need new technologies to help us construct meaningful queries and then explore, filter, combine, and analyze the results.[58] We need to see the past from many different vantage points in order to understand it, and that cannot happen in front of an infinite tabular spreadsheet.[59] Visualization has proven to be an extremely effective tool for representing data and navigating data across many disciplines.[60] ArcGIS, Tableau, R, and Gephi are some of the most powerful desktop applications and programming languages for data visualization used in humanities research today. And yet the quantitative bias that defines much of the functionality of those tools does little to support the kind of qualitative analysis practiced in our case studies. We require different tools that can accommodate ambiguities, paradoxes, and contingencies to help us process historical and more generally humanistic data effectively.

Designing digital tools is a long-term, expensive, complicated process, one that typically involves working with grant officers, data scientists, programmers, and interface designers. Needless to say, we had no idea what we were getting into when we set off down this path. This phase of our collaboration arose from the growing realization that we simply could not begin to explore, to any serious degree, the questions that we wanted to pose without experimenting with different approaches to visualizing historical data. We could borrow or build a database or two, but what did we want to do with it? This became the crucial question that led us in the direction that ultimately resulted in the creation of a web-based data visualization platform, Palladio.[61]

[58] For a similar point, see Guldi and Armitage, *The History Manifesto*, chap. 4.

[59] For a fascinating historical reflection on this problem that has inspired aspects of our own historical visualizations, see Daniel Rosenberg and Anthony Grafton, *Cartographies of Time: A History of the Timeline* (New York, 2010).

[60] Evidence of this can be found in the development of the commercial visualization software packages mentioned below, as well as in the popularity of recent books and websites on the topic from the fields of statistics, design, business analytics, and computer science. For some prominent examples, see Edward Tufte, http://www.edwardtufte.com; Manuel Lima, http://www.visualcomplexity.com/vc/; Stephen Few, http://perceptualedge.com; Alberto Cairo, http://www.thefunctionalart.com; David McCandless, http://www.informationisbeautiful.net.

[61] Palladio is a data-visualization platform that allows users to upload and analyze their data in the browser; funds for its development came from the National Endowment for the Humanities and Stanford University. It brings together much of what we learned in the design and development of prototype tools over the past six years as members of the Mapping the Republic of Letters project at Stanford University, primarily in collaboration with the Density Design research lab at the Politecnico di Milano, and in conversation with allied initiatives such as the Cultures of Knowledge project at Oxford University, as well as the Circulation of Knowledge project jointly sponsored by Huygens ING in The Hague and the Des-

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Since our project has always had humanistic pursuits at heart, it is no surprise that our underlying philosophy of tool development would seek to enhance, not replace, traditional historical methods. Big data are not useful only to scholars who adopt statistical methods. The large scale also allows us to pursue traditional research in a much broader space of possibilities. When we construct visual models with data, we consider those visualizations to be powerful heuristics, not proofs. We wield our data with the assumption that they are incomplete, ambiguous, and uncertain. We visualize them to help guide us, to uncover patterns and provoke questions that we otherwise might not ever have considered. Our publications include online access precisely so that readers can explore and manipulate the data of our case studies. These use cases of Palladio allow direct insight into the process of heuristic visualizations, and on these same web pages readers can download the data to pursue their own research questions as well.[62] But we do not believe that the data in and of themselves provide answers. The interpretive work is done outside of the computer, in the decisions that go into building our datasets, in the design of the instruments we use to visualize the data, and finally in making sense of the very imprecise maps, graphs, and diagrams we produce. We still need to read those shelf-loads of books and boxes of manuscripts—in the archive or online—to make sense of our findings.

As a result of dealing with data visualizations, we found ourselves thinking anew about space. Many of the patterns that interest us, and that our visualizations help to uncover, concern geographical relations. This puts our project in dialogue with spatial history, an area in which digital research has had a great impact. But our material and questions have inspired a distinct methodology for our research. Many spatial history projects analyze the relationships between different categories of space—for instance, between spatial practice and represented space, to borrow Henri Lefebvre's terms.[63] This approach often involves comparing *layers* of space—quite literally in the case of the HyperCities project, for example.[64] By contrast, our project has been much more concerned with the relationships between different areas within a single layer of space. In fleeing Germany, Kircher was cast adrift from his native land, and his correspondence captures beautifully the essence of Jesuit mobility as well as how people maintained a strong sense of identity at a distance. Locke's exile in France belonged to one of those moments when a portion of the "English" world was on the Continent, as would later occur on a much larger scale when the Grand Tour created a world in

cartes Centre at the University of Utrecht. See http://hdlab.stanford.edu/projects/palladio/. We count it as a success of our project that Palladio has now been used widely in a variety of projects well beyond our own within Mapping the Republic of Letters. For a recent assessment, see Graham, Milligan, and Weingart, *Exploring Big Historical Data*, 112, 268–269.

[62] See http://republicofletters.stanford.edu/publications, which hosts the online interactive visualizations, databases, and data schemas for our case studies. Note that the datasets for each case study are also stored in SDR (the Stanford Digital Repository), https://sdr.stanford.edu/collections/druid:zn653qj0117, to ensure their longevity in what might otherwise at times seem the quite ephemeral afterlife of digital projects, and where they can be accessed openly by the scholarly community.

[63] See Richard White, "What Is Spatial History?," Spatial History Lab: Working Paper, submitted February 1, 2010, http://web.stanford.edu/group/spatialhistory/cgi-bin/site/pub.php?id=29. White draws on Henri Lefebvre, *The Production of Space*, trans. Donald Nicholson-Smith (Cambridge, Mass., 1991).

[64] HyperCities: Thick Mapping in the Digital Humanities, http://hypercities.com/.

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between England and Italy for multiple generations of Britons. Benjamin Franklin, remembered today as the archetypal American, in fact spent much of the last third of his life largely in England and the Continent, often doing business in which the lines between what was "British" and what was "American" were blurred to the point of meaninglessness. When we reflect on the connections among England, France, and North America in the eighteenth century, for example, we are essentially asking how an "English space" was situated in relation to a "French space," or an American one within a British world. Instead of superimposing layers, we look at a patchwork of spaces, all connected on the same plane by the human interactions that constituted the early modern Republic of Letters.[65]

The kind of spaces we study might best be described as "cultural zones."[66] Such zones exist, almost by definition, in the plural: to a large degree, they are defined relative to and in distinction from one another. Our project requires us to identify these zones spatially and conceptually, but also to reflect on their interactions.

At the geographical level, we do not place much importance on exact boundaries or even precise locations; for this reason, we have not used GIS (geographic information systems) in our visualizations, and we tend to prefer abstract maps over historical ones. We take as a given that cultural zones have fuzzy borders and can overlap: eighteenth-century Geneva, for instance, could be seen as belonging to both a Swiss Protestant and a French cultural zone. We use the term "cultural zone" rather than "cultural space" in order to stress its imprecise, often amorphous, edges. A zone also covers a large area, which is why we chose this term over "place." It can indeed capture connections between places in a given period.

Much of spatial theory, from Georg Simmel and Henri Lefebvre to David Harvey, has roots in urban studies.[67] Cities are also central to our thinking about cultural zones, but in our case we look *extra muros*. Indeed, a cultural zone can often be defined as the area affected by the dominance of a city (usually a capital). The French cultural zone radiates around Paris; in the eighteenth century, its reach could be felt as far away as St. Petersburg. The English cultural zone extended around London, and reached as far as Philadelphia and the Indian subcontinent. Rome's reputation as *caput mundi* was certainly enhanced by its role as the nerve center of overlapping global missionary networks whose reach extended to every corner of the world. Of course, capitals did not always play this role: not every region created cultural zones

[65] On place vs. space, see Edward S. Casey, *The Fate of Place: A Philosophical History* (Berkeley, Calif., 1997); Philip J. Ethington, "Placing the Past: 'Groundwork' for a Spatial Theory of History," *Rethinking History* 11, no. 4 (2007): 465–493.

[66] The concept of "zone," to the extent that it is used in spatial theory, tends to be deployed by literary scholars in relation either to cities (see, for instance, David Bell, Jon Binnie, Ruth Holliday, Robyn Longhurst, and Robin Peace, *Pleasure Zones: Bodies, Cities, Spaces* [Syracuse, N.Y., 2001]) or to climatic "zones" (see Felicity Nussbaum, *Torrid Zones: Maternity, Sexuality, and Empire in Eighteenth-Century English Narratives* [Baltimore, 1995]). There has also been some reflection on this spatial category in postcolonial scholarship; see Imre Szeman, *Zones of Instability: Literature, Postcolonialism, and the Nation* (Baltimore, 2003). Among historians, Peter Sahlins considers zones in relation to frontiers in *Boundaries: The Making of France and Spain in the Pyrenees* (Berkeley, Calif., 1991). We use the term in the sense provided by the *Oxford English Dictionary*: "A definite region or area of the earth, or of any place or space, distinguished from adjacent regions by some special quality or condition (indicated by a defining word or phrase)."

[67] See Georg Simmel, "The Metropolis and Mental Life" (1903), in Gary Bridge and Sophie Watson, eds., *The Blackwell City Reader* (Oxford, 2002), 11–19; Lefebvre, *The Urban Revolution*, trans. Robert Bononno (Minneapolis, 2003); David Harvey, *Social Justice and the City*, revised ed. (Athens, Ga., 2009).

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around a single capital city. But in these instances, cultural zones still formed primarily around a constellation of cities.

Our general focus on capitals is not meant to deny the importance of other centers in the circulation or creation of knowledge and goods.[68] For example, Rome's centrality in Grand Tour travels gains meaning only in relation to other Italian destinations—and in any case travelers could not even reach Rome without journeying through other cities and regions. But for definitional purposes, the periphery of a cultural zone is extremely hard to identify spatially (i.e., to map). The edges of a cultural zone tend to be unstable, and can vary depending on the contingencies of war, empire-building, and governmental policy. The Jesuit diaspora was constantly expanding and contracting as new opportunities emerged and others became unstable or vanished entirely. For instance, pockets of a French cultural zone emerged in London and the Dutch Republic after the revocation of the Edict of Nantes in 1685. Cultural zones are rarely demarcated by a wall, a line, or a river. At the edges, they just fade away, as other cultural zones take their place. Cities, by contrast, are much more easily localized. They anchor the cultural zone in cartographic space.

At the conceptual level, what makes these zones cultural? Here we can only point to a wide array of features, most of which have, in themselves, little to do with space. A language, a religion, a system of social hierarchy, literary and artistic tastes, fashions, food, constitutional theories, and so forth: all these features, and others, contribute to the production of a cultural zone, and each zone will be more or less determined by different features. That said, these practices and ideas are nonetheless spatial in that they prevail in a certain zone, and beyond that zone, different practices and ideas prevail. Political institutions are obviously important for establishing differences between cultural zones, but they are not sufficient: the American states remained in an English cultural zone well after independence. In other words, a cultural zone and a political sphere are not necessarily the same thing.

THE ARGUMENTS ADVANCED HERE are mainly intended to answer questions about our methodology; they are not meant to serve as a manifesto, and there are of course slight but telling differences in how each of our case studies maps an episode in the Republic of Letters. Since every database is different, and since the needs of different scholarly agendas are best met by different technologies, we do not think it possible or even desirable to dictate a unique methodological approach for all experiments in digital history. One objective of these reflections is to encourage other scholars to venture out into these still largely uncharted waters, using what we have learned along the way as either a guide or a foil.

But we also have our own pipe dreams about the future of digital history, particularly where intellectual and cultural history are concerned. Given that our own collection of metadata was pooled from different sources, we experienced firsthand the advantages of studying historical networks from multiple angles.[69] This multiplicity can

[68] See, e.g., Londa Schiebinger, *Plants and Empire: Colonial Bioprospecting in the Atlantic World* (Cambridge, Mass., 2004).

[69] Our largest source of metadata was the EEP; the Humanities Packard Institute gave us the metadata for the Benjamin Franklin correspondence; Stanford Libraries, for the Kircher correspondence; the

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uncover networks that exist at one or two (if not six) degrees of separation. Attempting to make our databases converge has not been an explicit goal of the project, since it is not yet clear that doing so would yield any strong insights. Franklin and Voltaire never corresponded directly (and we know of only one meeting in person), but because we are in possession of the metadata for both their correspondences, we can identify everyone with whom they both corresponded.[70] Kircher and Locke had even less reason to meet, and in fact never did meet, and they certainly never corresponded, so perhaps there is nothing particularly revealing to be gained by exploring the degree to which their correspondence intersects, since they shared so little. Instead, each illuminates a largely distinctive dimension of seventeenth-century intellectual life and its political, religious, and commercial entanglements. By contrast, the Grand Tour Project thrives on identifying points of intersection that emerge as people meet in Rome, Florence, Venice, and Naples, among other places, form temporary communities forged by travel, and potentially renew their acquaintance around an architectural commission that reminds a patron and an architect that they once shared the experience of Italy together. These examples highlight the potential benefits of having overlapping metadata: it allows us to explore possible connections by following the trail.

The metadata in our own databases is heterogeneous and somewhat arbitrary; it reflects our varied interests and the availability of such information, so we have been opportunistic in selecting projects where it was already possible to do this kind of work. Its limitations also showcase, *a contrario*, what a more systematic approach could yield. Imagine a database even larger in scale than the one currently under construction by Oxford's Cultures of Knowledge project—Early Modern Letters Online (EMLO), which contains the entire Kircher database, among many others—in which all metadata relating to scholarly communication could be pooled.[71] Each person would have a unique identifier, connecting all of his or her interactions, and embedded in a much larger pool of information that spanned two hundred, three hundred, or four hundred years of intellectual activity. Add to this the possible bibliographic dimensions, with each author linked to the relevant holdings in major libraries; and throw in other assortments of metadata, such as membership in learned academies, participation in salons or clubs, university education, occupation, travels, and so on. This would indeed be a multifaceted archive for all of us to explore.

No doubt such a database is a fantasy, and we are not suggesting that it could ever exist in complete form, notably for the reasons advanced above. But a partial realization of this encyclopedia of metadata is not beyond our abilities. A pan-European re-

data for the Grand Tour came from Ingamells, *A Dictionary of British and Irish Travellers in Italy*. For an interesting reflection on the advantages of pooling data, see Gingras, "Mapping the Structure of the Intellectual Field Using Citation and Co-Citation Analysis of Correspondences." We also warmly thank Robert Hatch at the University of Florida for generously sharing his experience constructing and working simultaneously with correspondence databases for his research on the relationship between the Republic of Letters and the Scientific Revolution.

[70] Franklin met Voltaire when the latter was initiated into a Masonic Lodge; see Nicholas Hans, "UNESCO of the Eighteenth Century: *La Loge des Neuf Sœurs* and Its Venerable Master, Benjamin Franklin," *Proceedings of the American Philosophical Society* 97, no. 5 (1953): 513–524. The list of their shared correspondents includes Jean Le Rond d'Alembert, Pierre-Jean-Georges Cabanis, Octavie Durey de Mesnières, Mme Helvétius, David Hume, André Morellet, and Anne-Robert-Jacques Turgot.

[71] The EMLO catalogue at http://emlo.bodleian.ox.ac.uk/ is a promising step in this direction.

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search network, Reassembling the Republic of Letters, has just set out with a similar vision, and in fact, we can draw inspiration from the remarkable work undertaken by classicists to create a "cyberinfrastructure" that enables and encourages interoperability between digital projects.[72] However, we also need tools, developed for and by humanists, to explore and analyze these data, to make them meaningful. And even in this ideal scenario, the time will invariably come when it will be necessary to turn away from the screen, pick up a book, and try to figure out what these piles of historical information mean. That is the exercise that all of our publications seek to do. We have not attempted to map the Republic of Letters without forgetting why we began this project: because the figures who populate our project are interesting for many reasons and left behind a rich trail of documentation whose content we need to read with care if we hope that we will ever be able to understand them.

[72] See http://www.republicofletters.net/ for more information on the Reassembling the Republic of Letters project, with which Mapping the Republic of Letters is now affiliated, and which counts collaborators from more than thirty European countries, as well as from Canada and Australia. For the "cyberinfrastructure," see *Changing the Center of Gravity: Transforming Classical Studies through Cyberinfrastructure*, ed. Gregory Crane and Melissa Terras, Special Issue, *Digital Humanities Quarterly* 3, no. 1 (2009).

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